

Generalized Grade-of-Membership Estimation for High-dimensional Locally Dependent Data

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Abstract

This work focuses on the mixed membership models for multivariate categorical data widely used for analyzing survey responses and population genetics data. These grade of membership (GoM) models offer rich modeling power but present significant estimation challenges for high-dimensional polytomous data. Popular existing approaches, such as Bayesian MCMC inference, are not scalable and lack theoretical guarantees in high-dimensional settings. To address this, we first observe that data from this model can be reformulated as a three-way (quasi-)tensor, with many subjects responding to many items with varying numbers of categories. We introduce a novel and simple approach that flattens the three-way quasi-tensor into a “fat” matrix, and then perform a singular value decomposition of it to estimate parameters by exploiting the singular subspace geometry. Our fast spectral method can accommodate a broad range of data distributions with arbitrarily locally dependent noise, which we formalize as the generalized-GoM models. We establish finite-sample entrywise error bounds for the generalized-GoM model parameters. This is supported by a new sharp two-to-infinity singular subspace perturbation theory for locally dependent and flexibly distributed noise, a contribution of independent interest. Simulations and applications to data in political surveys, population genetics, and single-cell sequencing demonstrate our method’s superior performance.

Keywords: Grade of Membership Model; Local Dependence; Mixed Membership; Spectral Method; Two-to-infinity Singular Subspace Perturbation.

1 Introduction

Multivariate categorical data are prevalent in various fields, including social science surveys (Clinton et al., 2004), population genetics (Pritchard et al., 2000), health sciences (Manton

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et al., 1994), text analysis (Blei et al., 2003), among others. These data consist of high-dimensional categorical variables, with many subjects responding to a large number of items with varying numbers of categories. Identifying interpretable latent patterns to capture heterogeneity in such data is of great interest. Among latent variable models for multivariate categorical data, the grade of membership (GoM) model (Woodbury et al., 1978; Erosheva et al., 2007) stands out as a popular tool that assumes each subject’s latent variables live in a simplex. The GoM model falls within the broad family of mixed membership models for individual-level mixtures (Airoldi et al., 2015). Unlike traditional population-level mixture models that restrict each individual to belong to a single latent class, the GoM model more flexibly allows each individual to partially belong to multiple latent classes. In a GoM model with K extreme latent classes, each individual has K nonnegative membership scores that sum to one, quantifying the extent to which the individual belongs to each class.

The flexibility of GoM models, however, brings significant challenges in estimation, particularly for high-dimensional polytomous data. Due to the complex hierarchical modeling nature of GoM models, most existing studies adopt the Bayesian approach with Markov Chain Monte Carlo (MCMC) algorithms for posterior inference (Erosheva, 2002; Erosheva et al., 2007; Bhattacharya and Dunson, 2012; Gu et al., 2023). Despite their popularity, these methods face critical challenges in high-dimensional settings. MCMC methods often suffer from high computational cost, making them hard to scale to big data. Also, Bayesian or likelihood-based methods are usually tailored to a specific probabilistic model, limiting their applicability across different data distributions. Extending GoM model estimation methods developed for dichotomous data (Robitzsch and Robitzsch, 2017; Chen and Gu, 2024) to polytomous data also requires careful new development. Furthermore, the theoretical guarantees for estimating the GoM model in high-dimensional settings remain unknown.

Another challenge in GoM models is the local dependence issue among items, meaning that a subject’s observed multivariate responses exhibit additional dependencies that are

not fully explained by the latent variables. Local dependence has long been known as a practical issue when applying latent variable measurement models in social science (Chen and Thissen, 1997; Chen et al., 2018). For instance, in the American National Election Studies (ANES) dataset, blocks of items concerning voting participation, presidential approval, economics, and other topics may display local dependence within each topic. In addition to social science, in genome-wide association studies (GWAS), linkage disequilibrium (LD) – where neighboring alleles are correlated – represents a form of local dependence (Uffelmann et al., 2021). In single-cell data, gene co-expression that reflects dependencies between genes (Van Der Wijst et al., 2018) also constitutes local dependence. It is crucial to devise accurate estimation methods for locally dependent GoM models, especially in high-dimensional settings where such dependencies are often more pronounced.

To address the above challenges, we propose a fast spectral method based on singular value decomposition (SVD) for high-dimensional data with potential local dependence. We first observe that the multivariate categorical data can be reformulated as a three-way (quasi-)tensor, with many subjects responding to many items with varying numbers of categories. We employ a simple yet novel technique to flatten this quasi-tensor into a “fat” matrix and then perform SVD to estimate parameters by exploiting the singular subspace geometry. We further introduce the generalized-GoM models, which encompass a wide range of data distributions with potential local dependence and cover the traditional GoM model as a special case. We establish finite-sample entrywise estimation error bounds for generalized-GoM parameters. To achieve this, we develop a novel sharp two-to-infinity singular subspace perturbation theory for arbitrary locally dependent data from flexible distributions, which is of independent interest. Extensive simulation studies demonstrate that our method efficiently and accurately estimates the generalized-GoM parameters with local dependence that either results from the flattening technique or is a consequence of the intrinsic data generation mechanism. We apply our method to a diverse range of real-world datasets in political

surveys, population genetics, and single-cell sequencing, and uncover interesting insights.

The rest of the paper is structured as follows. Section 2 explains the background and motivation for this work and introduces the model setup. Section 3 discusses model identifiability and presents a spectral method for generalized-GoM models. Section 4 develops a two-to-infinity singular subspace perturbation theory for locally dependent data from flexible distributions and establishes entrywise error bounds for parameter estimation. Section 5 reports the results of simulation studies and Section 6 presents three real-data applications. Section 7 concludes the paper. The R codes implementing the proposed method are available at <https://github.com/lscientific/gGoM>. All technical proofs of the theoretical results and additional computation details are provided in the Supplementary Material.

2 Motivation and Model Setup

2.1 GoM for Multivariate Categorical Data and Existing Methods

We first review the classical GoM model for multivariate categorical data (Woodbury et al., 1978; Manton et al., 1994; Erosheva, 2002). Consider categorical responses from N subjects to L items, and denote the response of subject i to item l by $\tilde{R}_{i,l} \in \{1, \dots, C_l\}$, for $i \in [N]$ and $l \in [L]$, where $C_l \geq 2$ denotes the number of categories of item l .

The GoM model uses latent mixed membership structures to model multivariate categorical data. It has two levels of modeling: the population level and the individual level. On the population level, there are K *extreme latent profiles*, each representing a prototypical response pattern. For each $k \in [K]$, the extreme latent profile k corresponds to an item parameter vector $\tilde{\boldsymbol{\theta}}_k = (\tilde{\boldsymbol{\theta}}_{1,k}^\top, \dots, \tilde{\boldsymbol{\theta}}_{L,k}^\top)^\top$ with $\tilde{\boldsymbol{\theta}}_{l,k} = (\tilde{\theta}_{l,k,1}, \dots, \tilde{\theta}_{l,k,C_l})^\top \in [0, 1]^{C_l}$, and

$$\tilde{\theta}_{l,k,c} = \mathbb{P}(\tilde{R}_{i,l} = c \mid \text{subject } i \text{ solely belongs to the } k\text{-th extreme latent profile})$$

for $c \in [C_l]$, $l \in [L]$. Note that $\sum_{c=1}^{C_l} \tilde{\theta}_{l,k,c} \equiv 1$ always holds for each $l \in [L]$ and $k \in [K]$. On the individual level, each subject i is associated with a vector of mixed membership scores

$\boldsymbol{\pi}_i^* = (\pi_{i1}^*, \dots, \pi_{iK}^*)$, with $\pi_{ik}^* \geq 0$ and $\sum_{k=1}^K \pi_{ik}^* = 1$. Each membership score π_{ik}^* indicates the extent to which subject i partially belongs to extreme latent profile k . The item response probability for the i -th subject's response to the l -th item is a convex combination:

$$\mathbb{P}(\tilde{R}_{i,l} = c) = \sum_{k=1}^K \pi_{i,k}^* \tilde{\theta}_{l,k,c}, \text{ for } c \in [C_l].$$

We collect all item parameters in matrix $\tilde{\boldsymbol{\Theta}} = (\tilde{\boldsymbol{\vartheta}}_1, \dots, \tilde{\boldsymbol{\vartheta}}_K) \in [0, 1]^{(\sum_{l=1}^L C_l) \times K}$.

Most existing studies on the GoM model treat the membership scores as random effects coming from some distribution on the simplex. It was also typically assumed that a subject's responses $\tilde{R}_{i,1}, \dots, \tilde{R}_{i,L}$ are conditionally independent given the membership scores $\boldsymbol{\pi}_i^*$, which is called the local independence assumption. Under this assumption, if $\boldsymbol{\pi}_i^*$ are considered to be i.i.d random vectors, for example, $\boldsymbol{\pi}_i^*$ follows the Dirichlet distribution $D_{\boldsymbol{\alpha}^*}(\cdot)$ parameterized with $\boldsymbol{\alpha}^* \in \mathbb{R}_+^K$, then the marginal likelihood function of a GoM model is

$$L(\tilde{\boldsymbol{\Theta}}, \boldsymbol{\alpha}^* | \tilde{\mathbf{R}}) = \prod_{i=1}^N \int_{\Delta_{K-1}} \prod_{l=1}^L \prod_{c=1}^{C_l} \left(\sum_{k=1}^K \pi_{i,k}^* \tilde{\theta}_{l,k,c} \right)^{\mathbb{1}(\tilde{R}_{i,l}=c)} dD_{\boldsymbol{\alpha}^*}(\boldsymbol{\pi}_i), \quad (1)$$

where $\Delta_{K-1} = \{\mathbf{x} = (x_1, \dots, x_K) : x_i \geq 0, \sum_{i=1}^K x_i = 1\}$ denotes the probability simplex. The marginal maximum likelihood (MML) approach estimates the population parameters $(\tilde{\boldsymbol{\Theta}}, \boldsymbol{\alpha}^*)$ by maximizing the marginal likelihood function (1). The integrals over the membership scores are hard to evaluate, posing challenges for MML. Currently, Bayesian inference with MCMC algorithms is arguably the most popular estimation paradigm for the GoM model and its variants (e.g., [Airolidi et al., 2015](#); [Erosheva, 2002](#); [Erosheva et al., 2004](#); [Bhattacharya and Dunson, 2012](#); [Gu et al., 2023](#); [Kang and Gu, 2024](#)). But MCMC algorithms often suffer from high computational cost and do not scale to high-dimensional data.

In addition to Bayesian MCMC methods, there exist several optimization-based estimation methods for the GoM model. [Wang and Erosheva \(2015\)](#) developed a heuristic variational inference method for the GoM model. Another optimization-based estimation approach is the joint maximum likelihood (JML) method ([Manton et al., 1994](#)). JML treats

the mixed membership scores as fixed parameters and maximizes the joint likelihood function to simultaneously estimate $\mathbf{\Pi}^*$ and $\tilde{\mathbf{\Theta}}$. Another study Zhao et al. (2018) proposed a moment estimation method that leverages the second and third order moments to estimate the population parameters in Dirichlet latent variable models with mixed data types.

2.2 Mixed Membership Structures for Other Data Types

Related mixed membership (MM) models have also been used to model multivariate data from the Binomial and Poisson distributions in some applications. For example, the number of minor allele of a single nucleotide polymorphism (SNP) is usually assumed to follow a Binomial distribution. Let $\mathbf{\Theta}^* = (\theta_{j,k}^*)$ and $\theta_{j,k}^* \in (0, 1)$ represents the binomial parameter for the j -th SNP and the k -th *ancestry profile*, then the genotype $R_{i,j}$ follows $\text{Binom}(2, \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^*)$ for $i \in [N], j \in [J]$. The marginal likelihood function for a Binomial MM model is

$$L(\mathbf{\Theta}^*, \mathbf{\alpha}^* | \mathbf{R}) = \prod_{i=1}^N \int_{\Delta_{K-1}} \prod_{j=1}^J \binom{2}{R_{i,j}} \left(\sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{R_{i,j}} \left(1 - \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{2-R_{i,j}} dD_{\mathbf{\alpha}^*}(\boldsymbol{\pi}_i)$$

when assuming $\pi_{i,k}^*$ follows a distribution $D_{\mathbf{\alpha}^*}(\cdot)$ parametrized by $\mathbf{\alpha}^*$. The influential MCMC algorithm STRUCTURE (Pritchard et al., 2000) imposes a Dirichlet prior on $\pi_{i,k}^*$ and estimates this *admixture model* for Binomial distributed genotype data. As an example for count data, let $R_{i,j}$ represent the expression count for the gene j in the cell i in single-cell sequencing data. In this case, $\theta_{j,k}^* \in \mathbb{R}_+$ denotes the Poisson rate and $R_{i,j} \stackrel{\text{ind.}}{\sim} \text{Poisson}(\sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^*)$ for $i \in [N], j \in [J]$. The joint likelihood function for a Poisson MM model is given by

$$L(\mathbf{\Theta}^*, \mathbf{\Pi}^* | \mathbf{R}) = \prod_{i=1}^N \prod_{j=1}^J \frac{1}{R_{i,j}!} \left(\sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{R_{i,j}} \exp \left(- \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right). \quad (2)$$

To maximize (2), nonnegative matrix factorization (NMF) with iterative multiplicative updates is a popular approach (Lee and Seung, 1999, 2000). This algorithm depends on the Poisson distribution assumption and is sensitive to initializations.

2.3 Generalized-GoM Models and Our Assumptions

To facilitate statistical analysis of high-dimensional polytomous data, we introduce a simple but novel approach that flattens the three-way quasi-tensor into a “fat” binary matrix (see Figure 1), with columns corresponding to the response categories of all items. Define $J := \sum_{l=0}^L C_l$ with $C_0 = 0$ and consider the binary matrix $\mathbf{R} = (R_{i,j}) \in \mathbb{R}^{N \times J}$ with

$$R_{i, \sum_{m=0}^{l-1} C_m + c} = \mathbb{1}(\tilde{R}_{i,l} = c), \quad \text{for } c \in [C_l]. \quad (3)$$

Similarly, we define the corresponding flattened item parameter matrix $\Theta^* = (\theta_{jk}^*) \in [0, 1]^{J \times K}$ with $\theta_{\sum_{m=0}^{l-1} C_m + c, k}^* = \tilde{\theta}_{l,k,c}$ for $c \in [C_l], l \in [L], k \in [K]$. The columns of $\mathbf{R}$ consist of L blocks, with the size of the l -th block equal to the number of categories for the l -th item. Within the l -th block, we always have $\sum_{c=1}^{C_l} R_{i, \sum_{m=0}^{l-1} C_m + c} \equiv 1$. Apparently, this flattening technique introduces local dependence within each block that cannot be accounted for by the latent variables. Figure 1 visualizes flattening a three-way quasi-tensor into a fat binary matrix. We interpret this figure using the ANES data (see Section 6) as an illustration. Each of the five rows represents a subject that responds to four items regarding illegal immigration, climate, political participation, and taxes, respectively. The number of response categories for the four items are 5, 3, 2, 3, as reflected as the number of columns for the cubes in the middle of the figure. For instance, item 1 asks “*How important is the illegal immigration issue in the country today?*” and the 5 possible responses vary from “Extremely important” to “Not at all important”. Item 3 asks “*During the past 12 months, have you posted a message or comment online about a political issue or campaign?*” and the possible responses are “Yes” and “No”. Similar concatenated data structures have been considered in the multilayer random graph models (Jones and Rubin-Delanchy, 2020; Gallagher et al., 2021), and Arroyo et al. (2021) considers such structures in the spectral subspace.

To accommodate flexible local dependence structures and various data types, we introduce a class of *generalized-GoM* models that support general data distributions, including those with sub-exponential tails and covering the classical GoM model as a special case.

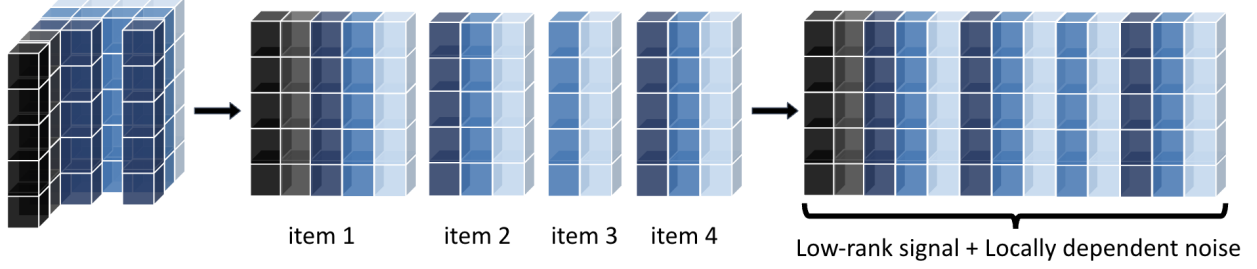


Figure 1: Visualization of the flattening technique. Cubes on the left represent the three-way quasi-tensor $\tilde{\mathbf{R}}$, and those on the right represent the flattened binary matrix $\mathbf{R}$.

Write the (flattened) data matrix $\mathbf{R}$ as the sum of a mean matrix $\mathbf{R}^*$ and a noise matrix $\mathbf{E}$,

$$\mathbf{R}^* := \mathbb{E}[\mathbf{R}] = \underbrace{\mathbf{\Pi}^*}_{N \times K} \underbrace{\mathbf{\Theta}^{*\top}}_{K \times J}, \quad \mathbf{E} := \mathbf{R} - \mathbf{R}^*,$$

where $\mathbf{\Pi}^* = (\pi_{i,k}^*) \in [0, 1]^{N \times K}$ includes the membership scores, and $\mathbf{\Theta}^* = (\theta_{j,k}^*) \in \mathbb{R}^{J \times K}$ collects all the item parameters. Such expectation-noise decomposition is commonly used in the topic modeling and network modeling literature, generally assuming independent entries in the noise matrix $\mathbf{E}$. In contrast, we impose the following Assumption 1 to characterize the general block dependence structure of the noise entries, along with their tail behavior.

Assumption 1 (Block Dependent Noise under Flexible Distributions). *Matrix $\mathbf{E}$ satisfies:*

- (a) *There exists a partition $S_1, S_2, \dots, S_L$ of $[J]$ with $|S_l| \leq M$ for each $l \in [L]$, such that the vectors $\{\mathbf{E}_{i,S_l}\}_{l=1}^L$ are mutually independent for different $i \in [N]$ and $l \in [L]$.*
- (b) *Either $|E_{i,j}| \leq B$ for all $i \in [N], j \in [J]$, or there exists a random matrix $\mathbf{E}' = (E'_{i,j}) \in \mathbb{R}^{N \times J}$ obeying the same dependence structure in (a), such that for any $i \in [N], j \in [J]$, it holds that $\|E'_{i,j}\|_\infty \leq B$, $\mathbb{E}[E'_{i,j}] = 0$, $\|\text{Cov}(\mathbf{E}'_{i,:})\| \lesssim \|\text{Cov}(\mathbf{E}_{i,:})\|$, and $\Pr(E_{i,j} = E'_{i,j}) \geq 1 - O(d^{-22})$.*

The generalized-GoM models satisfying Assumption 1 are suitable for handling highly asymmetric and sparse data, and are applicable across diverse fields, including social science surveys, admixture modeling in population genetics, single-cell sequencing data, among others. The flexible local dependence in Assumption 1(a) effectively captures the dependence

in the flattened data from multivariate polytomous responses and accommodates other more complex dependence structures. The entries in each block S_l can be arbitrarily dependent on each other, which can either consist of those flattened response categories of a single item and/or concatenated responses across different intrinsically locally dependent items. Related local dependence considerations are explored in [Lei \(2019\)](#) and [Modell et al. \(2024\)](#). [Agterberg et al. \(2022\)](#) relaxes the entrywise independence assumption using a linear-transformed entrywise-independent noise assumption, but their assumption cannot capture the dependence structure of flattened polytomous data in our motivating setting.

Notably, the high-probability boundedness Assumption 1(b) encompasses a wide range of data types, covering all sub-exponential distributions and substantially extending beyond the flattened polytomous data case in the traditional GoM model. For example, the mixed membership models for Binomial and Poisson data mentioned in Section 2.2 both belong to our generalized-GoM modeling framework. Specifically, one can write the data arising from these models as the sum of a low-rank signal part $\mathbf{\Pi}^* \mathbf{\Theta}^{*\top}$ and a mean-zero Binomial/Poisson noise part $\mathbf{E}$, where $\mathbf{\Theta}^*$ are Binomial or Poisson parameters with their respective range.

Consider the SVD of the signal matrix $\mathbf{R}^* = \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top}$, where $\mathbf{\Lambda}^* = \text{diag}(\sigma_1^*, \dots, \sigma_K^*)$ with $\sigma_1^* \geq \dots \geq \sigma_K^* > 0$ collects the K singular values. Orthonormal matrices $\mathbf{U}_{N \times K}^*$, $\mathbf{V}_{J \times K}^*$ respectively consist of the left and right singular vectors. Similarly, we consider the top- K SVD for the observed matrix $\mathbf{R} \approx \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$ where $\mathbf{\Lambda} = \text{diag}(\sigma_1, \dots, \sigma_K)$ with $\sigma_1 \geq \dots, \sigma_K \geq 0$, $\mathbf{U} \in \mathbb{R}^{N \times K}$, $\mathbf{V} \in \mathbb{R}^{J \times K}$, and $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_K$.

Write the ratio of J and N as $r := J/N$. Let $[B] = \{1, 2, \dots, B\}$ for any positive integer B . For a matrix $\mathbf{A}$, let $\lambda_k(\mathbf{A})$ denote the k -th largest eigenvalue and $\sigma_k(\mathbf{A})$ denote the k -th largest singular value; define its condition number $\kappa(\mathbf{A}) := \sigma_1(\mathbf{A})/\sigma_{\min}(\mathbf{A})$ with $\sigma_{\min}(\mathbf{A})$ being the minimum non-zero singular value. Specifically, define $\kappa^* := \sigma_1^*/\sigma_K^*$ as the condition number of $\mathbf{R}^*$. The $\ell_{2,\infty}$ norm of $\mathbf{A}$ is $\|\mathbf{A}\|_{2,\infty} := \max_i \|\mathbf{A}_{i,:}\|_2$, which is the maximum ℓ_2 norm of the rows of $\mathbf{A}$. Let $\mathbf{A}_{\mathbf{S},:}$ denote the submatrix consisting of the rows indexed by $\mathbf{S}$

and similarly $\mathbf{A}_{:,s}$ for columns. Define the matrix incoherence degrees, which are standard in entrywise matrix perturbation theory: $\mu_1 := (N/K) \|\mathbf{U}^*\|_{2,\infty}^2$, and $\mu_2 := (J/K) \|\mathbf{V}^*\|_{2,\infty}^2$.

Finally, we define the following notations similar to those in [Chen et al. \(2021\)](#): (a) $f(n) = O(g(n))$ or $f(n) \lesssim g(n)$ means there is a universal constant $c > 0$ such that $|f(n)| \leq c|g(n)|$ for all n large enough, and $\tilde{O}(\cdot)$ differs from $O(\cdot)$ with up to poly-logarithmic factors; (b) $f(n) \asymp g(n)$ indicates that there exists universal constants $c_1, c_2 > 0$ such that $c_1|g(n)| \leq |f(n)| \leq c_2|g(n)|$ for all n large enough; (c) $f(n) = o(g(n))$ denotes that $f(n)/g(n) \xrightarrow{n \rightarrow \infty} 0$; (d) $f(n) \gg g(n)$ (resp. $f(n) \ll g(n)$) is used when there exists universal constant $c > 0$ large (resp. small) enough such that $|f(n)| \geq c|g(n)|$ (resp. $|f(n)| \leq c|g(n)|$).

3 A Spectral Method for Parameter Estimation

Model identifiability is a crucial prerequisite for valid and reproducible statistical analysis. Conventionally, a statistical model is said to be identifiable if there is a one-to-one mapping between the population parameters and the marginal likelihood function. [Gu et al. \(2023\)](#) established such a notion of *population identifiability* for a dimension-grouped variant of the GoM model, by investigating when the probability mass function (PMF) of the observed random vector (see (1)) uniquely determines the item parameters Θ and Dirichlet parameters α . However, this notion does not concern the identifiability of the latent membership scores π_i , which are also often of interest. To address this problem for a special case of the GoM model with binary responses, [Chen and Gu \(2024\)](#) considered another simpler notion of identifiability, called *expectation identifiability*. With this notion, a GoM model with parameter set (Π^*, Θ^*) is said to be identifiable, if for any other valid parameter set $(\tilde{\Pi}, \tilde{\Theta})$, the equality of expectation $\mathbb{E}[\mathbf{R}] = \tilde{\Pi}\tilde{\Theta}^\top = \Pi^*\Theta^{*\top}$ holds if and only if (Π^*, Θ^*) and $(\tilde{\Pi}, \tilde{\Theta})$ are identical up to a permutation of the K extreme profiles. To establish expectation identifiability for generalized-GoM models, we define subject i to be a pure subject for extreme profile k if the only positive entry of π_i^* is located at index k ; i.e., $\pi_i^* = \mathbf{e}_k$. We

impose Assumption 2 for expectation identifiability of the generalized-GoM models.

Assumption 2. Π^* satisfies that every extreme latent profile has at least one pure subject. In addition, $\text{rank}(\Theta^*) = K$.

The pure subject condition in Assumption 2 is analogous to the anchor word assumption in topic modeling (Ke and Wang, 2024) and the pure node assumption in mixed membership stochastic block model for networks (Jin et al., 2024; Mao et al., 2021). Under Assumption 2, denote $\mathbf{S} = (S_1, \dots, S_K)$ as the index vector of one set of K pure subjects such that $\Pi_{\mathbf{S},:}^* = \mathbf{I}_K$. Lemma 1 gives the model parameter expressions in terms of $\mathbf{S}$ and the population SVD.

Lemma 1. Under Assumption 2, the generalized-GoM model is identifiable and $\Pi^* = \mathbf{U}^* \mathbf{U}_{\mathbf{S},:}^{*-1}$, $\Theta^* = \mathbf{V}^* \Lambda^* \mathbf{U}_{\mathbf{S},:}^{*\top}$.

The first equation in Lemma 1 implies that the i -th row of the left singular matrix $\mathbf{U}^*$ can be written as a convex combination of $\mathbf{U}_{S_1}^*, \dots, \mathbf{U}_{S_K}^*$, with the weights being the membership scores π_i^* . This geometric property motivates us to use the successive projection algorithm (SPA, Gillis and Vavasis, 2013) to find the indices $\mathbf{S}$ of the pure subjects. SPA successively finds the vertices with the largest norm in a sequentially defined orthogonal subspace. After estimating $\hat{\mathbf{S}} = (\hat{S}_1, \dots, \hat{S}_K)$, we can estimate the parameters using Lemma 1. Algorithm 1 summarizes the proposed procedure for the polytomous-GoM (i.e., traditional GoM) model. The algorithm for Poisson-GoM is given in Supplementary Material S.3.2. Algorithm 1 can be easily generalized to accommodate generalized-GoM models with other data types by modifying the post-processing on the item parameter estimate $\hat{\Theta}$ (line 11) to the appropriate range. These post-processing steps truncate the estimates to valid values and can only decrease the estimation error.

Related spectral methods have been studied for network models and topic models with mixed membership structures. The geometric properties in the spectral domain have been exploited for the MM network model using the eigenvalue decomposition of the symmetric

Algorithm 1 Spectral Method for Estimating the GoM Model with Categorical Responses

Input: Categorical response matrix $\tilde{\mathbf{R}} \in \mathbb{R}^{N \times L}$, number of extreme latent profiles K

Output: Estimated $\hat{\mathbf{S}} \in \mathbb{R}^K$, $\hat{\mathbf{\Theta}}^{(\text{post})} \in \mathbb{R}^{J \times K}$, $\hat{\mathbf{\Pi}}^{(\text{post})} \in \mathbb{R}^{N \times K}$

- 1: Obtain the flattened binary matrix $\mathbf{R} \in \mathbb{R}^{N \times J}$ from $\tilde{\mathbf{R}}$ according to (3).
 - 2: Get the top K singular value decomposition of $\mathbf{R}$ as $\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top$
 - 3: $\mathbf{Y} = \mathbf{U}$
 - 4: **for** $k \in \{1, \dots, K\}$ **do**
 - 5: $\hat{S}_k = \arg \max(\{\|\mathbf{Y}_{i,:}\|_2 : i \in [N] \setminus \hat{\mathbf{P}}\})$
 - 6: $\mathbf{u} = \mathbf{Y}_{\hat{S}_k,:} / \|\mathbf{Y}_{\hat{S}_k,:}\|_2$
 - 7: $\mathbf{Y} = \mathbf{Y}(\mathbf{I}_K - \mathbf{u}\mathbf{u}^\top)$
 - 8: **end for**
 - 9: $\hat{\mathbf{\Pi}} = \mathbf{U}(\mathbf{U}_{\hat{\mathbf{S}},:})^{-1}$, $\hat{\mathbf{\Pi}}^{(\text{post})} = \text{diag}(\hat{\mathbf{\Pi}}_+ \mathbf{1}_K)^{-1} \hat{\mathbf{\Pi}}_+$
 - 10: $\hat{\mathbf{\Theta}} = \mathbf{V}\mathbf{\Lambda}\mathbf{U}_{\hat{\mathbf{S}},:}^\top$
 - 11: Let $\hat{\mathbf{\Theta}}^{(\text{post})} = (\hat{\boldsymbol{\theta}}_{j,k}^{(\text{post})})$ with $\hat{\boldsymbol{\theta}}_{j,k}^{(\text{post})} = \begin{cases} \hat{\theta}_{j,k} & \text{if } 0 \leq \hat{\theta}_{j,k} \leq 1 \\ 0 & \text{if } \hat{\theta}_{j,k} < 0 \\ 1 & \text{if } \hat{\theta}_{j,k} > 1 \end{cases}$
 - 12: **for** $l \in \{1, \dots, L\}$ **do**
 - 13: **for** $k \in \{1, \dots, K\}$ **do**
 - 14: $\boldsymbol{\theta}^b = \hat{\mathbf{\Theta}}_{(\sum_{t=1}^{l-1} C_t + 1):(\sum_{t=1}^l C_t), k}^{(\text{post})}$
 - 15: $\hat{\mathbf{\Theta}}_{(\sum_{t=1}^{l-1} C_t + 1):\sum_{t=1}^l C_t, k}^{(\text{post})} = \boldsymbol{\theta}^b / \mathbf{1}_K^\top \boldsymbol{\theta}^b$
 - 16: **end for**
 - 17: **end for**
-

adjacency matrix (Jin et al., 2024; Ke and Jin, 2023; Jin et al., 2021; Mao et al., 2021; Zhang et al., 2020). Also, SVD-based methods have been used to estimate the topic models (Ke and Wang, 2024; Wu et al., 2023). Compared to these existing works, we are motivated by asym-

metric mixed membership models, especially the popular GoM model for high-dimensional categorical (polytomous) responses. Data arising from this model cannot be directly written as a signal matrix plus an *entrywise-independent* noise matrix, in contrast to the usual network model and topic model. For our motivating GoM models, Bayesian methods and likelihood-based optimization methods are the predominant paradigms for parameter estimation. In contrast, our spectral method offers a unifying scalable estimation approach that comes with strong theoretical guarantees under local dependence; see Section 4.

4 Uniform Estimation Error Bounds for Parameters

To establish entrywise estimation error bounds for our spectral estimator $\hat{\Pi}$ and $\hat{\Theta}$, it is necessary to develop fine-grained perturbation theory for the empirical singular matrices $\mathbf{U}$ and $\mathbf{V}$ according to Lemma 1. In Section 4.1, we establish a novel $\ell_{2,\infty}$ perturbation theory for the left and right singular subspaces for the truncated SVD under locally dependent and flexibly distributed noise. In Section 4.2, we present uniform entrywise convergence rates for parameters using the new singular subspace perturbation theory. Finally, in Section 4.3, we discuss an application to GoM models with the popular Dirichlet priors.

4.1 New $\ell_{2,\infty}$ Singular Subspace Perturbation Theory

We parse the impact of local dependence by measuring the covariances of the noise blocks:

$$\sigma^2 := \max_{i \in [N], j \in [J]} \text{Var}(E_{i,j}), \quad \tilde{\sigma}^2 := \max_{i \in [N], l \in [L]} \|\text{Cov}(\mathbf{E}_{i,S_l})\|.$$

One can verify that $\sigma^2 \leq \tilde{\sigma}^2 \leq M\sigma^2$. In the special case of independent entries in $\mathbf{E}$, $\tilde{\sigma}^2 = \sigma^2$; in the another special case of exactly equal entries within each block, $\tilde{\sigma}^2 = M\sigma^2$.

Assumption 3. *We assume that $K \ll \min\{N, L\}$, $ML \asymp J$, and*

$$\sigma_K(\Pi^*)\sigma_K(\Theta^*) \gtrsim \sigma_{\kappa^*}(M\sqrt{N} + \sqrt{J}), \quad (4)$$

$$MB \log^2(N \vee J) \max \left\{ \sqrt{\frac{\mu_1 K}{N}}, \sqrt{\frac{\mu_2 K}{J}} \right\} \lesssim \sigma. \quad (5)$$

Assumption 3 assumes a low-rank signal matrix and balanced block size in the dependence structure. In addition, (4) assumes a lower bound on signal strength, and (5) assumes that the incoherence degrees cannot be too large. Theorem 1 below gives our novel $\ell_{2,\infty}$ singular subspace perturbation bounds. These results apply to general low-rank matrices with locally dependent noise from flexible distributions (see Assumption 1), and are of independent interest to other latent variable models beyond the mixed membership models.

Theorem 1 (Two-to-infinity Singular Subspace Perturbation Under Local Dependence).

Assume that Assumptions 1, 3 hold. Then with probability at least $1 - O((N \vee J)^{-10})$,

$$\begin{aligned} & \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\ & \lesssim \frac{\tilde{\sigma} \sqrt{N \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma MB \log(N \vee J) \sqrt{MN + J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} =: \xi_1 \end{aligned} \quad (6)$$

$$\begin{aligned} & \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \lesssim \frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\kappa^* \sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma B \log(N \vee J) \sqrt{MN + J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} =: \xi_2 \end{aligned} \quad (7)$$

$$\begin{aligned} & \|\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top - \mathbf{U}^*\mathbf{\Lambda}^*\mathbf{V}^{*\top}\|_\infty \\ & \lesssim \kappa^* K \sigma \log(N \vee J) \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + MB \log(N \vee J) \left(\frac{\mu_2 K}{J} + \frac{\mu_1 K}{MN} \right) =: \xi_3. \end{aligned} \quad (8)$$

Remark 1. By imposing additional assumptions, we can derive simpler expressions for the upper bounds presented in Theorem 1. First, provided that $MB/\sigma \lesssim \log(N \vee J)$ and $\mu_2/\mu_1 \lesssim J/(\log(N \vee J))^3$, then the third term in (6) is dominated by the first term and

$$\|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \lesssim \left(\frac{\tilde{\sigma} \sqrt{N \log(N \vee J)}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \right) \sqrt{\frac{\mu_1 K}{N}}. \quad (9)$$

Similarly, if $B/\sigma \lesssim \log(N \vee J)$ and $\mu_1/\mu_2 \lesssim N/(\log(N \vee J))^3$ hold, (7) is simplified to

$$\|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \lesssim \left(\frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 MN}{\sigma_K^{*2}} \right) \sqrt{\frac{\mu_2 K}{J}}. \quad (10)$$

Lastly, if $\sqrt{MB} \lesssim \kappa^* \sigma \sqrt{\frac{\mu_1}{\mu_2} \wedge \frac{\mu_2}{\mu_1}} \sqrt{\frac{M^2 N^2}{J} \wedge \frac{J^2}{MN}}$ holds, the entrywise bound (8) becomes

$$\left\| \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \right\|_\infty \lesssim \kappa^* K \sigma \sqrt{\mu_1 \mu_2} \log(N \vee J) \sqrt{\frac{M}{J} + \frac{1}{N}}. \quad (11)$$

Remark 2. We give an interpretation of (10) in Remark 1, and similar interpretation holds for (9). We employ the expansion formula in (Xia, 2021, Theorem 1):

$$\begin{aligned} & \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \\ &= (\mathbf{I}_J - \mathbf{V}^* \mathbf{V}^{*\top}) \underbrace{(\mathbf{E}^\top \mathbf{R}^* + \mathbf{E}^\top \mathbf{E} + \mathbf{R}^{*\top} \mathbf{E})}_{\mathbf{R}^\top \mathbf{R} - \mathbf{R}^{*\top} \mathbf{R}^*} \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \text{higher order terms} \\ &= \mathbf{E}^\top \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \text{negligible terms} + \text{higher order terms}. \end{aligned} \quad (12)$$

The two terms of the upper bound in (10) correspond to the decomposition above as follows:

$$\begin{aligned} \left\| \mathbf{E}^\top \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-2} \right\|_{2,\infty} &= \left\| \mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-1} \right\|_{2,\infty} \lesssim \frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}}, \\ \left\| \mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2} \right\|_{2,\infty} &\lesssim \frac{\sigma \sqrt{K \log(N \vee J)}}{\sigma_K^*} + \frac{\sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} \\ &\lesssim \frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\kappa^* \sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}}, \end{aligned}$$

hold with probability at least $1 - O((N \vee J)^{-10})$. The first line follows from inequality (S.8) in Lemma S.3 and the second line holds by Lemma S.7 in the Supplementary Material.

Allowing for arbitrarily flexible locally dependent noise is a key distinguishing feature of our theory. In recent years, there has been significant progress in $\ell_{2,\infty}$ eigenspace perturbation theory for random matrices (Fan et al., 2018; Cape et al., 2019; Mao et al., 2021; Chen et al., 2021; Cai et al., 2021; Abbe et al., 2022; Yan et al., 2024). However, in this literature, most studies consider independent entries in the noise matrix, with only a few exceptions including Lei (2019), Modell et al. (2024), and Agterberg et al. (2022). To facilitate the interpretation, we first delve further into the simplified forms presented in Remark 1. On the one hand, our upper bound (9) for the left singular subspace $\mathbf{U}$ involves $\tilde{\sigma}$ as a critical quantity to measure the noise level, and provides a sharp characterization of the SVD-based method's

performance (in light of a similar decomposition to (12) for $\mathbf{U}$). On the other hand, it is inevitable to involve the block size M in our leave-one-block-out argument for deriving the bound for the right singular subspace $\mathbf{V}$. Actually, we highlight that the dependence on M in our bounds (10) and (11) is *sharp*, by considering the following special case. Imagine the scenario where each block size is exactly M , and the entries $R_{i,j}$, $j \in S_l$ in each block S_l are duplicates of a single variable $\check{R}_{i,l} \sim \mathcal{N}(0, \sigma^2)$ for each $i \in [N]$. Invoking (12), the fluctuation $\mathbf{V}_{j,:} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}_{j,:}^*$ is partially attributed to the component $\mathbf{E}_{:,j}^\top \mathbf{E}_{:,S_l} \mathbf{V}_{S_l,:}^* \mathbf{\Lambda}^{*-2}$, given $j \in S_l$. Applying elementary concentration inequalities (e.g. Hoeffding's inequality for sub-Gaussian entries), we find that $(\mathbf{E}^\top \mathbf{E})_{S_l, S_l}$ is concentrated around $\sum_{i \in [N]} \text{Var}(\check{R}_{i,l}) \mathbf{1}_M \mathbf{1}_M^\top$. This leads to the result that $\max_{j \in [J]} \|\mathbf{E}_{:,j}^\top \mathbf{E}_{:,S_l} \mathbf{V}_{S_l,:}^* \mathbf{\Lambda}^{*-2}\|_2$ is lower bounded by $\frac{\sigma^2 M N}{\sigma_1^{*2}} \|\mathbf{V}^*\|_{2,\infty}$ with high probability. This confirms that the dependence of (10) on M is sharp and cannot be improved. Moreover, under the same setting, the minimax lower bound for the entry-wise error, as informed by Koltchinskii et al. (2011), is $\sigma \sqrt{K(M/J + 1/N)}$. Provided that $\mu_1, \mu_2, \kappa^*, K = O(1)$, our upper bound (11) is optimal up to a logarithmic term.

We further clarify how our bounds differ from previous results. In the literature, Lei (2019) and Modell et al. (2024) studied eigenspace perturbation to accommodate certain dependence structures. To adapt these results from symmetric random matrices to asymmetric matrices, a common approach is to embed the asymmetric matrix as two blocks within a larger symmetric matrix, a technique known as the ‘‘symmetric dilation’’ trick. However, this symmetrization technique often falls short of delivering a sharp dependence on the ratio J/N and the incoherence degrees μ_1 and μ_2 , because the corresponding singular subspace perturbation bounds for asymmetric random matrices resulting from symmetrization typically involve $J \vee N$ and $\mu_1 \vee \mu_2$. However, distinguishing N from J and distinguishing μ_1 from μ_2 become particularly important when encountering unbalanced observation matrices ($N = o(J)$ or $J = o(N)$) with unbalanced incoherence degrees ($\mu_1 \ll \mu_2$ or $\mu_1 \gg \mu_2$). To see the importance of distinguishing μ_1 between μ_2 , we note that $\mu_1 \leq \kappa^2(\mathbf{\Pi}^*)/K$ when Assump-

tion 2 holds by Lemmas 1 and S.7, thus $\mu_1 = O(1)$ if $\kappa(\mathbf{\Pi}^*) = O(\sqrt{K})$. Section 4.3 will show that $\kappa(\mathbf{\Pi}^*) = O(\sqrt{K})$ holds with high probability if the membership scores are generated from a Dirichlet distribution, a common assumption in Bayesian inference for GoM models. In contrast, μ_2 often exhibits much larger values due to the prevalence of sparse patterns and heterogeneity among the items. This distinction is supported by numerous real-world datasets, including the single-cell data in Section 6 where the empirical value of μ_1 is around 2.9 while for μ_2 it is 55.7. We address this issue by establishing perturbation bounds for the left and right singular vectors *separately*. Our technique is conceptually similar to that of the arXiv version of Yan et al. (2024), but with a nuanced treatment of local dependence.

Lastly, we remark on the application of matrix concentration universality in our proof to overcome potential dependence issues. Specifically, given an N -by- J noise matrix $\mathbf{E}$ with block dependence in Assumption 1, we employ Corollary 2.15 in Brailovskaya and van Handel (2024) to derive that $\|\mathbf{E}\| \lesssim \tilde{\sigma}\sqrt{N} + \sigma\sqrt{J}$ with high probability. To compare this with the preceding approaches, consider the bound provided by the matrix Bernstein inequality, which often yields $\|\mathbf{E}\| \lesssim (\tilde{\sigma}\sqrt{N} + \sigma\sqrt{J})\sqrt{\log(N \vee J)}$ with high probability. This result is $\sqrt{\log(N \vee J)}$ -looser than our universality-based concentration. Moreover, this elimination of the $\sqrt{\log(N \vee J)}$ factor also extends to the bounds on $\|\mathbf{E}\mathbf{V}^*\|$ and $\|\mathbf{U}^{*\top}\mathbf{E}\|$.

4.2 Uniform Consistency of Parameter Estimation

We establish the error bounds for the estimators $\hat{\mathbf{\Pi}} = \mathbf{U}\mathbf{U}_{\hat{\mathbf{S}},:}^{-1}$, $\hat{\mathbf{\Theta}} = \mathbf{V}\mathbf{\Lambda}\mathbf{U}_{\hat{\mathbf{S}},:}^\top$ in Theorem 2.

Theorem 2 (Uniform Consistency of Parameter Estimation). *Suppose Assumptions 1, 2, 3 hold, and additionally assume that $\sqrt{K}\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1 \ll 1$. Then there exists a permutation matrix $\mathbf{P}$ such that with probability at least $1 - O((N \vee J)^{-10})$,*

$$\left\| \hat{\mathbf{\Pi}} - \mathbf{\Pi}^*\mathbf{P} \right\|_{2,\infty} = O(\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1), \quad \left\| \hat{\mathbf{\Theta}} - \mathbf{\Theta}^*\mathbf{P} \right\|_\infty = O(\kappa^2(\mathbf{\Pi}^*)\xi_3).$$

The $\ell_{2,\infty}$ bound of $\hat{\mathbf{\Pi}}$ primarily depends on the left singular subspace perturbation level ξ_1 , as the estimate is derived from $\hat{\mathbf{U}}$. The additional term $\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)$ in the bound of

$\widehat{\Pi}$ arises from the vertex hunting step via the successive projection. Similarly, the ℓ_∞ error bound of $\widehat{\Theta}$ involves the upper bound of $\|\mathbf{R}^* - \mathbf{U}\mathbf{A}\mathbf{V}^\top\|_\infty$, which is reasonable given that $\widehat{\Theta}^\top$ is a submatrix of $\mathbf{U}\mathbf{A}\mathbf{V}^\top$, and $\kappa^2(\mathbf{\Pi}^*)$ again stems from estimating $\widehat{\mathbf{S}}$ via vertex hunting.

Corollary 1. *Suppose the assumptions in Theorem 2 hold and $\max\{\kappa(\mathbf{\Pi}^*), \kappa(\mathbf{\Theta}^*)\} = O(1)$. Then there exists a permutation matrix $\mathbf{P}$ such that with probability at least $1 - O((N \vee J)^{-10})$,*

$$\begin{aligned} & \left\| \widehat{\Pi} - \mathbf{\Pi}^* \mathbf{P} \right\|_{2,\infty} \\ & \lesssim \frac{\tilde{\sigma} \sqrt{\mu_1 K \log(N \vee J)}}{\sigma_K(\mathbf{\Theta}^*)} + \frac{\sigma^2 \sqrt{JK}}{\sigma_K(\mathbf{\Pi}^*) \sigma_K^2(\mathbf{\Theta}^*)} \left(\sqrt{\frac{J}{N}} \sqrt{\mu_1} + \sqrt{1 + \frac{MN}{J} \frac{MB \log(N \vee J)}{\sigma \sqrt{J}}} \sqrt{\mu_2} \right), \\ & \left\| \widehat{\Theta} - \mathbf{\Theta}^* \mathbf{P} \right\|_\infty \\ & \lesssim K \sigma \log(N \vee J) \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + MB \log(N \vee J) \left(\frac{\mu_1 K}{MN} + \frac{\mu_2 K}{J} \right). \end{aligned}$$

To see the rates of convergence with Corollary 1, consider the case where $\sigma_K^2(\mathbf{\Pi}^*) \gtrsim N/K, \sigma_K^2(\mathbf{\Theta}^*) \gtrsim J/(MK)$. Under the mild assumptions for (9) and (11) in Remark 1, $\widehat{\pi}_i$ converges to π_i^* with a rate of $\tilde{O}\left(\sigma K^2 M \sqrt{\mu_1} \left(\frac{1}{\sqrt{J}} \vee \frac{\sigma K}{N}\right)\right)$, and $\widehat{\theta}_j$ converges to θ_j^* with a rate of $\tilde{O}\left(\sigma K^2 \sqrt{\mu_1 \mu_2} \left(\sqrt{\frac{M}{J}} \vee \frac{1}{\sqrt{N}}\right)\right)$.

4.3 Application to GoM Models with Dirichlet Prior

Dirichlet prior on the membership scores is commonly adopted in Bayesian inference for GoM models (Erosheva, 2002; Erosheva et al., 2007; Gu et al., 2023). To illustrate the theoretical results, we discuss a Dirichlet prior application on the GoM model for polytomous data estimated using the flattening procedure. Assume that π_i^* is independently sampled from a Dirichlet distribution with parameters $\alpha = (\alpha_1, \dots, \alpha_K)$. For simplicity, we assume $C_l = M$ for all $l \in [L]$. Similarly, we assume that for each $l \in [L], k \in [K]$, vector $(\theta_{M(l-1)+c,k}^*)_{c=1}^M$ independently follows a Dirichlet distribution with parameters $\beta = (\beta_1, \dots, \beta_M)$.

Assumption 4. *Assume that $\max_k \alpha_k \leq C_1 \min_k \alpha_k$ for some $C_1 > 0$, $\max_c \beta_c \leq C_2 \min_c \beta_c$ for some $C_2 > 0$, $\max_k \alpha_k = O(1), \max_c \beta_c = O(1)$. In addition, $N \gtrsim M^2 K^5 \log^4(N \vee J), J \gtrsim$*

$M^4 K^6 \log^4(N \vee J)$ and

$$\nu \leq \frac{\sqrt{N/\log N}}{6\sqrt{3}(1+\alpha_0)}, \quad c_1 \geq 4K\sqrt{15\log L/\bar{L}}. \quad (13)$$

Note that (13) is similar to Assumption 3.3 in Mao et al. (2021).

Corollary 2. *Let $\pi_i^* \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\alpha})$ and $(\theta_{M(l-1)+c,k}^*)_{c=1}^M \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\beta})$. Suppose Assumption 4 holds. Then with probability at least $1 - O(K(N \wedge L)^{-3})$, one has*

$$\begin{aligned} \|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}^* \mathbf{P}\|_{2,\infty} &\lesssim \frac{M^{3/2} K^{5/2} \sqrt{\log(N \vee J)}}{\sqrt{J}} + \frac{\sqrt{M^2 K^{9/2}}}{N}, \\ \|\hat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}^* \mathbf{P}\|_{\infty} &\lesssim \frac{M^{5/2} K^{9/2} \log(N \vee J)}{\sqrt{J}} + \frac{M^2 K^{9/2} \log(N \vee J)}{\sqrt{N}}. \end{aligned}$$

Corollary 2 gives strong entrywise guarantees for our spectral estimator for the GoM model for categorical data. To our best knowledge, this is the first uniform estimation error control for any method applied to the GoM model in high-dimensional settings.

5 Simulation Studies

We conduct three simulation studies to assess the efficiency and accuracy of the proposed method. Section 5.1 presents results for the GoM model for multivariate polytomous data, where the item responses are polytomous and local dependence is introduced by the flattening technique. Section 5.2 presents results for the GoM model for multivariate binary data, where the binary item responses are designed to have intrinsic blockwise dependent entries. In the third study in Section 5.3, we consider the generalized-GoM model for Poisson data and compare our method with the popular NMF algorithm (Lee and Seung, 2000). For all three studies, we consider $K = 3$ and generate 100 independent replications for each simulation setting. Within each setting and each replication, the rows of $\boldsymbol{\Pi}$ are independently simulated from the $\text{Dirichlet}(\mathbf{1}_K)$ distribution, and the first K rows of $\boldsymbol{\Pi}$ are set to the identity matrix $\mathbf{I}_K$ to satisfy Assumption 2. For both the simulation studies and real-data applications, we

employ a pruning step on $\mathbf{U}$ before SPA in Algorithm 1 to de-noise (Mao et al., 2021), where we set the tuning parameters as $r = 10, q = 0.4, e = 0.2$ as suggested in Chen and Gu (2024).

5.1 Local Dependence from Flattening Polytomous Responses

We first consider local dependence introduced by the flattening technique for polytomous data, with the number of response categories set to $C_l \equiv C = 3$ for all $l \in [L]$. Consider sample size $N \in \{200, 1000, 2000, 3000, 4000, 5000\}$, and $J = N/5$. For each $j \in [J]$ and $k \in [K]$, $\Theta_{(C(j-1)+1):(Cj),k}^*$ is independently simulated from $\text{Dirichlet}(0.2 \cdot \mathbf{1}_C)$. For each pair (i, j) , the non-zero index in $\mathbf{R}_{i,(C(j-1)+1):(Cj)}$ is independently generated from the categorical distribution with the probability vector being $\Theta_{(C(j-1)+1):(Cj),:}^* \Pi_{i,:}^{*\top}$.

Figure 2 displays the boxplots of the two-to-infinity estimation error for $\hat{\Pi}$ and the maximum absolute estimation error for $\hat{\Theta}$ obtained from the proposed method. We observe that the error decreases and approaches 0 as N, J grow for both estimates, which verifies our uniform consistency results for parameter estimation.

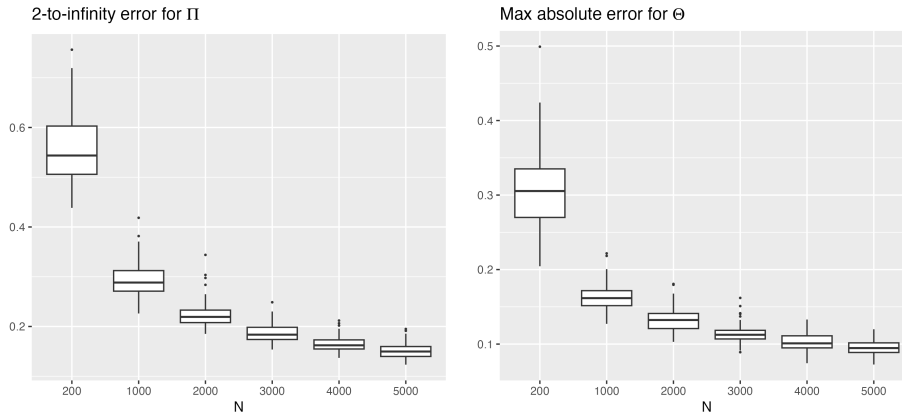


Figure 2: Simulation study I, estimation error for the proposed method using the flattening technique. The left panel presents the two-to-infinity error for $\hat{\Pi}$ and the right presents the maximum absolute error for $\hat{\Theta}$.

We compare our method with the Gibbs sampling method and the variational inference method for the polytomous GoM model. The parameters in the Gibbs sampler are initialized with their generating distributions. The derivation of each updating step in the Gibbs

	$N = 200$	$N = 1000$	$N = 2000$
Proposed	0.07	2.25	12.84
VI	6.63	85.14	264.40
Gibbs	709.44	16458.22	65464.17

Table 1: Simulation study I, average computation time in seconds for polytomous responses. The flattening technique is used in the proposed method. VI stands for the variational inference algorithm implemented in the `mixedMem` R package.

sampler can be found in Section S.3.1 in the Supplementary Material. We set the first 5000 samples as burn-in and take the average of 2000 samples after the burn-in phase as the Gibbs sampling estimates. We employ the variational inference (VI) algorithm proposed in Wang and Erosheva (2015), implemented in the `mixedMemModel` function in R package `mixedMem`. As Gibbs sampling is not scalable to large datasets and would take a very long time to compute when N, J are large, we only consider $N \in \{200, 1000, 2000\}$ to compare the three methods’ performances. Table 1 presents the average computation time in seconds across the 100 replications. When $N = 2000$, our proposed method is about 10,000 times faster compared to Gibbs sampling and 30 times faster compared to VI. Table 2 summarizes the average two-to-infinity error of $\hat{\Pi}$ and the max absolute error of $\hat{\Theta}$. In comparison to Gibbs sampling, the proposed method delivers comparable results in accuracy. We also find that the VI method in this case does not yield consistent results, and it is very sensitive to initialization. In summary, our proposed method is efficient in computation, especially for large-scale datasets, and is comparably accurate with popular existing methods.

5.2 Intrinsic Local Dependence across Different Items

In this section, we generate binary responses with a more general intrinsic local dependence structure. Now $\Theta^* \in [0, 1]^{J \times K}$ is no longer a flattened parameter matrix, but rather contains the item response probabilities as the Bernoulli parameters $\theta_{jk}^* \stackrel{i.i.d}{\sim} \text{Beta}(0.2, 0.2)$. The

Θ	$N = 200$	$N = 1000$	$N = 2000$	Π	$N = 200$	$N = 1000$	$N = 2000$
Proposed	0.057	0.030	0.024	Proposed	0.098	0.047	0.035
VI	0.252	0.265	0.272	VI	0.204	0.198	0.198
Gibbs	0.072	0.032	0.023	Gibbs	0.080	0.038	0.028

Table 2: Simulation study I, average mean absolute error for Θ and Π for polytomous responses. The flattening technique is used in the proposed method. VI stands for the variational inference algorithm implemented in the `mixedMem` R package.

membership scores π_i^* are generated with the same mechanism as stated in Section 5.1. To generate the locally dependent responses, we first generate latent variables $\eta_i \in \mathbb{R}^J$ independently and identically from the Gaussian distribution $N(\mathbf{0}, \Sigma)$, with Σ being blockwise diagonal with J/M copies of a $M \times M$ block covariance matrix Σ_0 . Here $\Sigma_0 \in \mathbb{R}^{M \times M}$ has an autoregressive structure with parameter ρ , that is, $(\Sigma_0)_{ij} = \rho^{|i-j|}$. We then define a threshold $d_{ij} = \Phi^{-1}(\Pi_{i,:}^\top \Theta_{j,:})$ where Φ is the cumulative distribution function for the standard Normal distribution. Finally, each binary response is defined as $R_{i,j} = \mathbb{1}(\eta_{ij} < d_{ij})$. From this construction, we have $\mathbb{E}[R_{i,j}] = \Pr(R_{i,j} = 1) = \Pi_{i,:}^\top \Theta_{j,:}$, and the elements in $\mathbf{R}_{i,:}$ are blockwisely dependent with a block size M . Specifically, we set $\rho = 0.5$ and $M = 10$, which induces a relatively strong level of dependence. We still set $N \in \{200, 1000, 2000, 3000, 4000, 5000\}$ and let $J = N/5$. Figure 3 presents the error boxplots in this scenario. We can see that the estimates from our method are consistent as N and J grow, as expected from the theory.

5.3 Generalized-GoM Estimation for Poisson Data

We simulate data with independent Poisson entries, and compare our method with the NMF algorithm of multiplicative update proposed in Lee and Seung (2000). The Poisson rate parameters $\theta_{j,k}^*$ are independently sampled from $\text{Gamma}(1, 2)$. To see the effect of the ratio between N and J , we consider both $N = 5J$ with $N \in \{200, 1000, 2000, 3000\}$ and $J = 5N$ with $N \in \{200, 400, 600, 800\}$. We use the R package `NMF` to implement the NMF algorithm

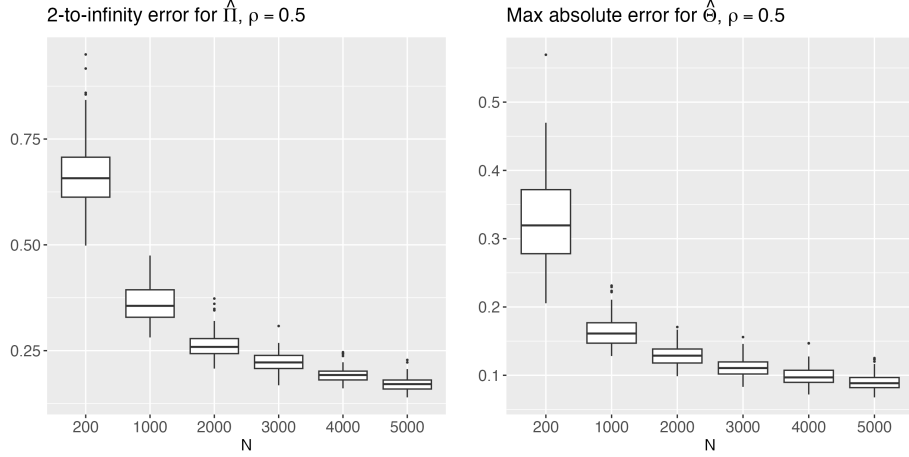


Figure 3: Simulation study II with block-wise dependent data. The left panel presents the two-to-infinity error for $\hat{\Pi}$ and the right presents the maximum absolute error for $\hat{\Theta}$.

(Gaujoux and Seoighe, 2010), where most of the built-in algorithms have been optimized in C++. We compare the two methods in terms of computation time and the scaled Frobenius error $\frac{1}{\sqrt{NJ}} \left\| \mathbf{R}^* - \hat{\Pi} \hat{\Theta}^\top \right\|_F$ in Figure 4. The proposed method yields slightly more accurate estimates across different simulation settings. The computation time of the two methods is comparable when $N = 5J$, while the proposed method is much more efficient in the high-dimensional setting with $J = 5N$. It is worth noting that the matrix decomposition estimate from NMF requires post-processing to locate each row of $\hat{\Pi}$ on a simplex. Additionally, the NMF algorithm itself does not give consistent estimates of parameters Π and Θ due to the lack of identifiability constraints. Therefore, we have only compared the estimation errors of the low-rank matrix itself in Figure 4.

6 Real-data Applications

In this section, we apply the proposed method to three real-world datasets arising from diverse applications, including political survey data ($N > J$, polytomous data), population genetics data ($N \ll J$, binomial data), and single-cell sequencing data ($N \ll J$, nonnegative count data). These datasets exhibit various different local dependence behaviors, either

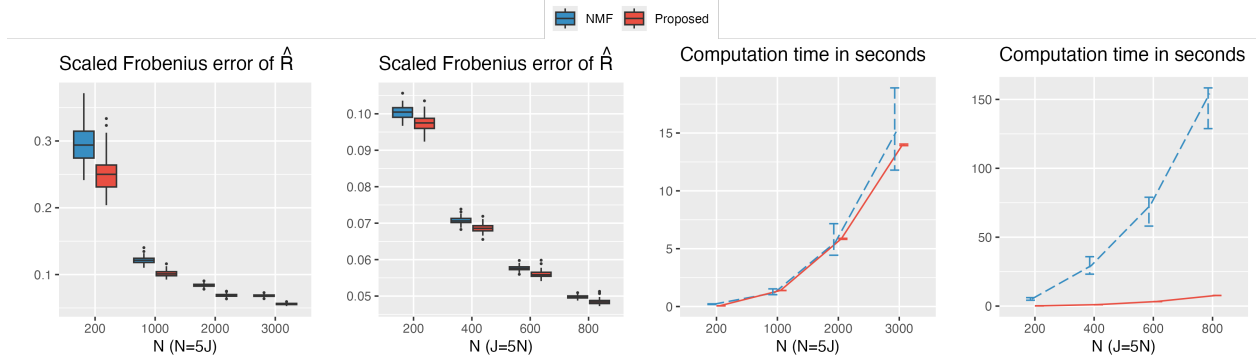


Figure 4: Simulation study III, comparing the estimation error and computation time for the proposed method and NMF with $N = 5J$ and $J = 5N$.

resulting from the flattening technique, or from intrinsic dependence, or both.

6.1 Political Survey Data

The 2022 pilot survey from the American National Election Studies (ANES) provides data about voting and public opinion after the 2022 midterm elections in the United States (publicly available at <https://electionstudies.org/>). We use a subset of $J = 145$ items after excluding followup questions, numeric response questions, and questions that directly ask about party or political ideology. A total $N = 1,511$ respondents that answered all these items is included in the analysis. The number of categories for these items ranges from 2 to 7, with the majority being 5. After flattening the polytomous response matrix, the column dimension of the data matrix becomes $J = 648$. We choose the number of extreme profiles to be $K = 3$, which is the convention in the literature for analyzing US political data (Gross and Manrique-Vallier, 2012; Wang and Erosheva, 2015). The computation time for the whole flattening and parameter estimation procedure takes 3.8 seconds.

By inspecting the estimated item response probability matrix $\hat{\Theta}$, we can draw very intuitive interpretations on the three extreme profiles. Figure 5 gives the heatmap of the estimated response probabilities for a selected subset of items with a representative category of each item. Note that these items are not necessarily binary and are indeed mostly

polytomous with more than two categories. We can see that the three extreme profiles are well-differentiated by the three columns of $\hat{\Theta}$. We interpret profile $k = 1$ to be conservative in their political ideology. Respondents belonging to this profile are much more likely to vote for Donald Trump, tend to care more about illegal immigration and gun rights, and also much more in favor of the overturn of Roe v. Wade. On the contrary, profile $k = 3$ are the liberals and are much more likely to vote for Joe Biden. They also think the Democrats would do a better job handling a variety of issues such as taxes, voter fraud, Covid-19, jobs and employment, etc. Profile $k = 2$ are interpreted to be indifferent. They tend not to follow what is going on in government and public affairs, are less likely to be registered to vote.

The ANES dataset also contains self-reported party information, which we do not use for estimating the parameters. To explore how the estimated latent membership scores align with this held-out information, we present the ternary plot of $\hat{\Pi}$ color coded by the party information in the right panel in Figure 7. In this figure, each dot corresponds to a

Which political party would do a better job handling taxes? (Democrats)	0	0.018	1
Which political party would do a better job handling voter fraud? (Democrats)	0	0.004	1
Which political party would do a better job handling illegal immigration? (Democrats)	0	0	0.99
Which political party would do a better job handling the cost of living and rising prices? (Democrats)	0	0	0.987
Which political party would do a better job handling voting rights? (Democrats)	0	0.062	1
Which political party would do a better job handling Covid-19? (Democrats)	0	0.05	1
Which political party would do a better job handling jobs and employment? (Democrats)	0	0.036	1
How much responsibility do you think Donald Trump bears for the violence that occurred on 1/6/2021 at the US Capitol? (A great deal)	0	0.026	0.921
Which political party would do a better job handling health care? (Democrats)	0	0.143	1
Which political party would do a better job handling abortion? (Democrats)	0	0.13	1
Which political party would do a better job handling climate change? (Democrats)	0.003	0.107	1
Which political party would do a better job handling gun policy? (Democrats)	0	0.086	1
In American society, do you think that being White comes with advantages, disadvantages, or doesn't it matter? (Advantages)	0.069	0.188	1
In the 2020 presidential election, who do you believe to be the legitimate winner? (Biden)	0	0.78	1
Regarding abortion, do you usually think of yourself as pro-choice, pro-life, or something else? (Pro-choice)	0	0.56	1
How important is illegal immigration in the country today? (Extremely important)	0.948	0	0.147
How important to you is protecting the right to own guns? (Extremely important)	0.909	0	0.073
Do you favor, oppose, or neither favor nor oppose the Supreme Court's decision to overturn Roe v. Wade? (Favor)	1	0	0.033
Would you vote for Donald Trump, Joe Biden, someone else, or probably not vote? (Donald Trump)	0.916	0.076	0
How important is what's being taught in public schools in the country today? (Extremely important)	0.911	0	0.299
Does Roe v Wade being overturned make you feel outraged? (Not at all)	0.976	0.277	0
Does Roe v Wade being overturned make you feel angry? (Not at all)	0.969	0.277	0
Does Roe v Wade being overturned make you feel afraid? (Not at all)	0.956	0.314	0.041
Does Roe v Wade being overturned make you feel worried? (Not at all)	0.916	0.203	0
Do you follow what's going on in government and public affairs? (Most of the time)	0.884	0	0.863
During the past 12 months, have you posted a message or comment online about a political issue or campaign? (Yes)	0.577	0	0.7
Are you registered to vote, or not? (Yes, registered to vote at my current address)	1	0.469	0.965

Conservative Indifferent Liberal

Figure 5: Heatmap of a selected subset of the estimated item response parameters with $K = 3$ for the ANES data. Each value corresponds to the estimated probability of choosing the corresponding response in the parenthesis.

respondent’s $\hat{\pi}_i$, whose color represents his or her self-reported party information. We can see that Democrats are closer to the liberal profile, Republicans are closer to the conservative profile, and Independents are closer to the indifferent profile. Since we did not use the party information in the estimation procedure, this result indicates that our method extracts meaningful and interpretable latent representations of people’s political ideology.

To demonstrate the local dependence structure, we provide a covariance matrix heatmap of the first 50 columns of the estimated error matrix $\hat{\mathbf{E}} = \mathbf{R} - \hat{\Pi}\hat{\Theta}^\top$ in the right panel in Figure 6. Except for the weak local dependence introduced by the flattening technique (consecutive diagonal blocks of size 3), we also observe bigger blocks of dependence. It turns out these blocks correspond to different topics of questions in the survey, such as voting participation, global emotion battery, and presidential approval. The answers to the questions within each topic are naturally dependent of each other, giving rise to the bigger blocks in the heatmap.

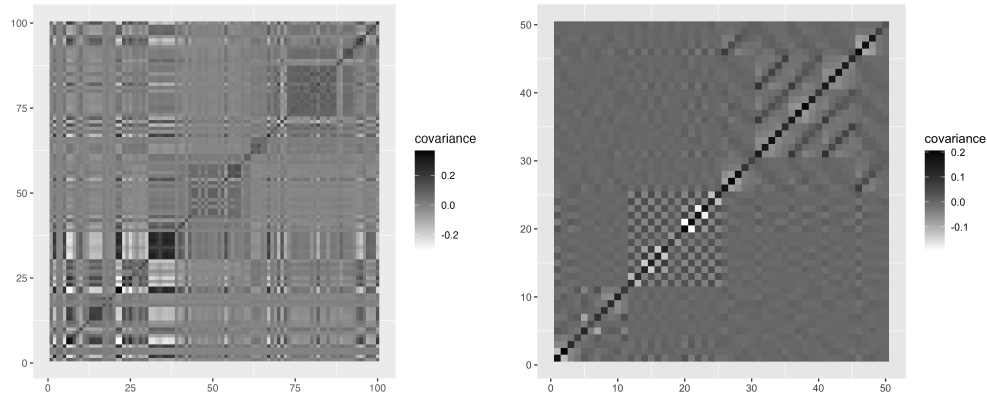


Figure 6: Heatmap of the covariance matrix of the estimated noise matrix $\mathbf{R} - \hat{\Pi}\hat{\Theta}^\top$. The left panel corresponds to the first 100 columns of the HapMap3 data and the right panel corresponds to the first 50 columns of the ANES data.

6.2 Population Genetics Data

We consider a genotyping dataset from the HapMap3 project (Consortium et al., 2010, publicly available at <https://www.broadinstitute.org/medical-and-population-genetics/hapmap-3>). A subset of 4 sub-populations are considered, including the African ancestry

in Southwest USA (ASW), Utah residents with Northern and Western European ancestry (CEU), Mexican ancestry in Los Angeles, California (MEX), and Yoruba in Ibadan, Nigeria (YRI). We follow the conventional data preprocessing procedure to remove SNPs with minor allele frequency smaller than 5% and SNPs with missing data (Kranzler et al., 2019). The pre-processed dataset consists of $N = 467$ individuals and $J = 274,128$ SNPs.

The observed genotype R_{ij} for individual i at locus j takes value in $\{0, 1, 2\}$, which is the count of mutations relative to the reference allele. The genotype of each SNP is often assumed to follow a binomial distribution (Jiang et al., 2024). So, we have $\mathbf{R}^* = \mathbb{E}[\mathbf{R}] = 2\mathbf{\Pi}^*\mathbf{\Theta}^{*\top}$ with $\mathbf{\Theta}^*$ containing the binomial parameters. Linkage disequilibrium (LD) is an important and well-studied phenomena for SNP genotype data, which refers to correlations among neighboring alleles (Reich et al., 2001). Therefore, we consider a binomial model for this dataset, and directly employ our proposed method on $\mathbf{R}/2$ to estimate $\mathbf{\Pi}^*$ and $\mathbf{\Theta}^*$. The left panel in Figure 7 is the ternary plot based on the estimated membership scores with our proposed method. Our method is able to reveal that the ASW sample is a mixture of the YRI and CEU samples, and the MEX sample is a mixture between the CEU sample and a third ancestral population. These observations align with the established biological understanding of human population genetics (Jakobsson et al., 2008). Furthermore, the left panel in Figure 6 confirms the existence of local dependence among the SNPs in the data. As demonstrated in Figure S.1 in the Supplementary Material, the covariance matrix of the estimated error matrix $\hat{\mathbf{E}}$ is a good estimate of the true error covariance matrix.

We compare the proposed method with a highly influential method called STRUCTURE (Pritchard et al., 2000; Falush et al., 2003). STRUCTURE is a method for multilocus genotype data to infer population structure and assign admixed proportions (membership scores) to each individual. Essentially, STRUCTURE falls within the generalized-GoM model, where the admixture proportions are assumed to be sampled from $\text{Dirichlet}(\alpha, \dots, \alpha)$, and the item parameters from $\text{Dirichlet}(\lambda_1, \dots, \lambda_{C_j})$. The hyper-parameter α is assumed to have a uni-

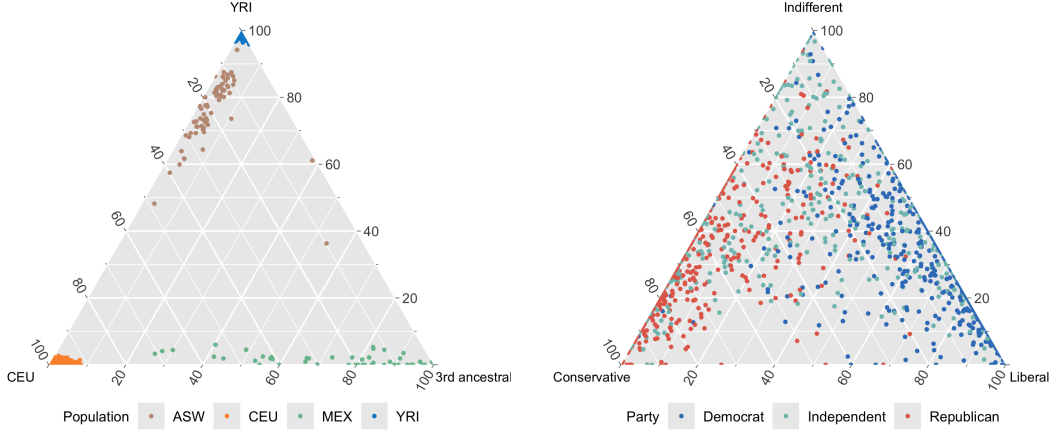


Figure 7: Ternary plots of the estimated membership scores for the HapMap3 data (left) and ANES data (right). For the Hapmap3 dataset, ASW denotes African ancestry in Southwest USA; CEU denotes Utah residents with Northern and Western European ancestry; MEX denotes Mexican ancestry in Los Angeles, California; YRI denotes Yoruba in Ibadan, Nigeria. For the ANES data, each dot is colored by the subject’s self-reported party information.

form hyper-prior $\text{Unif}[0, 10]$. STRUCTURE uses an MCMC algorithm to estimate the model parameters and employs Metropolis-Hasting to update α at each iteration. Figure 8 displays the comparisons of our proposed method with STRUCTURE in terms of the membership score estimates. The figures are generated with the **Distruct** software (Rosenberg, 2004). Each vertical slice represents an individual, and each color represents one extreme latent profile. The lengths of the three color bars for each individual i are proportional to the estimated membership score $\hat{\pi}_{i,k}$ for $k = 1, 2, 3$. We can see that both methods capture rather similar admixture structures in the sample. However, STRUCTURE took over 44 hours and 22 minutes to run with default settings, yet the proposed method takes only about 9 seconds. Notably, our method is able to generate similar results with a huge computation advantage. On another note, if we treat the $\{0, 1, 2\}$ responses as polytomous data and use the flattening technique before estimating parameters, it turns out that it gives very similar estimation results of $\hat{\Pi}$ to those given by the binomial model reported above.

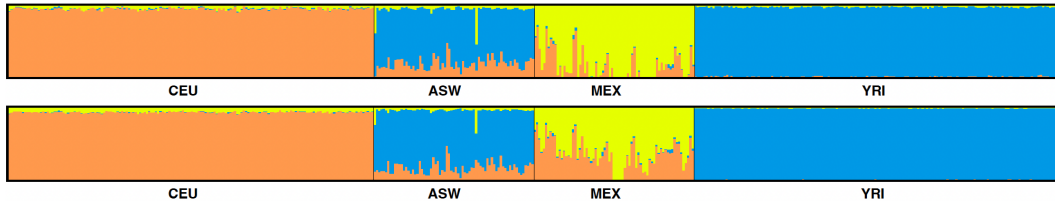


Figure 8: Membership results from STRUCTURE (upper) and the proposed method (lower) with $K = 3$ for the HapMap3 dataset. STRUCTURE takes more than 44 hours to finish while our method takes only 9 seconds. Each vertical slice represents an individual, with the lengths of its three color bars proportional to the estimated $\hat{\pi}_{i,k}$.

6.3 Single-cell Sequencing Data

We finally demonstrate an application to single-cell sequencing data, where the observed features are nonnegative counts. The dataset contains RNA-seq read counts for single cells of mouse embryo development with $N = 259$ cells and $J = 22,431$ genes (Deng et al., 2014, publicly available at <https://github.com/kkdey/CountClust>). This dataset also provides labels of 10 pre-implantation development phase categories for each cell.

The gene expression counts in our dataset range from 0 to 693,974, and we assume a Poisson model. In single-cell sequencing data, the phenomenon of local dependence is commonly referred to as gene co-expression (Van Der Wijst et al., 2018). We choose $K = 6$ according to the analysis on the same dataset conducted in Dey et al. (2017). Figure 9 presents our estimated membership scores $\hat{\Pi}$, with each vertical slice corresponding to a cell. From left (zygote) to right (late blast) are the 10 cell development phases. We can see that cells belonging to the same phase have relatively similar mixed membership allocations, and that the early stage cells are less mixed compared to cells at later stages. Furthermore, we observe a continuous evolution in the membership scores as the cells progress through the developmental stages. These findings align with those reported in Dey et al. (2017). Additionally, it takes 0.5 second to run the proposed method on the count matrix, while the computation time is reported to be 12 minutes in Dey et al. (2017), where they use R package `maptp` (Taddy, 2012) to fit the GoM model.

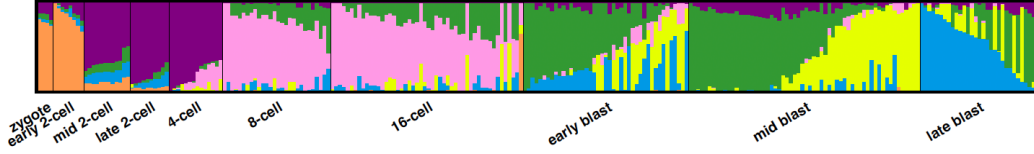


Figure 9: Generalized-GoM estimated memberships for the single cell data with $K = 6$. Each color represents an extreme latent profile, and each vertical slice represents a cell. The cells are ordered by their preimplantation development phase, and within each phase sorted by the membership score of the dominating profile in this phase.

7 Discussion

This paper proposes a fast spectral method for generalized-GoM models with potential local dependence. For the GoM model for high-dimensional polytomous data, we flatten the three-way quasi-tensor of data into a fat binary matrix and exploit its singular subspace geometry to estimate parameters. We establish fine-grained entrywise estimation error bounds for parameters in a broad family of generalized-GoM models. We also develop a novel $\ell_{2,\infty}$ singular subspace perturbation theory for arbitrary locally dependent and flexibly distributed noise, which is of independent interest and can be useful for other latent factor models. Our method has a significant computational advantage over popular Bayesian MCMC methods and outputs competitively accurate parameter estimates. Even if one ultimately desires to perform Bayesian posterior inference, we recommend initializing the MCMC algorithm using our spectral method. This sufficiently accurate initialization can help speed up the convergence of the MCMC chain to its stationary distribution.

There are several future directions worth exploring. First, Algorithm 1 requires the input of K , which is often unknown in practice. There are various tools to determine the choice of K . The simple yet widely-used approach is to inspect the scree plot of the singular values and find the “elbow point”. Other methods include the gap statistic (Tibshirani et al., 2001), parallel analysis (Hayton et al., 2004), eigen selection (Han et al., 2023). It is of special interest to apply the proposed method on an estimated K and still obtain theoretical guarantees, similar to the work in Fan et al. (2022) and Jin et al. (2023). Second, missing data

can appear in real-world applications especially with big data scenarios. Matrix completion using spectral methods has been studied (Cai et al., 2021) and could be potentially extended to our setting. Third, to further increase the modeling power of the generalized-GoM model, a potential extension is to incorporate the degree-correction parameter similar to the degree-corrected mixed membership model for networks (Ke and Jin, 2023; Jin et al., 2024; Lyu et al., 2024). Then one can perform certain normalizations in the spectral domain to correct the degree effects before estimating other parameters. Finally, it would be interesting to develop statistical inference results for locally dependent generalized-GoM models, similar to the considerations in Fan et al. (2022) for network models.

Supplementary Material. The Supplementary Material contains all technical proofs of the theoretical results and additional computation details.

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Supplementary Material

This Supplementary Material is organized as follows. Section S.1 includes the theoretical results and proofs for row-wise singular subspace perturbation bounds for asymmetric matrices with local dependence (Theorem 1). Section S.2 includes the proof for the parameter estimation error bound (Theorem 2). Section S.3 contains additional computational details.

S.1 Row-wise Singular Subspace Perturbation Bounds for Asymmetric Matrices with Local Dependence

S.1.1 Notations and Assumptions

Recall that we observe the data matrix

$$\mathbf{R} = \mathbf{R}^* + \mathbf{E} \in \mathbb{R}^{N \times J},$$

where the deterministic part $\mathbf{R}^*$ has rank K and the noise matrix $\mathbf{E}$ has mean zero. The rows of the error matrix $\mathbf{E}$ are independent. The following Assumption S.5 (Assumption 1 in the main paper) allows local dependence structures on the noise matrix and enables us to treat the error matrix entries as bounded variables with high probability. We will verify that this assumption holds for sub-exponential noises in Section S.1.4.

Assumption S.5 (Block Dependent Noise under Flexible Distributions). *Matrix $\mathbf{E}$ satisfies:*

- (a) *There exists a partition $S_1, S_2, \dots, S_L$ of $[J]$ with $|S_l| \leq M$ for each $l \in [L]$, such that the vectors $\{\mathbf{E}_{i,S_l}\}_{l=1}^L$ are mutually independent for different $i \in [N]$ and $l \in [L]$.*
- (b) *Either $|E_{i,j}| \leq B$ for all $i \in [N], j \in [J]$, or there exists a random matrix $\mathbf{E}' = (E'_{i,j}) \in \mathbb{R}^{N \times J}$ obeying the same dependence structure in (a), such that for any $i \in [N], j \in [J]$, it holds that $\|\mathbf{E}'_{i,:}\|_\infty \leq B$, $\mathbb{E}[E'_{i,j}] = 0$, $\|\text{Cov}(\mathbf{E}'_{i,:})\| \lesssim \|\text{Cov}(\mathbf{E}_{i,:})\|$, and $\Pr(E_{i,j} = E'_{i,j}) \geq 1 - O(d^{-22})$.*

Assumption S.6 (Pure Subject). $\mathbf{\Pi}^*$ satisfies that every extreme latent profile has at least one pure subject. In addition, $\text{rank}(\mathbf{\Theta}^*) = K$.

Consider the SVD of $\mathbf{R}^*$: $\mathbf{R}^* = \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top}$ where $\mathbf{\Lambda}^* = \text{diag}(\sigma_1^*, \dots, \sigma_K^*)$ with $\sigma_1^* \geq \dots \geq \sigma_K^* > 0$ collects the K singular values of $\mathbf{R}^*$. Matrices $\mathbf{U}_{N \times K}^*$, $\mathbf{V}_{J \times K}^*$ respectively consist of the left and right singular vectors and satisfy $\mathbf{U}^{*\top} \mathbf{U}^* = \mathbf{V}^{*\top} \mathbf{V}^* = \mathbf{I}_K$. Similarly, we consider the top- K SVD for the data matrix $\mathbf{R}$: $\mathbf{R} \approx \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$, where $\mathbf{\Lambda} = \text{diag}(\sigma_1, \dots, \sigma_K)$ with $\sigma_1 \geq \dots, \sigma_K \geq 0$, $\mathbf{U} \in \mathbb{R}^{N \times K}$, $\mathbf{V} \in \mathbb{R}^{J \times K}$, and $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_K$. Let $(\mathbf{U}^\perp, \mathbf{\Lambda}^\perp, \mathbf{V}^\perp)$ be the SVD of $\mathbf{R} - \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$. We introduce the canonical projection operator $\mathcal{P}_S : \mathbb{R}^{N \times J} \rightarrow \mathbb{R}^{N \times J}$, $P_S(\mathbf{E}) \mapsto (E_{i,j} \mathbf{1}_{\{(i,j) \in S\}})$ where $S \subset [N] \times [J]$. We also denote a tuple returned by the top- k singular value decomposition of a matrix $\mathbf{A} \in \mathbb{R}^{n_1 \times n_2}$ by $\text{svd}(\mathbf{A}; k) := (\mathbf{U}_\mathbf{A}, \mathbf{\Lambda}_\mathbf{A}, \mathbf{V}_\mathbf{A})$, where $\mathbf{U}_\mathbf{A} \in \mathbb{R}^{n_1 \times k}$ and $\mathbf{V}_\mathbf{A} \in \mathbb{R}^{n_2 \times k}$ respectively collect the top- k left and right singular vectors, and $\mathbf{\Lambda}_\mathbf{A}$ is a diagonal matrix with its diagonal entries being the top- k singular values in non-increasing order. We define additional notations in Table S.3, which will be used frequently.

Elementary Facts. One can easily verify that $\sigma^2 \leq \tilde{\sigma}^2 \leq M\sigma^2$.

Notation	Definition	Notation	Definition
Σ_{i,S_l}	$\text{Cov}(\mathbf{E}_{i,S_l})$	d	$N \vee J$
Σ_i	$\text{Cov}(\mathbf{E}_{i,:})$	r	$\frac{J}{N}$
σ^2	$\max_{i \in [N], j \in [J]} \text{Var}(E_{i,j})$	$[B]$	$\{1, 2, \dots, B\}$
$\tilde{\sigma}^2$	$\max_{i \in [N], l \in [L]} \ \Sigma_{i,S_l}\ $	κ^*	$\frac{\sigma_1^*}{\sigma_K^*}$
μ_1	$\frac{\ \mathbf{U}^*\ _{2,\infty}^2 N}{K}$	μ_2	$\frac{\ \mathbf{V}^*\ _{2,\infty}^2 J}{K}$

Table S.3: Notations.

Assumption S.7. We assume that $K \ll \min\{N, L\}$, $ML \asymp J$, and

$$\sigma_K^* \geq \sigma_K(\mathbf{\Pi}^*)\sigma_K(\mathbf{\Theta}^*) \gtrsim \kappa^* \sigma(M\sqrt{N} + \sqrt{J}), \quad (\text{S.1})$$

$$MB(\log d)^2 \max \left\{ \sqrt{\frac{\mu_1 K}{N}}, \sqrt{\frac{\mu_2 K}{J}} \right\} \lesssim \sigma, \quad (\text{S.2})$$

When $M = 1$, our setting reduces to the independent noise case. With Assumption S.7, we assume a low-rank signal matrix and balanced block size in the dependence structure. Comparing the first line (S.1) to Theorem 9 in Yan et al. (2021), even in the independent noise case, our assumption is weaker than Theorem 9 by removing the log factor in the lower bound of σ_K^* . The second line (S.2) is about the assumption on block size B . Compared to (6.5) in Assumption 3 in Yan et al. (2021), this is slightly stronger in the power of the $\log d$ term, changing from $1/2$ to 2 .

S.1.2 Tools for the Proof of Theorem 1

Note that we treat the entries of $\mathbf{E}$ as bounded variables hereinafter, given that all subsequent high-probability controls have an exceptional probability not greater than $O(d^{-20})$. According to Assumption S.5, replacing $\mathbf{E}$ with its truncated version does not alter the concentration inequalities.

Intuitively, we can establish concentration inequalities for $\mathbf{E}$ if the local dependence block is adequately small. For a vector $\mathbf{a}$ of length C and a subset $S \subseteq [C]$, denote $\mathbf{a}_S = (a_c)_{c \in S}$. For example, to use the matrix Bernstein inequality to give an upper bound for $\mathbf{E}_{1,:}\mathbf{A}$ with $\mathbf{A} \in \mathbb{R}^{J \times q}$, we can write $\mathbf{E}_{1,:}\mathbf{A}$ as $\sum_{l=1}^L \tilde{\mathbf{E}}_l^\top \mathbf{A}$ where $\tilde{\mathbf{E}}_l$ is a J -dimensional vector that has the l th block equal to $\mathbf{E}_{1,S_l}$ and other blocks equal to zero. Then

$$\|\mathbf{E}_{1,:}\mathbf{A}\|_2 \lesssim \sqrt{(\log J)V} + S \log J,$$

where

$$S = \max_{l \in [L]} \left\| \tilde{\mathbf{E}}_l \mathbf{A} \right\|_2 \leq MB \|\mathbf{A}\|_{2,\infty}$$

and

$$V = \max \left\{ \left\| \sum_j \mathbb{E} \left[\tilde{\mathbf{E}}_j^\top \mathbf{A} \mathbf{A}^\top \tilde{\mathbf{E}}_j \right] \right\|, \left\| \sum_j \mathbb{E} \left[\mathbf{A}^\top \tilde{\mathbf{E}}_j \tilde{\mathbf{E}}_j^\top \mathbf{A} \right] \right\| \right\} \leq \|\Sigma\| \|\mathbf{A}\|_F^2 \leq J \|\Sigma\| \|\mathbf{A}\|_{2,\infty}^2.$$

Remark 3. The perturbation theories in [Abbe et al. \(2022\)](#) also allow dependency on the columns of $\mathbf{E}$, assuming that $\mathbf{E}_{i,:}$ are sub-Gaussian vectors. However, their proofs depend on Hoeffding's inequality instead of Bernstein's inequality, with the help of sub-Gaussianity.

Lemma S.2. For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with rank K and $\mathbf{B} \in \mathbb{R}^{n \times k}$, $\mathbf{C} \in \mathbb{R}^{k \times l}$, we have

$$\|\mathbf{A}\|/\sqrt{m} \leq \|\mathbf{A}\|_{2,\infty} \leq \|\mathbf{A}\| \leq \|\mathbf{A}\|_F \leq \sqrt{m} \|\mathbf{A}\|_{2,\infty}, \quad (\text{S.3})$$

$$\|\mathbf{AB}\|_{2,\infty} \leq \|\mathbf{A}\|_{2,\infty} \|\mathbf{B}\|, \quad (\text{S.4})$$

$$\|\mathbf{ABC}\|_\infty \leq \|\mathbf{A}\|_{2,\infty} \|\mathbf{B}\| \|\mathbf{C}\|_{\infty,2}. \quad (\text{S.5})$$

Proof of Lemma S.2. Refer to [Cape et al. \(2019\)](#) for the first three inequalities in (S.3).

The last inequality of (S.3) is because $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \|\mathbf{A}_{i,:}\|^2} \leq \sqrt{m \|\mathbf{A}\|_{2,\infty}^2} = \sqrt{m} \|\mathbf{A}\|_{2,\infty}$.

Inequality (S.4) holds since

$$\|\mathbf{AB}\|_{2,\infty} = \max_i \|\mathbf{A}_{i,:} \mathbf{B}\| \leq \max_i \|\mathbf{A}_{i,:}\| \|\mathbf{B}\| = \|\mathbf{A}\|_{2,\infty} \|\mathbf{B}\|.$$

And inequality (S.5) yields from

$$\begin{aligned} |(\mathbf{ABC})_{ij}| &= |\mathbf{A}_{i,:} \mathbf{B} \mathbf{C}_{:,j}| \leq \|\mathbf{A}_{i,:}\| \|\mathbf{B}\| \|\mathbf{C}_{:,j}\| \\ &\leq \|\mathbf{A}\|_{2,\infty} \|\mathbf{B}\| \|\mathbf{C}\|_{\infty,2}. \end{aligned}$$

□

Lemma S.3. For any matrices $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{B} \in \mathbb{R}^{d_1 \times d_3}$, we have

$$\text{tr}(\mathbf{A} \mathbf{A}^\top \mathbf{B}) \leq \|\mathbf{A}\|_F^2 \|\mathbf{B}\|.$$

Proof of Lemma S.3. It follows directly from

$$\text{tr}(\mathbf{A} \mathbf{A}^\top \mathbf{B}) \leq \sum_{j \in [d_1]} \sigma_j(\mathbf{A} \mathbf{A}^\top \mathbf{B}) \leq \sum_{j \in [d_1]} \sigma_j(\mathbf{A} \mathbf{A}^\top) \|\mathbf{B}\| = \|\mathbf{A}\|_F^2 \|\mathbf{B}\|.$$

□

Lemmas S.4 and S.5 below give matrix concentration results for the block dependent noise matrix.

Lemma S.4 (Bernstein inequalities for block dependent noise matrix). *Suppose Assumption S.5 holds. Given any deterministic matrix $\mathbf{A} \in \mathbb{R}^{J \times q}$, $\mathbf{B} \in \mathbb{R}^{N \times p}$ and a positive constant $\alpha > 0$, with probability at least $1 - O(d^{-\alpha})$ one has*

$$\|\mathbf{E}_{i,:}\mathbf{A}\| \lesssim \tilde{\sigma}\sqrt{\alpha \log d} \|\mathbf{A}\|_F + \alpha MB \log d \|\mathbf{A}\|_{2,\infty}, \quad (\text{S.6})$$

$$\|\mathbf{E}_{:,j}^\top \mathbf{B}\| \lesssim \sigma\sqrt{\alpha \log d} \|\mathbf{B}\|_F + \alpha B \log d \|\mathbf{B}\|_{2,\infty}, \quad (\text{S.7})$$

$$\|\mathbf{E}\mathbf{A}\| \lesssim \tilde{\sigma}\sqrt{\alpha \log d} (\|\mathbf{A}\|_F + \sqrt{N} \|\mathbf{A}\|) + \alpha MB \log d \|\mathbf{A}\|_{2,\infty},$$

$$\|\mathbf{E}^\top \mathbf{B}\| \lesssim \sigma\sqrt{\alpha \log d} (\sqrt{M} \|\mathbf{B}\|_F + \sqrt{J} \|\mathbf{B}\|) + \alpha \sqrt{M} B \log d \|\mathbf{B}\|_{2,\infty}.$$

Proof of Lemma S.4. The proof is an application of the matrix Bernstein inequality (for example, Theorem 6.1.1 in Tropp (2015); Corollary 3.3 in Chen et al. (2021)). We write matrices as a summation of a sequence of independent random matrices: $\mathbf{E}_{i,:} = \sum_{l \in [L]} \mathcal{P}_{:,S_l}(\mathbf{E}_{i,:})$, $\mathbf{E}_{:,j} = \sum_{i \in [N]} \mathcal{P}_{i,:}(\mathbf{E}_{:,j})$, $\mathbf{E}_{:,S_j} = \sum_{i \in [N]} \mathcal{P}_{i,:}(\mathbf{E}_{:,S_j})$, $\mathbf{E} = \sum_{i \in [N]} \sum_{l \in [L]} \mathcal{P}_{i,S_l}(\mathbf{E})$. Note that for $\forall i \in [N], l \in [L], j \in [J]$, we can upper bound each independent block matrix: $\|\mathbf{E}_{i,S_l} \mathbf{A}_{S_l,:}\| = \|\sum_{j \in S_l} \mathbf{E}_{i,j} \mathbf{A}_{j,:}\| \leq MB \|\mathbf{A}\|_{2,\infty}$, $\|\mathbf{E}_{i,j} \mathbf{B}_{i,:}\| \leq B \|\mathbf{B}\|_{2,\infty}$, and $\|\mathbf{E}_{i,S_l}^\top \mathbf{B}_{i,:}\| \leq \|\mathbf{E}_{i,S_l}\| \|\mathbf{B}\|_{2,\infty} \leq \sqrt{MB} \|\mathbf{B}\|_{2,\infty}$.

For the first inequality we consider $\mathbf{E}_{i,:}\mathbf{A}$,

$$\|\mathbb{E} [\mathbf{A}^\top \mathbf{E}_{i,:}^\top \mathbf{E}_{i,:} \mathbf{A}] \| = \|\mathbf{A}^\top \mathbb{E} [\mathbf{E}_{i,:}^\top \mathbf{E}_{i,:}] \mathbf{A}\| \leq \|\mathbb{E} [\mathbf{E}_{i,:}^\top \mathbf{E}_{i,:}]\| \|\mathbf{A}\|^2 \leq \max_{l \in [L]} \|\Sigma_{i,S_l}\| \|\mathbf{A}\|^2 \leq \tilde{\sigma}^2 \|\mathbf{A}\|^2,$$

$$\mathbb{E} [\mathbf{E}_{i,:} \mathbf{A} \mathbf{A}^\top \mathbf{E}_{i,:}^\top] = \text{tr} (\mathbf{A} \mathbf{A}^\top \mathbb{E} [\mathbf{E}_{i,:}^\top \mathbf{E}_{i,:}]) \leq \|\mathbf{A}\|_F^2 \|\Sigma_i\| \leq \|\mathbf{A}\|_F^2 \max_{l \in [L]} \|\Sigma_{i,S_l}\| \leq \tilde{\sigma}^2 \|\mathbf{A}\|_F^2,$$

where the second line uses Lemma S.3.

For the second inequality, $\mathbf{E}_{:,j}^\top \mathbf{B}$ satisfies

$$\|\mathbb{E} [\mathbf{B}^\top \mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top \mathbf{B}]\| \leq \|\mathbb{E} [\mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top]\| \|\mathbf{B}\|^2 \leq \sigma^2 \|\mathbf{B}\|^2,$$

$$\mathbb{E}[\mathbf{E}_{:,j}^\top \mathbf{B} \mathbf{B}^\top \mathbf{E}_{:,j}] = \text{tr}(\mathbf{B} \mathbf{B}^\top \mathbb{E}[\mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top]) \leq \sigma^2 \|\mathbf{B}\|_F^2,$$

where the second line uses Lemma S.3.

For the third inequality,

$$\|\mathbb{E}[\mathbf{A}^\top \mathbf{E}^\top \mathbf{E} \mathbf{A}]\| \leq \sum_{i \in [N]} \|\mathbf{A}^\top \mathbb{E}[\mathbf{E}_{i,:}^\top \mathbf{E}_{i,:}] \mathbf{A}\| \leq N \tilde{\sigma}^2 \|\mathbf{A}\|^2,$$

$$\|\mathbb{E}[\mathbf{E} \mathbf{A} \mathbf{A}^\top \mathbf{E}^\top]\| = \|\text{diag}(\mathbb{E}[\mathbf{E}_{i,:} \mathbf{A} \mathbf{A}^\top \mathbf{E}_{i,:}^\top])\| \leq \tilde{\sigma}^2 \|\mathbf{A}\|_F^2.$$

For the last inequality,

$$\begin{aligned} \|\mathbb{E}[\mathbf{E}^\top \mathbf{B} \mathbf{B}^\top \mathbf{E}]\| &= \max_{l \in [L]} \|\mathbb{E}[\mathbf{E}_{:,S_l}^\top \mathbf{B} \mathbf{B}^\top \mathbf{E}_{:,S_l}]\| \leq M \max_{l \in [L]} \|\mathbb{E}[\mathbf{E}_{:,S_l}^\top \mathbf{B} \mathbf{B}^\top \mathbf{E}_{:,S_l}]\|_\infty \\ &\leq M \max_{j \in [J]} \text{tr}(\mathbf{B} \mathbf{B}^\top \mathbb{E}[\mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top]) \leq M \|\mathbf{B}\|_F^2 \|\mathbb{E}[\mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top]\| \leq \sigma^2 M \|\mathbf{B}\|_F^2, \\ \|\mathbb{E}[\mathbf{B}^\top \mathbf{E} \mathbf{E}^\top \mathbf{B}]\| &= \left\| \sum_{i \in [N]} \mathbf{B}_{i,:}^\top \mathbb{E}[\mathbf{E}_{i,:} \mathbf{E}_{i,:}^\top] \mathbf{B}_{i,:} \right\| \leq \sigma^2 J \left\| \sum_{i \in [N]} \mathbf{B}_{i,:}^\top \mathbf{B}_{i,:} \right\| \leq \sigma^2 J \|\mathbf{B}\|^2. \end{aligned}$$

□

Proposition 1 (Corollary 2.17 in Brailovskaya and van Handel (2024)). *Let $\mathbf{Y} = \sum_{i=1}^n \mathbf{Z}_i$, where $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ are independent (possibly non-self-disjoint) $d \times d$ random matrices with $\mathbb{E}[\mathbf{Z}_i] = 0$. Then*

$$\begin{aligned} \mathbb{P}\left[\|\mathbf{Y}\| \geq \|\mathbb{E}[\mathbf{Y}^* \mathbf{Y}]\|^\frac{1}{2} + \|\mathbb{E}[\mathbf{Y} \mathbf{Y}^*]\|^\frac{1}{2} + \right. \\ \left. C\left\{v(\mathbf{Y})^\frac{1}{2} \sigma(\mathbf{Y})^\frac{1}{2} (\log d)^\frac{3}{4} + \sigma_*(\mathbf{Y}) t^\frac{1}{2} + R(\mathbf{Y})^\frac{1}{3} \sigma(\mathbf{Y})^\frac{2}{3} t^\frac{2}{3} + R(\mathbf{Y}) t\right\}\right] \leq 4de^{-t} \end{aligned}$$

for a universal constant C and all $t > 0$, where

$$\begin{aligned} \sigma(\mathbf{Y}) &= \max\left\{\|\mathbb{E}[\mathbf{Y}^* \mathbf{Y}]\|^\frac{1}{2}, \|\mathbb{E}[\mathbf{Y} \mathbf{Y}^*]\|^\frac{1}{2}\right\}, \\ v(\mathbf{Y}) &= \|\text{Cov}(\mathbf{Y})\|^\frac{1}{2}, \\ \sigma_*(\mathbf{Y}) &= \sup_{\|\mathbf{v}\|=\|\mathbf{w}\|=1} \left\{\mathbb{E}[|\langle \mathbf{v}, \mathbf{Y} \mathbf{w} \rangle|^2]\right\}^\frac{1}{2}, \\ R(\mathbf{Y}) &= \left\| \max_{1 \leq i \leq n} \|\mathbf{Z}_i\| \right\|_\infty, \end{aligned}$$

with $\text{Cov}(\mathbf{Y})$ denotes the $d^2 \times d^2$ covariance matrix of the entries of $\mathbf{Y}$.

Remark 4. Note that the definitions of $\sigma(\mathbf{Y})$, $v(\mathbf{Y})$, and $\sigma_*(\mathbf{Y})$ do not depend on the decomposition of $\mathbf{Y}$. The results in Proposition 1 also hold for rectangular matrices, as we can consider a square matrix by padding zeros to a rectangular matrix.

Lemma S.5 (Concentration inequalities for locally dependent noise matrix). *Suppose Assumption S.5 holds. Then we have*

$$\begin{aligned}
\|\mathbf{E}\| &\lesssim \sigma\sqrt{J} + \tilde{\sigma}\sqrt{N} + M^{\frac{1}{6}}B^{\frac{1}{3}}\left(\sigma^{\frac{2}{3}}J^{\frac{1}{3}} + \tilde{\sigma}^{\frac{2}{3}}N^{\frac{1}{3}}\right)(\log d)^{\frac{2}{3}} + \sqrt{MB}\log d, \\
\|\mathbf{E}_{i,:}\| &\lesssim \sigma\sqrt{J} + M^{\frac{1}{6}}B^{\frac{1}{3}}\sigma^{\frac{2}{3}}J^{\frac{1}{3}}(\log d)^{\frac{2}{3}} + \sqrt{MB}\log d, \quad \forall i \in [N], \\
\|\mathbf{E}_{:,j}\| &\lesssim \sigma\sqrt{N} + B\log d, \quad \forall j \in [J], \\
\|\mathbf{E}\mathbf{V}^*\| &\lesssim \tilde{\sigma}\sqrt{N} + M^{\frac{1}{6}}B^{\frac{1}{3}}\tilde{\sigma}^{\frac{2}{3}}N^{\frac{1}{3}}(\log d)^{\frac{2}{3}} + MB\|\mathbf{V}^*\|_{2,\infty}\log d, \\
\|\mathbf{E}^\top \mathbf{U}^*\| &\lesssim \sigma\sqrt{J+MK} + \sqrt{M}\sigma\sqrt{\log d} + M^{\frac{1}{6}}B^{\frac{1}{3}}\sigma^{\frac{2}{3}}(J+mK)^{\frac{1}{3}}\|\mathbf{U}^*\|_{2,\infty}^{\frac{1}{3}}(\log d)^{\frac{2}{3}} \\
&\quad + \sqrt{MB}\|\mathbf{U}^*\|_{2,\infty}\log d,
\end{aligned}$$

with probability at least $1 - O(d^{-20})$.

Moreover, if Assumption S.7 hold, we have

$$\|\mathbf{E}\| \lesssim \sigma\sqrt{J} + \tilde{\sigma}\sqrt{N}, \tag{S.8}$$

$$\|\mathbf{E}_{i,:}\| \lesssim \sigma\sqrt{J}, \quad \forall i \in [N], \tag{S.9}$$

$$\|\mathbf{E}_{:,j}\| \lesssim \sigma\sqrt{N}, \quad \forall j \in [J], \tag{S.10}$$

$$\|\mathbf{E}\mathbf{V}^*\| \lesssim \tilde{\sigma}\sqrt{N}, \tag{S.11}$$

$$\|\mathbf{E}^\top \mathbf{U}^*\| \lesssim \sigma\sqrt{J}, \tag{S.12}$$

with probability at least $1 - O(d^{-20})$.

Proof of Lemma S.5. Using the facts established in the proof of Lemma S.4, we have

$$\|\mathbb{E}[\mathbf{E}^\top \mathbf{E}]\|^{\frac{1}{2}} \leq \tilde{\sigma}\sqrt{N}, \quad \|\mathbb{E}[\mathbf{E}\mathbf{E}^\top]\|^{\frac{1}{2}} \leq \sigma\sqrt{J}, \quad \sigma(\mathbf{E}) \leq \sigma\sqrt{J} + \tilde{\sigma}\sqrt{N},$$

$$v(\mathbf{E}) \leq \max_{i \in [N]} \max_{l \in [L]} \|\mathbb{E}[\mathbf{E}_{i,S_l}^\top \mathbf{E}_{i,S_l}]\|^{\frac{1}{2}} \leq \tilde{\sigma},$$

$$\sigma_*(\mathbf{E}) \leq \sup_{\|\mathbf{w}\|=1} \|\mathbb{E}[\mathbf{E}\mathbf{w}\mathbf{w}^\top \mathbf{E}^\top]\|^\frac{1}{2} \leq \max_{i \in [N]} \sup_{\|\mathbf{w}\|=1} \|\mathbb{E}[\mathbf{E}_{i,:} \mathbf{w}\mathbf{w}^\top \mathbf{E}_{i,:}^\top]\|^\frac{1}{2} \leq \tilde{\sigma}, \quad (\text{S.13})$$

$$R(\mathbf{E}) = \left\| \max_{i \in [N], l \in [L]} \|\mathbf{E}_{i,S_l}\| \right\|_\infty \leq \sqrt{MB}.$$

Then the first inequality in Lemma S.5 follows by plugging these conditions into Proposition 1 with $t = c \log d$ with a sufficiently large constant c .

Similarly, for the second and third inequalities we use

$$\begin{aligned} \sigma(\mathbf{E}_{i,:}) &= \max\{\|\mathbb{E}[\mathbf{E}_{i,:}^\top \mathbf{E}_{i,:}]\|^\frac{1}{2}, \|\mathbb{E}[\mathbf{E}_{i,:} \mathbf{E}_{i,:}^\top]\|^\frac{1}{2}\} \leq \max\{\sigma\sqrt{J}, \tilde{\sigma}\} = \sigma\sqrt{J}, \\ v(\mathbf{E}_{i,:}) &= \|\text{Cov}(\mathbf{E}_{i,:})\|^\frac{1}{2} \leq \tilde{\sigma}, \quad \sigma_*(\mathbf{E}_{i,:}) \leq \tilde{\sigma}, \quad R(\mathbf{E}_{i,:}) \leq \sqrt{MB}, \end{aligned}$$

and

$$\begin{aligned} \sigma(\mathbf{E}_{:,j}) &= \max\{\|\mathbb{E}[\mathbf{E}_{:,j}^\top \mathbf{E}_{:,j}]\|^\frac{1}{2}, \|\mathbb{E}[\mathbf{E}_{:,j} \mathbf{E}_{:,j}^\top]\|^\frac{1}{2}\} \leq \sigma\sqrt{N}, \\ v(\mathbf{E}_{:,j}) &\leq \sigma, \quad \sigma_*(\mathbf{E}_{:,j}) \leq \sigma, \quad R(\mathbf{E}_{:,j}) \leq B. \end{aligned}$$

Furthermore, for the fourth and fifth inequalities,

$$\begin{aligned} \sigma(\mathbf{E}\mathbf{V}^*) &= \max\left\{\|\mathbb{E}[\mathbf{E}\mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top]\|^\frac{1}{2}, \|\mathbb{E}[\mathbf{V}^{*\top} \mathbf{E}^\top \mathbf{E}\mathbf{V}^*]\|^\frac{1}{2}\right\} \leq \tilde{\sigma}\sqrt{N}, \\ v(\mathbf{E}\mathbf{V}^*) &\leq \tilde{\sigma}, \quad \sigma_*(\mathbf{E}\mathbf{V}^*) \leq \sigma_*(\mathbf{E}) \stackrel{(\text{S.13})}{\leq} \tilde{\sigma}, \end{aligned}$$

and with $MK \lesssim ML \asymp J$ we have

$$\begin{aligned} \|\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}]\| &= \max_{l \in [L]} \|\mathbb{E}[\mathbf{E}_{:,S_l}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}_{:,S_l}]\| \leq \max_{l \in [L]} \mathbb{E} \|\mathbf{E}_{:,S_l}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}_{:,S_l}\| \\ &= M \max_{j \in [J]} \mathbb{E}[\mathbf{E}_{:,j}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}_{:,j}] \leq M\sigma^2 \|\mathbf{U}^*\|_F^2 = MK\sigma^2, \\ \|\mathbb{E}[\mathbf{U}^{*\top} \mathbf{E} \mathbf{E}^\top \mathbf{U}^*]\| &\leq \|\mathbb{E}[\mathbf{E} \mathbf{E}^\top]\| \leq \sigma^2 J, \\ \sigma(\mathbf{E}^\top \mathbf{U}^*) &\leq \sigma\sqrt{J} + \sigma\sqrt{MK}, \\ R(\mathbf{E}^\top \mathbf{U}^*) &= \left\| \max_{i \in [N], l \in [L]} \|\mathbf{E}_{i,S_l}^\top \mathbf{U}_{i,:}^*\| \right\|_\infty \leq \sqrt{MB} \|\mathbf{U}^*\|_{2,\infty}, \\ v(\mathbf{E}^\top \mathbf{U}^*) &\leq \max_{l \in [L]} \|\mathbb{E}[\mathbf{E}_{:,S_l}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}_{:,S_l}]\|^\frac{1}{2} \leq \sqrt{MK}\sigma, \\ \sigma_*(\mathbf{U}^{*\top} \mathbf{E})^2 &\leq \sigma_*(\mathbf{E})^2 \stackrel{\text{by (S.13)}}{\leq} \tilde{\sigma}^2 \leq M\sigma^2. \end{aligned}$$

□

Lemma S.6 below utilizes a leave-one-out analysis that decouples the dependency between $\mathbf{R}$ and $\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*$, and a leave-one-block-out analysis on $\mathbf{R}$ and $\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*$.

Lemma S.6. *Suppose Assumptions S.5, S.7 hold. Then with probability at least $1 - O(d^{-10})$, one has*

$$\begin{aligned} & \|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top\mathbf{V}^*\|_{2,\infty} \\ & \lesssim \frac{\tilde{\sigma}\sigma\sqrt{K\log d(MN+J)}}{\sigma_K^*} + MB\log d \|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \frac{\sigma^2 J}{\sigma_K^*} \|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\ & \quad + \frac{\kappa^*\sigma^2(MN+J)}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma MB\log d\sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty}, \end{aligned} \quad (\text{S.14})$$

$$\begin{aligned} & \|\mathbf{R}^\top\mathbf{U}^* - \mathbf{R}^\top\mathbf{U}\mathbf{U}^\top\mathbf{U}^*\|_{2,\infty} \\ & \lesssim \frac{\sigma^2\sqrt{K\log d(MN+J)}}{\sigma_K^*} + B\log d \|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \frac{\sigma^2 MN}{\sigma_K^*} \|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \quad + \frac{\kappa^*\sigma^2(MN+J)}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma B\log d\sqrt{MN}}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty}. \end{aligned} \quad (\text{S.15})$$

Proof of Lemma S.6. We first establish some facts using the matrix perturbation theory as a tool for later use. By the perturbation theory under the spectral norm (for example, Theorem 2.9 in Chen et al. (2021), Lemma 2 in Yan et al. (2021)), (S.1) in Assumption S.7, and inequalities (S.8), (S.11), (S.12) in Lemma S.5, we know that

$$\max \left\{ \|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|, \|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\| \right\} \lesssim \frac{\tilde{\sigma}\sqrt{N} + \sigma\sqrt{J}}{\sigma_K^*}, \quad (\text{S.16})$$

$$\max \left\{ \left\| \mathbf{U}^{*\top}\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{I}_K \right\|, \left\| \mathbf{V}^{*\top}\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{I}_K \right\| \right\} \lesssim \frac{\tilde{\sigma}^2 N + \sigma^2 J}{\sigma_K^{*2}}. \quad (\text{S.17})$$

We first give an overview of the proof. In Part I, we upper bound $\|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top\mathbf{V}^*\|_{2,\infty}$ by separately bounding $\beta_1 := \|\mathbf{R}^*(\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*)\|_{2,\infty}$ and $\beta_2 := \|\mathbf{E}(\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*)\|_{2,\infty}$. To bound β_2 that involves the error matrix $\mathbf{E}$ and the right singular space perturbation, we use the “leave-one-out” technique since the rows of $\mathbf{E}$ are independent. In Part II, we upper bound $\|\mathbf{R}^\top\mathbf{U}^* - \mathbf{R}^\top\mathbf{U}\mathbf{U}^\top\mathbf{U}^*\|_{2,\infty}$ by the summation of $\beta'_1 := \|\mathbf{R}^{*,\top}(\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*)\|_{2,\infty}$ and

$\beta'_2 := \|\mathbf{E}^\top(\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*)\|_{2,\infty}$. Similarly, to bound β'_2 we use the “leave-one-block-out technique” since the columns of $\mathbf{E}$ are block dependent.

Part I: upper bounding $\|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top\mathbf{V}^*\|_{2,\infty}$.

We define the following leave-one-row-out matrix for every $i \in [N]$:

$$\mathbf{R}^{(-i)} := \mathbf{R}^* + \mathcal{P}_{-i,:}(\mathbf{E}).$$

Let $\sigma_1^{(-i)} \geq \dots \geq \sigma_{N \wedge J}^{(-i)}$ be the singular values of $\mathbf{R}^{(-i)}$, and $\mathbf{u}_k^{(-i)}, \mathbf{v}_k^{(-i)}$ be the left and right singular vectors associated with $\sigma_k^{(-i)}$. Write $\mathbf{U}^{(-i)} = (\mathbf{u}_1^{(-i)}, \dots, \mathbf{u}_K^{(-i)}) \in \mathbb{R}^{N \times K}$ and $\mathbf{V}^{(-i)} = (\mathbf{v}_1^{(-i)}, \dots, \mathbf{v}_K^{(-i)}) \in \mathbb{R}^{J \times K}$. We first establish some facts about these quantities. Note that $\|\mathcal{P}_{-i,:}(\mathbf{E})\| \leq \|\mathbf{E}\|$. By Weyl’s inequality we know $\sigma_K^{(-i)} \geq \sigma_K^* - \|\mathcal{P}_{-i,:}(\mathbf{E})\|$ and $\sigma_{K+1}^{(-i)} \leq \|\mathcal{P}_{-i,:}(\mathbf{E})\|$, thus

$$\sigma_K^{(-i)} - \sigma_{K+1}^{(-i)} - \|\mathcal{P}_{-i,:}(\mathbf{E})\| \geq \sigma_K^* - 3\|\mathcal{P}_{-i,:}(\mathbf{E})\| \geq \sigma_K^* - 3\|\mathbf{E}\| \gtrsim \sigma_K^*, \quad (\text{S.18})$$

where the last inequality is from (S.8) in Lemma S.5 and (S.1) in Assumption S.7.

It follows by the triangle inequality that

$$\begin{aligned} \|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top\mathbf{V}^*\|_{2,\infty} &= \|(\mathbf{R}^* + \mathbf{E})(\mathbf{V}^* - \mathbf{V}\mathbf{V}^\top\mathbf{V}^*)\|_{2,\infty} \\ &\leq \underbrace{\|\mathbf{R}^*(\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*)\|_{2,\infty}}_{\beta_1} + \underbrace{\|\mathbf{E}(\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*)\|_{2,\infty}}_{\beta_2}. \end{aligned} \quad (\text{S.19})$$

We first notice

$$\begin{aligned} \beta_1 &= \|\mathbf{R}^*(\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*)\|_{2,\infty} = \left\| \mathbf{U}^* \mathbf{\Lambda}^* \left(\mathbf{V}^{*\top} \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{I}_K \right) \right\|_{2,\infty} \\ &\leq \sigma_1^* \|\mathbf{U}^*\|_{2,\infty} \left\| \mathbf{V}^{*\top} \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{I}_K \right\| \\ &\lesssim \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty}, \end{aligned} \quad (\text{S.20})$$

where the last inequality yields from (S.17).

For β_2 , we use the leave-one-out argument to deal with the dependency between $\mathbf{E}_{i,:}$ and $\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*$. Applying the triangle inequality gives

$$\beta_2 \leq \underbrace{\max_{i \in [N]} \left\| \mathbf{E}_{i,:} \left(\mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^* \right) \right\|}_{\gamma_1} + \underbrace{\max_{i \in [N]} \left\| \mathbf{E}_{i,:} \left(\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* \right) \right\|}_{\gamma_2} \quad (\text{S.21})$$

Note that $\mathbf{E}_{i,:}$ is independent of $\mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^*$, which allows us to apply the matrix Bernstein inequality to control γ_1 conditional on $\mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^*$.

Upper bounding γ_1 in (S.21). We condition on $\mathbf{V}^{(-i)}$ and apply Lemma S.4 on γ_1 . With probability at least $1 - O(d^{-10})$, one has

$$\begin{aligned} \gamma_1 &\lesssim \tilde{\sigma} \sqrt{\log d} \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_F + MB \log d \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\ &\leq \tilde{\sigma} \sqrt{\log d} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_F + \tilde{\sigma} \sqrt{\log d} \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}\mathbf{V}^\top \mathbf{V}^* \right\|_F \\ &\quad + MB \log d \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + MB \log d \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}\mathbf{V}^\top \mathbf{V}^* \right\|_{2,\infty} \\ &\leq \tilde{\sigma} \sqrt{\log d} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_F + \tilde{\sigma} \sqrt{\log d} \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} - \mathbf{V}\mathbf{V}^\top \right\|_F \\ &\quad + MB \log d \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + MB \log d \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} - \mathbf{V}\mathbf{V}^\top \right\|_F \\ &\leq \tilde{\sigma} \sqrt{\log d} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_F + MB \log d \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\ &\quad + MB \log d \max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} - \mathbf{V}\mathbf{V}^\top \right\|_F, \end{aligned} \quad (\text{S.22})$$

where the last line follows since $\tilde{\sigma} \leq \sqrt{MB}$.

We next bound $\max_{i \in [N]} \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} - \mathbf{V}\mathbf{V}^\top \right\|_F$. Combining the Wedin theorem (Theorem 2.9 in [Chen et al. \(2021\)](#)) and (S.18) gives

$$\begin{aligned} &\max \left\{ \left\| \mathbf{V}\mathbf{V}^\top - \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \right\|_F, \left\| \mathbf{U}\mathbf{U}^\top - \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \right\|_F \right\} \\ &\lesssim \frac{\max \left\{ \left\| (\mathbf{R} - \mathbf{R}^{(-i)}) \mathbf{V}^{(-i)} \right\|_F, \left\| \mathbf{U}^{(-i)\top} (\mathbf{R} - \mathbf{R}^{(-i)}) \right\|_F \right\}}{\sigma_K^*}. \end{aligned} \quad (\text{S.23})$$

For the numerator of the RHS above, we have with probability that at least $1 - O(d^{-12})$ that

$$\left\| (\mathbf{R} - \mathbf{R}^{(-i)}) \mathbf{V}^{(-i)} \right\|_F = \left\| \mathbf{E}_{i,:} \mathbf{V}^{(-i)} \right\| \stackrel{(i)}{\leq} 2 \left\| \mathbf{E}_{i,:} \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* \right\|$$

$$\begin{aligned}
&\leq 2 \|\mathbf{E}_{i,:} \mathbf{V}^*\| + 2 \|\mathbf{E}_{i,:} (\mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* - \mathbf{V}^*)\| \\
&\stackrel{(ii)}{\lesssim} \tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \gamma_1,
\end{aligned} \tag{S.24}$$

$$\begin{aligned}
\left\| \mathbf{U}^{(-i)\top} (\mathbf{R} - \mathbf{R}^{(-i)}) \right\|_F &= \left\| \mathbf{U}_{i,:}^{(-i)\top} \mathbf{E}_{i,:} \right\|_F \stackrel{(iii)}{\leq} 2 \left\| \left(\mathbf{U}_{i,:}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* \right)^\top \mathbf{E}_{i,:} \right\|_F \\
&\leq 2 \left\| \mathbf{U}_{i,:}^{*\top} \mathbf{E}_{i,:} \right\|_F + 2 \left\| \left(\mathbf{U}_{i,:}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}_{i,:}^{*\top} \right)^\top \mathbf{E}_{i,:} \right\|_F \\
&\stackrel{(iv)}{\lesssim} \sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \sigma \sqrt{J} \left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty}.
\end{aligned} \tag{S.25}$$

To show that (i) holds, see that with probability at least $1 - O(d^{-20})$ one has

$$\begin{aligned}
\sigma_K^2 (\mathbf{V}^{(-i)\top} \mathbf{V}^*) &\geq 1 - \left\| \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} - \mathbf{V}^* \mathbf{V}^{*\top} \right\|^2 \\
&\geq 1 - \left(\sqrt{2} \frac{\|\mathcal{P}_{-i,:}(\mathbf{E})\|}{\sigma_K^* - \|\mathcal{P}_{-i,:}(\mathbf{E})\|} \right)^2 \geq 1 - \left(\sqrt{2} \frac{\|\mathbf{E}\|}{\sigma_K^* - \|\mathbf{E}\|} \right)^2 \gg \frac{1}{4},
\end{aligned}$$

where the first inequality is from Lemma 2.5 in [Chen et al. \(2021\)](#), the second inequality is from Wedin's Theorem, and the last inequality uses (S.8), and (S.1) in Assumption S.7. In addition, (ii) uses (S.6) in Lemma S.4, (iii) uses a similar argument to that of (i), and (iv) uses (S.9) in Lemma S.5. Furthermore, it holds with probability at least $1 - O(d^{-12})$ that

$$\begin{aligned}
&\left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
&\leq \left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U} \mathbf{U}^\top \mathbf{U}^* \right\|_{2,\infty} + \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
&\leq \left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} - \mathbf{U} \mathbf{U}^\top \right\|_{2,\infty} + \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
&\stackrel{(v)}{\lesssim} \frac{1}{\sigma_K^*} \left(\tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \gamma_1 \right) \\
&\quad + \frac{1}{\sigma_K^*} \left(\sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \sigma \sqrt{J} \left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \right) \\
&\quad + \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty},
\end{aligned} \tag{S.26}$$

where (v) uses (S.23), (S.24), (S.25). Rearranging the terms in (S.26) yields that

$$\begin{aligned}
&\left\| \mathbf{U}^{(-i)} \mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
&\stackrel{(vi)}{\lesssim} \frac{1}{\sigma_K^*} \left(\tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \gamma_1 \right)
\end{aligned}$$

$$+ \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \quad (\text{S.27})$$

holds with probability at least $1 - O(d^{-12})$ where (vi) uses $\sigma\sqrt{J} = o(\sigma_K^*)$ from Assumption S.7. Plugging (S.24), (S.25), (S.27) into (S.23) leads to

$$\begin{aligned} & \max \left\{ \left\| \mathbf{V}\mathbf{V}^\top - \mathbf{V}^{(-i)}\mathbf{V}^{(-i)\top} \right\|_F, \left\| \mathbf{U}\mathbf{U}^\top - \mathbf{U}^{(-i)}\mathbf{U}^{(-i)\top} \right\|_F \right\} \\ & \lesssim \frac{1}{\sigma_K^*} \left(\tilde{\sigma}\sqrt{K\log d} + MB\log d \|\mathbf{V}^*\|_{2,\infty} + \gamma_1 \right) \\ & \quad + \frac{1}{\sigma_K^*} \left(\sigma\sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \sigma\sqrt{J} \left\| \mathbf{U}^{(-i)}\mathbf{U}^{(-i)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \right) \\ & \stackrel{(vii)}{\lesssim} \frac{1}{\sigma_K^*} \left(\tilde{\sigma}\sqrt{K\log d} + MB\log d \|\mathbf{V}^*\|_{2,\infty} + \sigma\sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \sigma\sqrt{J} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \gamma_1 \right) \end{aligned} \quad (\text{S.28})$$

with probability at least $1 - O(d^{-12})$, where (vii) again uses $\sigma\sqrt{J} = o(\sigma_K^*)$ from Assumption S.7. Plugging (S.28) back into (S.22), we have with probability at least $1 - O(d^{-10})$

$$\begin{aligned} \gamma_1 & \lesssim \tilde{\sigma}\sqrt{\log d} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_F + MB\log d \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \quad + \frac{MB\log d}{\sigma_K^*} \left(\tilde{\sigma}\sqrt{K\log d} + MB\log d \|\mathbf{V}^*\|_{2,\infty} + \sigma\sqrt{J} \|\mathbf{U}^*\|_{2,\infty} \right. \\ & \quad \left. + \sigma\sqrt{J} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \gamma_1 \right). \end{aligned} \quad (\text{S.29})$$

Since $MB\log d \ll \sigma_K^*$, rearranging the terms in (S.29) yields that

$$\begin{aligned} \gamma_1 & \lesssim \tilde{\sigma}\sqrt{\log d} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_F + MB\log d \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \quad + \frac{MB\log d}{\sigma_K^*} \left(\tilde{\sigma}\sqrt{K\log d} + MB\log d \|\mathbf{V}^*\|_{2,\infty} + \sigma\sqrt{J} \|\mathbf{U}^*\|_{2,\infty} \right. \\ & \quad \left. + \sigma\sqrt{J} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right), \\ & \stackrel{(xiii)}{\lesssim} \frac{\tilde{\sigma}\sigma\sqrt{K(MN+J)\log d}}{\sigma_K^*} + MB\log d \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \quad + \frac{MB\log d}{\sigma_K^*} \left(\tilde{\sigma}\sqrt{K\log d} + MB\log d \|\mathbf{V}^*\|_{2,\infty} + \sigma\sqrt{J} \|\mathbf{U}^*\|_{2,\infty} \right. \\ & \quad \left. + \sigma\sqrt{J} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right), \end{aligned}$$

where (xiii) is from (S.16).

Upper bounding γ_2 in (S.21). Using (S.9) in Lemma S.5 and (S.28), we have with probability at least $1 - O(d^{-10})$ that

$$\begin{aligned}
\gamma_2 &= \max_{i \in [N]} \left\| \mathbf{E}_{i,:} \left(\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \mathbf{V}^* \right) \right\| \\
&\lesssim \max_{i \in [N]} \left\| \mathbf{E}_{i,:} \right\| \left\| \mathbf{V} \mathbf{V}^\top - \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \right\| \\
&\lesssim \sigma \sqrt{J} \max_{i \in [N]} \left\| \mathbf{V} \mathbf{V}^\top - \mathbf{V}^{(-i)} \mathbf{V}^{(-i)\top} \right\| \\
&\lesssim \frac{\sigma \sqrt{J}}{\sigma_K^*} \left(\tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} + \right. \\
&\quad \left. \sigma \sqrt{J} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \gamma_1 \right).
\end{aligned}$$

Combining (S.16) with the upper bounds for γ_1 and γ_2 , with $\sigma \sqrt{J} = o(\sigma_K^*)$, we have with probability at least $1 - O(d^{-10})$ that

$$\begin{aligned}
\beta_2 &\lesssim \frac{\tilde{\sigma} \sigma \sqrt{K(MN + J) \log d}}{\sigma_K^*} + MB \log d \|\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{MB \log d}{\sigma_K^*} \left(\tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} \right. \\
&\quad \left. + \sigma \sqrt{J} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right) \\
&\quad + \frac{\sigma \sqrt{J}}{\sigma_K^*} \left(\tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} + \sigma \sqrt{J} \|\mathbf{U}^*\|_{2,\infty} \right. \\
&\quad \left. + \sigma \sqrt{J} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right) \\
&\lesssim \frac{\tilde{\sigma} \sqrt{K \log d} (\sigma \sqrt{MN + J} + MB \log d)}{\sigma_K^*} + MB \log d \|\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{(\sigma \sqrt{J} + MB \log d) MB \log d}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{\sigma \sqrt{J} (MB \log d + \sigma \sqrt{J})}{\sigma_K^*} (\|\mathbf{U}^*\|_{2,\infty} + \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}) \\
&\lesssim \frac{\tilde{\sigma} \sigma \sqrt{K \log d (MN + J)}}{\sigma_K^*} + MB \log d \|\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \frac{\sigma MB \log d \sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{\sigma^2 J}{\sigma_K^*} (\|\mathbf{U}^*\|_{2,\infty} + \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}), \tag{S.30}
\end{aligned}$$

where the last inequality is from $MB \log d \lesssim \sigma \sqrt{(MN) \wedge J}$ by (S.2) in Assumption S.7.

Now incorporating (S.30) and (S.20) into (S.19), we have with probability at least $1 - O(d^{-10})$ that

$$\begin{aligned}
\|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^*\|_{2,\infty} &\lesssim \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\tilde{\sigma} \sigma \sqrt{K \log d (MN + J)}}{\sigma_K^*} \\
&\quad + MB \log d \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \frac{\sigma MB \log d \sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{\sigma^2 J}{\sigma_K^*} \left(\|\mathbf{U}^*\|_{2,\infty} + \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right) \\
&\lesssim \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\tilde{\sigma} \sigma \sqrt{K \log d (MN + J)}}{\sigma_K^*} \\
&\quad + MB \log d \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \frac{\sigma MB \log d \sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{\sigma^2 J}{\sigma_K^*} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}.
\end{aligned}$$

Part II: upper bounding $\|\mathbf{R}\mathbf{V}^* - \mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^*\|_{2,\infty}$.

For the second part, we will adopt a “leave-one-block-out” technique on the noise matrix $\mathbf{E}$. We define the following leave-one-block-out matrix for every $j \in [J]$:

$$\mathbf{R}^{(j)} := \mathbf{R}^* + \mathcal{P}_{:, -S_{l_j}}(\mathbf{E}),$$

where $l_j \in [L]$ such that $j \in S_{l_j}$. Let $\sigma_1^{(j)} \geq \dots \geq \sigma_{N \wedge J}^{(j)}$ be the singular values of $\mathbf{R}^{(j)}$, and $\mathbf{u}_k^{(j)}, \mathbf{v}_k^{(j)}$ be the left and right singular vectors associated with $\sigma_k^{(j)}$. Write $\mathbf{U}^{(j)} = (\mathbf{u}_1^{(j)}, \dots, \mathbf{u}_K^{(j)}) \in \mathbb{R}^{N \times K}$ and $\mathbf{V}^{(j)} = (\mathbf{v}_1^{(j)}, \dots, \mathbf{v}_K^{(j)}) \in \mathbb{R}^{J \times K}$. With similar arguments that lead to (S.18), we know that

$$\sigma_K^{(j)} - \sigma_{K+1}^{(j)} - \|\mathcal{P}_{:, -S_{l_j}}(\mathbf{E})\| \gtrsim \sigma_K^*. \quad (\text{S.31})$$

To begin with, by the triangle inequality we have

$$\begin{aligned}
&\|\mathbf{R}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*)\|_{2,\infty} \\
&\leq \underbrace{\|\mathbf{R}^{*\top} (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*)\|_{2,\infty}}_{\beta'_1} + \underbrace{\|\mathbf{E}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*)\|_{2,\infty}}_{\beta'_2}. \quad (\text{S.32})
\end{aligned}$$

For β'_1 , with (S.17) we have with probability at least $1 - O(n^{-10})$

$$\begin{aligned}\beta'_1 &= \left\| \mathbf{R}^{*\top} (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*) \right\|_{2,\infty} \\ &\leq \sigma_1^* \|\mathbf{V}^*\|_{2,\infty} \left\| \mathbf{U}^{*\top} \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{I}_K \right\| \lesssim \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty}.\end{aligned}\quad (\text{S.33})$$

For β'_2 , introducing the "leave-one-block-out" alternative $\mathbf{U}^{(j)}$ yields that

$$\begin{aligned}\beta'_2 &= \max_{j \in [J]} \left\| \mathbf{E}_{:,j}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*) \right\|_2 \\ &\leq \underbrace{\max_{j \in [J]} \left\| \mathbf{E}_{:,j}^\top (\mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* - \mathbf{U}^*) \right\|}_{\gamma'_1} + \underbrace{\max_{j \in [J]} \left\| \mathbf{E}_{:,j}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^*) \right\|}_{\gamma'_2}.\end{aligned}\quad (\text{S.34})$$

Upper bounding γ'_1 in (S.34). Utilizing Lemma S.4, one has at least $1 - O(d^{-11})$ that,

$$\begin{aligned}\gamma'_1 &\leq \sigma \sqrt{\log d} \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_F + B \log d \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\ &\leq \sigma \sqrt{\log d} \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_F + \sigma \sqrt{\log d} \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* - \mathbf{U}\mathbf{U}^\top \mathbf{U}^* \right\|_F \\ &\quad + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} + B \log d \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* - \mathbf{U}\mathbf{U}^\top \mathbf{U}^* \right\|_{2,\infty} \\ &\leq \sigma \sqrt{\log d} \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_F + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\ &\quad + \left(\sigma \sqrt{\log d} + B \log d \right) \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} - \mathbf{U}\mathbf{U}^\top \right\|_F \\ &\lesssim \sigma \sqrt{\log d} \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_F + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\ &\quad + B \log d \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} - \mathbf{U}\mathbf{U}^\top \right\|_F,\end{aligned}\quad (\text{S.35})$$

where the last line holds since $\sigma \sqrt{\log d} \ll B \log d$.

To bound $\max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} - \mathbf{U}\mathbf{U}^\top \right\|_F$ in the RHS above, we apply Wedin's theorem with (S.31) and obtain

$$\begin{aligned}&\max \left\{ \left\| \mathbf{U}\mathbf{U}^\top - \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \right\|_F, \left\| \mathbf{V}\mathbf{V}^\top - \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \right\|_F \right\} \\ &\lesssim \frac{\max \left\{ \left\| \mathcal{P}_{:,S_{l_j}}(\mathbf{E})^\top \mathbf{U}^{(j)} \right\|_F, \left\| \mathcal{P}_{:,S_{l_j}}(\mathbf{E}) \mathbf{V}^{(j)} \right\|_F \right\}}{\sigma_K^*}.\end{aligned}\quad (\text{S.36})$$

Following similar arguments above, with probability at least $1 - O(d^{-10})$ we have

$$\begin{aligned}
\max_{j \in [J]} \left\| \mathcal{P}_{:,S_{l_j}}(\mathbf{E})^\top \mathbf{U}^{(j)} \right\|_F &\leq \sqrt{M} \max_{j \in [J]} \left\| \mathbf{E}_{:,j}^\top \mathbf{U}^{(j)} \right\| \leq 2\sqrt{M} \max_{j \in [J]} \left\| \mathbf{E}_{:,j}^\top \mathbf{U}^{(j)\top} \mathbf{U}^{(j)} \mathbf{U}^* \right\| \\
&\leq 2\sqrt{M} \max_{j \in [J]} \left(\left\| \mathbf{E}_{:,j}^\top \mathbf{U}^* \right\| + \left\| \mathbf{E}_{:,j}^\top \mathbf{U}^* - \mathbf{E}_{:,j}^\top \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \mathbf{U}^* \right\| \right) \\
&\stackrel{(ii)}{\lesssim} \sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1, \tag{S.37}
\end{aligned}$$

$$\begin{aligned}
\max_{j \in [J]} \left\| \mathcal{P}_{:,S_{l_j}}(\mathbf{E}) \mathbf{V}^{(j)} \right\|_F &= \max_{j \in [J]} \left\| \mathbf{E}_{:,S_{l_j}} \mathbf{V}_{S_{l_j},:}^{(j)} \right\|_F \leq 2 \max_{j \in [J]} \left\| \mathbf{E}_{:,S_{l_j}} \mathbf{V}_{S_{l_j},:}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* \right\|_F \\
&\leq 2 \max_{j \in [J]} \left\| \mathbf{E}_{:,S_{l_j}} \mathbf{V}_{S_{l_j},:}^* \right\|_F + 2 \max_{j \in [J]} \left\| \mathbf{E}_{:,S_{l_j}} \left(\mathbf{V}_{S_{l_j},:}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}_{S_{l_j},:}^* \right) \right\|_F \\
&\leq 2M \max_{j \in [J]} \left(\left\| \mathbf{E}_{:,j} \right\| \|\mathbf{V}^*\|_{2,\infty} + \left\| \mathbf{E}_{:,j} \right\| \left\| \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \right) \\
&\stackrel{(iii)}{\lesssim} \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} + \sigma M \sqrt{N} \left\| \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}, \tag{S.38}
\end{aligned}$$

where (ii) uses (S.7) in Lemma S.4, and (iii) uses (S.10) in Lemma S.5. Further, we note that

$$\begin{aligned}
&\left\| \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \leq \left\| \left(\mathbf{V} \mathbf{V}^\top - \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \right) \mathbf{V}^* \right\|_{2,\infty} + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\
&\leq \left\| \mathbf{V} \mathbf{V}^\top - \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \right\|_{2,\infty} + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\
&\stackrel{(iv)}{\lesssim} \frac{1}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 \right) \\
&\quad + \frac{1}{\sigma_K^*} \left(M \sigma \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} + \sigma M \sqrt{N} \left\| \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \right) + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\
&\stackrel{(v)}{\lesssim} \frac{1}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} \right) \\
&\quad + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}, \tag{S.39}
\end{aligned}$$

where (iv) uses (S.36), (S.37), (S.38), and (v) uses $M \sigma \sqrt{N} \ll \sigma_K^*$ from (S.1) in Assumption S.7.

Plugging (S.37), (S.38), (S.39) into (S.36), we have with probability at least $1 - O(d^{-10})$,

$$\begin{aligned}
&\left\| \mathbf{U} \mathbf{U}^\top - \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} \right\|_F \\
&\lesssim \frac{1}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 \right)
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{\sigma_K^*} \left(\sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} + \sigma M \sqrt{N} \left\| \mathbf{V}^{(j)} \mathbf{V}^{(j)\top} \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \right) \\
& \lesssim \frac{1}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} \right) \\
& \quad + \frac{\sigma M \sqrt{N}}{\sigma_K^{*2}} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} \right) \\
& \quad + \frac{\sigma M \sqrt{N}}{\sigma_K^*} \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\
& \stackrel{(vi)}{\lesssim} \frac{1}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} \right) \\
& \quad + \frac{\sigma M \sqrt{N}}{\sigma_K^*} \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}, \tag{S.40}
\end{aligned}$$

where (vi) is because $\sigma M \sqrt{N} \ll \sigma_K^*$ from (S.1) in Assumption S.7.

Combining (S.35) and (S.40) gives that with probability at least $1 - O(d^{-10})$ we have

$$\begin{aligned}
\gamma'_1 & \lesssim \sigma \sqrt{\log d} \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_F + B \log d \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
& \quad + B \log d \max_{j \in [J]} \left\| \mathbf{U}^{(j)} \mathbf{U}^{(j)\top} - \mathbf{U} \mathbf{U}^\top \right\|_F \\
& \stackrel{(vii)}{\lesssim} \frac{\sigma \sqrt{(\tilde{\sigma}^2 N + \sigma^2 J) K \log d}}{\sigma_K^*} + B \log d \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
& \quad + \frac{B \log d}{\sigma_K^*} \left(\sigma \sqrt{MK \log d} + \sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M} \gamma'_1 \right. \\
& \quad \left. + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} + \sigma M \sqrt{N} \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \right), \tag{S.41}
\end{aligned}$$

where (vii) uses (S.16).

Provided $B \sqrt{M} \log d \ll \sigma_K^*$ from (S.1) and (S.2) in Assumption S.7, we rearrange (S.41) and derive that

$$\begin{aligned}
\gamma'_1 & \lesssim \frac{\sigma \sqrt{K \log d} \left(\sigma \sqrt{MN} + J + \sqrt{M} B \log d \right)}{\sigma_K^*} + B \log d \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
& \quad + \frac{B \log d}{\sigma_K^*} \left(\sqrt{M} B \log d \|\mathbf{U}^*\|_{2,\infty} + \sigma M \sqrt{N} \|\mathbf{V}^*\|_{2,\infty} + \sigma M \sqrt{N} \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \right) \tag{S.42}
\end{aligned}$$

with probability at least $1 - O(d^{-10})$.

Upper bounding γ'_2 in (S.34). Analogous to the upper bounding of γ_2 , we have with probability at least $1 - O(d^{-10})$

$$\begin{aligned}
\gamma'_2 &\leq \max_{j \in [J]} \|\mathbf{E}_{:,j}\| \left\| \mathbf{U}\mathbf{U}^\top - \mathbf{U}^{(j)}\mathbf{U}^{(j)\top} \right\| \\
&\stackrel{(viii)}{\lesssim} \sigma\sqrt{N} \left\| \mathbf{U}\mathbf{U}^\top - \mathbf{U}^{(j)}\mathbf{U}^{(j)\top} \right\| \\
&\stackrel{(ix)}{\lesssim} \frac{\sigma\sqrt{N}}{\sigma_K^*} \left(\sigma\sqrt{MK \log d} + \sqrt{M}B \log d \|\mathbf{U}^*\|_{2,\infty} + \sqrt{M}\gamma'_1 + \sigma M\sqrt{N} \|\mathbf{V}^*\|_{2,\infty} \right) \\
&\quad + \frac{\sigma^2 MN}{\sigma_K^*} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}, \tag{S.43}
\end{aligned}$$

where (viii) uses (S.10) in Lemma S.5 and (ix) is by (S.40).

Combine (S.42) and (S.43) and use the fact that the coefficient $\sigma\sqrt{MN}/\sigma_K^*$ of γ'_1 in (S.43) is negligible, then with probability at least $1 - O(d^{-10})$

$$\begin{aligned}
\beta'_2 &\lesssim \frac{\sigma\sqrt{K \log d} \left(\sigma\sqrt{MN + J} + \sqrt{M}B \log d \right)}{\sigma_K^*} + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} \\
&\quad + \frac{\sqrt{M}B \log d \left(B \log d + \sigma\sqrt{N} \right)}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma M\sqrt{N} \left(B \log d + \sigma\sqrt{N} \right)}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} \\
&\quad + \frac{\sigma M\sqrt{N} \left(B \log d + \sigma\sqrt{N} \right)}{\sigma_K^*} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} \\
&\lesssim \frac{\sigma^2 \sqrt{K \log d} (MN + J)}{\sigma_K^*} + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} \\
&\quad + \frac{\sigma^2 MN}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma^2 MN}{\sigma_K^*} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}, \tag{S.44}
\end{aligned}$$

where the last inequality follow by $\sigma\sqrt{MN + J} \gtrsim \sqrt{M}B \log d$ and $\sigma\sqrt{N} \gtrsim B \log d$, from (S.2) in Assumption S.7.

By plugging (S.33) and (S.44) back into (S.32), we finally get with probability at least $1 - O(d^{-10})$ that

$$\begin{aligned}
&\left\| \mathbf{R}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*) \right\|_{2,\infty} \\
&\lesssim \frac{\sigma^2 \sqrt{K \log d} (MN + J)}{\sigma_K^*} + B \log d \left\| \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} \\
&\quad + \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma^2 MN}{\sigma_K^*} \left\| \mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty}.
\end{aligned}$$

□

Proposition 2. *The following equations hold for $(\mathbf{U}^*, \mathbf{\Lambda}^*, \mathbf{V}^*)$ and $(\mathbf{U}, \mathbf{\Lambda}, \mathbf{V})$*

$$\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* = -\mathbf{U}\mathbf{U}^\top \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1} + \mathbf{R}(\mathbf{V}\mathbf{V}^\top \mathbf{V}^* \mathbf{\Lambda}^{*-1} - \mathbf{V}^* \mathbf{\Lambda}^{*-1}) + \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1} \quad (\text{S.45})$$

$$\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* = -\mathbf{V}\mathbf{V}^\top \mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-1} + \mathbf{R}^\top (\mathbf{U}\mathbf{U}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-1} - \mathbf{U}^* \mathbf{\Lambda}^{*-1}) + \mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-1} \quad (\text{S.46})$$

Proof of Proposition 2. First notice that

$$\begin{aligned} \mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* &= \mathbf{U}\mathbf{U}^\top \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-1} - \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-1} \\ &= \mathbf{U}\mathbf{U}^\top (\mathbf{R} - \mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-1} - (\mathbf{R} - \mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-1} \\ &= -\mathbf{U}\mathbf{U}^\top \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1} + \mathbf{U}\mathbf{U}^\top \mathbf{R}\mathbf{V}^* \mathbf{\Lambda}^{*-1} - \mathbf{R}\mathbf{V}^* \mathbf{\Lambda}^{*-1} + \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1} \\ &= -\mathbf{U}\mathbf{U}^\top \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1} + \mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^* \mathbf{\Lambda}^{*-1} - \mathbf{R}\mathbf{V}^* \mathbf{\Lambda}^{*-1} + \mathbf{E}\mathbf{V}^* \mathbf{\Lambda}^{*-1}, \end{aligned}$$

where the last equality holds because $\mathbf{U}\mathbf{U}^\top \mathbf{R} = \mathbf{U}\mathbf{U}^\top (\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top + \mathbf{U}^\perp \mathbf{\Lambda}^\perp \mathbf{V}^{\perp\top}) = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top = (\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top + \mathbf{U}^\perp \mathbf{\Lambda}^\perp \mathbf{V}^{\perp\top}) \mathbf{V}\mathbf{V}^\top = \mathbf{R}\mathbf{V}\mathbf{V}^\top$. Eq. (S.46) holds following a similar reasoning. □

S.1.3 Proof of Theorem 1

We will prove Theorem 1 with Proposition 2 and Lemma S.6.

Part I: Bounding $\|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}$ and $\|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty}$. With inequality (S.6)

in Lemma S.4, we know with probability at least $1 - O(d^{-10})$ that,

$$\|\mathbf{E}\mathbf{V}^*\|_{2,\infty} \lesssim \tilde{\sigma} \sqrt{K \log d} + MB \log d \|\mathbf{V}^*\|_{2,\infty} \lesssim \tilde{\sigma} \sqrt{K \log d}, \quad (\text{S.47})$$

where the last inequality follows by (S.2) in Assumption (S.2). Using the triangle inequality on (S.45) in Proposition 2 gives that

$$\begin{aligned} &\|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\ &\lesssim \|\mathbf{U}\|_{2,\infty} \|\mathbf{E}\mathbf{V}^*\| \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{R}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{E}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| \end{aligned}$$

$$\begin{aligned}
& \stackrel{(i)}{\lesssim} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^*\|_{2,\infty} \|\mathbf{E}\mathbf{V}^*\| \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{R}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{E}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| \\
& \lesssim \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \|\mathbf{E}\mathbf{V}^*\| \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{U}^*\|_{2,\infty} \|\mathbf{E}\mathbf{V}^*\| \|\mathbf{\Lambda}^{*-1}\| \\
& \quad + \|\mathbf{R}\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{R}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{E}\mathbf{V}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| \\
& \stackrel{(ii)}{\lesssim} \underbrace{\frac{\tilde{\sigma}\sqrt{N}}{\sigma_K^*}}_{A_1} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sqrt{N}}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sigma\sqrt{K\log d(MN+J)}}{\sigma_K^{*2}} \\
& \quad + \frac{MB\log d}{\sigma_K^*} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \underbrace{\frac{\sigma^2 J}{\sigma_K^{*2}}}_{A_2} \|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\
& \quad + \frac{\kappa^* \sigma^2 (MN+J)}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma MB\log d\sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sqrt{K\log d}}{\sigma_K^*} \\
& \stackrel{(iii)}{\lesssim} \frac{\tilde{\sigma}\sqrt{N}}{\sigma_K^*} \|\mathbf{U}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sigma\sqrt{K\log d(MN+J)}}{\sigma_K^{*2}} + \frac{MB\log d}{\sigma_K^*} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& \quad + \frac{\kappa^* \sigma^2 (MN+J)}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma MB\log d\sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sqrt{K\log d}}{\sigma_K^*} \\
& \stackrel{(iv)}{\lesssim} \frac{\tilde{\sigma}\sqrt{N}\sigma_K^* + \kappa^* \sigma^2 J}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{MB\log d}{\sigma_K^*} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& \quad + \frac{\sigma MB\log d\sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma}\sqrt{K\log d}}{\sigma_K^*}, \tag{S.48}
\end{aligned}$$

holds with probability at least $1 - O(d^{-10})$. Here (i) uses Lemma 2.5 in [Chen et al. \(2021\)](#) combined with (S.16), (ii) uses (S.47), (S.11) in Lemma S.5, and (S.14) in Lemma S.6, (iii) is because $A_1 = o(1)$, $A_2 = o(1)$ by (S.1) in Assumption S.7, and (iv) holds by (S.1) in Assumption S.7.

We will use similar arguments on the right singular subspace. By (S.7) in Lemma S.4,

$$\|\mathbf{E}^\top \mathbf{U}^*\|_{2,\infty} \lesssim \sigma\sqrt{K\log d} + B\log d \|\mathbf{U}^*\|_{2,\infty} \lesssim \sigma\sqrt{K\log d} \tag{S.49}$$

holds with probability at least $1 - O(d^{-10})$, where the last inequality follows by (S.2). Therefore, (S.46) in Proposition 2 leads to

$$\begin{aligned}
& \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& \lesssim \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \|\mathbf{E}^\top \mathbf{U}^*\| \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{V}^*\|_{2,\infty} \|\mathbf{E}^\top \mathbf{U}^*\| \|\mathbf{\Lambda}^{*-1}\|
\end{aligned}$$

$$\begin{aligned}
& + \|\mathbf{R}^\top \mathbf{U}^* - \mathbf{R}^\top \mathbf{U} \mathbf{U}^\top \mathbf{U}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| + \|\mathbf{E}^\top \mathbf{U}^*\|_{2,\infty} \|\mathbf{\Lambda}^{*-1}\| \\
& \stackrel{(iv)}{\lesssim} \underbrace{\frac{\sigma\sqrt{J}}{\sigma_K^*}}_{B_1} \|\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + \frac{\sigma\sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma^2 \sqrt{K \log d (MN + J)}}{\sigma_K^{*2}} \\
& + \frac{B \log d}{\sigma_K^*} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + \underbrace{\frac{\sigma^2 MN}{\sigma_K^{*2}}}_{B_2} \|\mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& + \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*}, \\
& \stackrel{(v)}{\lesssim} \frac{\sigma\sqrt{J}}{\sigma_K^*} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma^2 \sqrt{K \log d (MN + J)}}{\sigma_K^{*2}} + \frac{B \log d}{\sigma_K^*} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\
& + \frac{\kappa^* \sigma^2 (MN + J)}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \\
& \stackrel{(vi)}{\lesssim} \frac{\sigma\sqrt{J} \sigma_K^* + \kappa^* \sigma^2 MN}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{B \log d}{\sigma_K^*} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\
& + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*}, \tag{S.50}
\end{aligned}$$

which holds with probability at least $1 - O(d^{-10})$. Here (iv) uses (S.49), (S.12) in Lemma S.5, and (S.15) in Lemma S.6; (v) holds since $B_1 = o(1)$, $B_2 = o(1)$ by (S.1) in Assumption S.7; and (vi) holds by (S.1) in Assumption S.7.

Now substitute (S.50) into (S.48), and with probability at least $1 - O(d^{-10})$ we have

$$\begin{aligned}
& \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\
& \lesssim \frac{\tilde{\sigma} \sqrt{N} \sigma_K^* + \kappa^* \sigma^2 J}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma M B \log d \sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} \\
& + \frac{M B \log d}{\sigma_K^*} \left(\frac{\sigma \sqrt{J} \sigma_K^* + \kappa^* \sigma^2 (MN + J)}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{B \log d}{\sigma_K^*} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \right. \\
& \left. + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \right) \\
& \stackrel{(vii)}{\lesssim} \frac{\tilde{\sigma} \sqrt{N} \sigma_K^* + \kappa^* \sigma^2 J}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma M B \log d \sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} \\
& + \frac{M B \log d (\sigma \sqrt{J} + \sigma \sqrt{MN + J})}{\sigma_K^{*2}} \|\mathbf{V}\|_{2,\infty} + \underbrace{\frac{M B^2 \log^2 d}{\sigma_K^{*2}}}_{C_1} \|\mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sigma MB^2 \log^2 d \sqrt{MN}}{\sigma_K^{*3}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma MB \sqrt{K \log^3 d}}{\sigma_K^{*2}} \\
& \stackrel{(viii)}{\lesssim} \frac{\tilde{\sigma} \sqrt{N} \sigma_K^* + \kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma MB \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} =: \xi'_1 \\
& \leq \frac{\tilde{\sigma} \sqrt{\mu_1 K \log d}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma MB \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} =: \xi_1
\end{aligned}$$

where (vii) uses (S.1) and (viii) holds because $C_1 = o(1)$ by (S.1), (S.2) in Assumption S.7.

On the other hand, substituting (S.48) into (S.50) gives that with probability at least $1 - O(d^{-10})$

$$\begin{aligned}
& \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& \lesssim \frac{\sigma \sqrt{J} \sigma_K^* + \kappa^* \sigma^2 MN}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \\
& + \frac{B \log d}{\sigma_K^*} \left(\frac{\tilde{\sigma} \sqrt{N} \sigma_K^* + \kappa^* \sigma^2 (MN+J)}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{MB \log d}{\sigma_K^*} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \right. \\
& \left. + \frac{\sigma MB \log d \sqrt{J}}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} \right) \\
& \stackrel{(ix)}{\lesssim} \frac{\sigma \sqrt{J} \sigma_K^* + \kappa^* \sigma^2 MN}{\sigma_K^{*2}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\sigma B \log d \sqrt{MN}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \\
& + \frac{\tilde{\sigma} B \log d \sqrt{N} + \sigma B \log d \sqrt{MN+J}}{\sigma_K^{*2}} \|\mathbf{U}^*\|_{2,\infty} + \underbrace{\frac{MB^2 \log^2 d}{\sigma_K^{*2}}}_{C_2} \|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\
& + \frac{\sigma MB^2 \log^2 d \sqrt{J}}{\sigma_K^{*3}} \|\mathbf{V}^*\|_{2,\infty} + \frac{\tilde{\sigma} B \log d \sqrt{K \log d}}{\sigma_K^{*2}} \\
& \stackrel{(x)}{\lesssim} \frac{\sigma \sqrt{J} \sigma_K^* + \kappa^* \sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma B \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} =: \xi'_2 \\
& \leq \frac{\sigma \sqrt{\mu_2 K \log d}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma B \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} =: \xi_2,
\end{aligned}$$

where (ix) uses (S.1) and (x) holds since $C_2 = o(1)$ by (S.1), (S.2) in Assumption S.7.

Part II: Bounding $\|\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top - \mathbf{U}^*\mathbf{\Lambda}^*\mathbf{V}^{*\top}\|_\infty$. For the last part, write $\mathbf{R}_U := \mathbf{U}_L \mathbf{V}_L^\top$ with $(\mathbf{U}_L, \mathbf{\Lambda}_L, \mathbf{V}_L) = \text{svd}(\mathbf{U}^\top \mathbf{U}^*; K)$ and $\mathbf{R}_V := \mathbf{U}_R \mathbf{V}_R^\top$ with $(\mathbf{U}_R, \mathbf{\Lambda}_R, \mathbf{V}_R) = \text{svd}(\mathbf{V}^\top \mathbf{V}^*; K)$.

Notice that $\mathbf{R}_U^\top \mathbf{R}_U = \mathbf{R}_U \mathbf{R}_U^\top = \mathbf{R}_V^\top \mathbf{R}_V = \mathbf{R}_V \mathbf{R}_V^\top = \mathbf{I}_K$. Then we have

$$\begin{aligned} & \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \\ &= (\mathbf{U} \mathbf{R}_U - \mathbf{U}^*) \mathbf{R}_U^\top \mathbf{\Lambda} \mathbf{V}^\top + \mathbf{U}^* \mathbf{R}_U^\top \mathbf{\Lambda} \mathbf{R}_V (\mathbf{R}_V^\top \mathbf{V}^\top - \mathbf{V}^{*\top}) + \mathbf{U}^* (\mathbf{R}_U \mathbf{\Lambda} \mathbf{R}_V - \mathbf{\Lambda}^*) \mathbf{V}^{*\top}. \end{aligned}$$

Using the triangle inequality and Lemma S.2 gives that

$$\begin{aligned} & \left\| \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \right\|_\infty \\ & \leq \left\| \mathbf{U} \mathbf{R}_U - \mathbf{U}^* \right\|_{2,\infty} \left\| \mathbf{V} \mathbf{\Lambda} \mathbf{R}_U \right\|_{2,\infty} + \left\| \mathbf{U}^* \right\|_{2,\infty} \left\| \mathbf{\Lambda} \right\| \left\| \mathbf{V} \mathbf{R}_V - \mathbf{V}^* \right\|_{2,\infty} \\ & \quad + \left\| \mathbf{U}^* \right\|_{2,\infty} \left\| \mathbf{R}_U^\top \mathbf{\Lambda} \mathbf{R}_V - \mathbf{\Lambda}^* \right\| \left\| \mathbf{V}^* \right\|_{2,\infty} \\ & \leq \left(\left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} + \left\| \mathbf{U} \right\|_{2,\infty} \left\| \mathbf{R}_U - \mathbf{U}^\top \mathbf{U}^* \right\| \right) \left\| \mathbf{V} \right\|_{2,\infty} \left\| \mathbf{\Lambda} \right\| \\ & \quad + \left\| \mathbf{U}^* \right\|_{2,\infty} \left\| \mathbf{\Lambda} \right\| \left(\left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + \left\| \mathbf{V} \right\|_{2,\infty} \left\| \mathbf{V}^\top \mathbf{V}^* - \mathbf{R}_V \right\| \right) \\ & \quad + \left\| \mathbf{U}^* \right\|_{2,\infty} \left\| \mathbf{R}_U^\top \mathbf{\Lambda} \mathbf{R}_V - \mathbf{\Lambda}^* \right\| \left\| \mathbf{V}^* \right\|_{2,\infty}. \end{aligned} \tag{S.51}$$

We then further break the norms of the empirical terms $\mathbf{V}, \mathbf{\Lambda}$ in (S.51) down into

$$\begin{aligned} \left\| \mathbf{V} \right\|_{2,\infty} & \leq \left\| \mathbf{V} (\mathbf{R}_V - \mathbf{V}^\top \mathbf{V}^*) \right\|_{2,\infty} + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + \left\| \mathbf{V}^* \right\|_{2,\infty} \\ & \leq \left\| \mathbf{V} \right\|_{2,\infty} \left\| \mathbf{R}_V - \mathbf{V}^\top \mathbf{V}^* \right\| + \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + \left\| \mathbf{V}^* \right\|_{2,\infty}, \end{aligned} \tag{S.52}$$

Additionally, we note that $\left\| \mathbf{R}_U - \mathbf{U}^\top \mathbf{U}^* \right\| = \left\| \mathbf{I} - \mathbf{\Lambda}_L \right\| \leq \left\| \mathbf{I} - \mathbf{\Lambda}_L^2 \right\| = \left\| \mathbf{U}_\perp^\top \mathbf{U}^* \right\|^2$ and similarly $\left\| \mathbf{R}_V - \mathbf{V}^\top \mathbf{V}^* \right\| \leq \left\| \mathbf{V}_\perp^\top \mathbf{V}^* \right\|^2$. Then Lemma 2.5 in Chen et al. (2021) combined with (S.16) yields that

$$\max \left\{ \left\| \mathbf{R}_U - \mathbf{U}^\top \mathbf{U}^* \right\|, \left\| \mathbf{R}_V - \mathbf{V}^\top \mathbf{V}^* \right\| \right\} \lesssim \frac{\tilde{\sigma}^2 N + \sigma^2 J}{\sigma_K^{*2}} \tag{S.53}$$

with probability at least $1 - O(d^{-10})$. Rearranging the term $\left\| \mathbf{V} \right\|_{2,\infty}$ in (S.52) with (S.1) in Assumption S.7, then with probability at least $1 - O(d^{-10})$ we have

$$\left\| \mathbf{V} \right\|_{2,\infty} \lesssim \left\| \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \right\|_{2,\infty} + \left\| \mathbf{V}^* \right\|_{2,\infty}, \tag{S.54}$$

and similarly,

$$\left\| \mathbf{U} \right\|_{2,\infty} \lesssim \left\| \mathbf{U} \mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^* \right\|_{2,\infty} + \left\| \mathbf{U}^* \right\|_{2,\infty}. \tag{S.55}$$

By substituting (S.54), and (S.55) into (S.51) and with $\|\mathbf{\Lambda}\| \leq \|\mathbf{\Lambda}^*\| + \|\mathbf{\Lambda} - \mathbf{\Lambda}^*\|$, we have with probability at least $1 - O(d^{-10})$ that

$$\begin{aligned}
& \|\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top - \mathbf{U}^*\mathbf{\Lambda}^*\mathbf{V}^*\|_{2,\infty} \\
& \lesssim \left(\|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} + (\|\mathbf{U}^*\|_{2,\infty} + \|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}) \|\mathbf{R}_U - \mathbf{U}^\top\mathbf{U}^*\| \right) \cdot \\
& \quad \left(\|\mathbf{V}^*\|_{2,\infty} + \|\mathbf{V}^* - \mathbf{V}\mathbf{V}^\top\mathbf{V}^*\|_{2,\infty} \right) (\|\mathbf{\Lambda}^*\| + \|\mathbf{\Lambda} - \mathbf{\Lambda}^*\|) \\
& \quad + \left(\|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} + (\|\mathbf{V}^*\|_{2,\infty} + \|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty}) \|\mathbf{V}^\top\mathbf{V}^* - \mathbf{R}_V\| \right) \cdot \\
& \quad \|\mathbf{U}^*\|_{2,\infty} (\|\mathbf{\Lambda}^*\| + \|\mathbf{\Lambda} - \mathbf{\Lambda}^*\|) + \|\mathbf{U}^*\|_{2,\infty} \|\mathbf{R}_U^\top\mathbf{\Lambda}\mathbf{R}_V - \mathbf{\Lambda}^*\| \|\mathbf{V}^*\|_{2,\infty}. \tag{S.56}
\end{aligned}$$

By Weyl's inequality and (S.8), $|\sigma_K - \sigma_K^*| \leq \|\mathbf{\Lambda} - \mathbf{\Lambda}^*\| \lesssim \tilde{\sigma}\sqrt{N} + \sigma\sqrt{J}$. In addition, one has

$$\begin{aligned}
& \|\mathbf{R}_U^\top\mathbf{\Lambda}\mathbf{R}_V - \mathbf{\Lambda}^*\| \\
& \leq \|(\mathbf{R}_U^\top - \mathbf{U}^{*\top}\mathbf{U})\mathbf{\Lambda}\mathbf{R}_V\| + \|\mathbf{U}^{*\top}\mathbf{U}\mathbf{\Lambda}(\mathbf{R}_V - \mathbf{V}^\top\mathbf{V}^*)\| + \|\mathbf{U}^{*\top}\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top\mathbf{V}^* - \mathbf{\Lambda}^*\| \\
& \leq \|(\mathbf{R}_U^\top - \mathbf{U}^{*\top}\mathbf{U})\mathbf{\Lambda}\mathbf{R}_V\| + \|\mathbf{U}^{*\top}\mathbf{U}\mathbf{\Lambda}(\mathbf{R}_V - \mathbf{V}^\top\mathbf{V}^*)\| \\
& \quad + \|\mathbf{U}^{*\top}\mathbf{R}\mathbf{V}^* - \mathbf{U}^{*\top}\mathbf{U}^*\mathbf{\Lambda}^*\mathbf{V}^{*\top}\mathbf{V}^*\| + \|\mathbf{U}^{*\top}\mathbf{U}^\perp\mathbf{\Lambda}^\perp\mathbf{U}^{\perp\top}\mathbf{V}^*\| \\
& \leq \|\mathbf{\Lambda}^*\| (\|\mathbf{R}_U^\top - \mathbf{U}^{*\top}\mathbf{U}\| + \|\mathbf{R}_V^\top - \mathbf{V}^{*\top}\mathbf{V}\|) + \|\mathbf{U}^{*\top}\mathbf{E}\mathbf{V}^*\| \\
& \quad + \|\mathbf{U}^{*\top}\mathbf{U}^\perp\| \|\mathbf{E}\| \|\mathbf{V}^{*\top}\mathbf{U}^\perp\|, \tag{S.57}
\end{aligned}$$

where we use $\|\mathbf{\Lambda}\| \leq \|\mathbf{\Lambda}^*\| + \|\mathbf{\Lambda} - \mathbf{\Lambda}^*\| \lesssim \|\mathbf{\Lambda}^*\|$ and $\|\mathbf{\Lambda}^\perp\| \leq \|\mathbf{E}\|$.

We separately look into the terms in (S.57) as follows:

1. Regarding the first term, (S.53) immediately implies that

$$\|\mathbf{\Lambda}^*\| (\|\mathbf{R}_U^\top - \mathbf{U}^{*\top}\mathbf{U}\| + \|\mathbf{R}_V^\top - \mathbf{V}^{*\top}\mathbf{V}\|) \lesssim \frac{\kappa^*(\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*}$$

holds with probability at least $1 - O(d^{-10})$.

2. For the second term, we first view $\mathbf{U}^{*\top}\mathbf{E}\mathbf{V}^*$ as the sum of independent random matrices

$\mathbf{U}^{*\top}\mathcal{P}_{i,S_l}(\mathbf{E})\mathbf{V}^*$ over $i \in [N], l \in [L]$ and calculate the following quantities:

$$\max \left\{ \left\| \mathbb{E}[\mathbf{U}^{*\top}\mathbf{E}\mathbf{V}^*\mathbf{V}^{*\top}\mathbf{E}^\top\mathbf{U}^*] \right\|, \left\| \mathbb{E}[\mathbf{V}^{*\top}\mathbf{E}^\top\mathbf{U}^*\mathbf{U}^{*\top}\mathbf{E}\mathbf{V}^*] \right\| \right\}$$

$$\begin{aligned}
&= \max \left\{ \left\| \mathbf{U}^{*\top} \mathbb{E}[\mathbf{E} \mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top] \mathbf{U}^* \right\|, \left\| \mathbf{V}^{*\top} \mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}] \mathbf{V}^* \right\| \right\} \\
&\stackrel{(i)}{\leq} \max \left\{ \sum_{i \in [N]} \left\| \mathbf{U}_{i,:}^* \right\|_2^2 (\mathbb{E}[\mathbf{E} \mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top])_{i,i}, \sum_{l \in [L]} \left\| \mathbf{V}_{S_l,:}^* \right\|_2^2 (\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}])_{S_l, S_l} \right\} \\
&\leq \max \left\{ \sum_{i \in [N]} \left\| \mathbf{U}_{i,:}^* \right\|_2^2 (\mathbb{E}[\mathbf{E} \mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top])_{i,i}, \right. \\
&\quad \left. \sum_{l \in [L]} \left(\sum_{j \in S_l} \left\| \mathbf{V}_{j,:}^* \right\|_2^2 \right) \left\| (\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}])_{S_l, S_l} \right\| \right\} \\
&\stackrel{(ii)}{\leq} MK\sigma^2,
\end{aligned}$$

where (i) holds since $\mathbb{E}[\mathbf{E} \mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top]$ is a diagonal matrix and $\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}]$ is a block-diagonal matrix. And for (ii) we use Lemma S.3 and

$$(\mathbb{E}[\mathbf{E} \mathbf{V}^* \mathbf{V}^{*\top} \mathbf{E}^\top])_{i,i} = \text{tr}(\mathbf{V}^* \mathbf{V}^{*\top} \text{Cov}(\mathbf{E}_{i,:})) \leq \|\mathbf{V}^*\|_F^2 \|\text{Cov}(\mathbf{E}_{i,:})\| = K\tilde{\sigma}^2 \leq KM\sigma^2,$$

$$\begin{aligned}
\left\| (\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}])_{S_l, S_l} \right\| &\leq \text{tr}((\mathbb{E}[\mathbf{E}^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{E}])_{S_l, S_l}) = \text{tr}(\mathbf{U}^* \mathbf{U}^{*\top} \mathbb{E}[\mathbf{E}_{:,S_l} \mathbf{E}_{:,S_l}^\top]) \\
&\leq \|\mathbf{U}^*\|_F^2 \|\mathbb{E}[\mathbf{E}_{:,S_l} \mathbf{E}_{:,S_l}^\top]\| \leq K \text{tr}(\mathbb{E}[\mathbf{E}_{:,S_l} \mathbf{E}_{:,S_l}^\top]) \leq KM\sigma^2.
\end{aligned}$$

In addition, $\|\mathbf{U}^{*\top} \mathcal{P}_{i,S_l}(\mathbf{E}) \mathbf{V}^*\| \leq \sqrt{M}B \|\mathbf{U}^*\|_{2,\infty} \|\mathbf{V}^*\|_{2,\infty} = \sqrt{M}BK \sqrt{\frac{\mu_1 \mu_2}{NJ}}$. With these inequalities, we apply the matrix Bernstein inequality which yields

$$\|\mathbf{U}^{*\top} \mathbf{E} \mathbf{V}^*\| \lesssim \sigma \sqrt{KM \log d} + \sqrt{M}BK \sqrt{\frac{\mu_1 \mu_2}{NJ}} \log d \lesssim \sigma \sqrt{KM \log d}$$

holds with probability at least $1 - O(d^{-10})$, where the last inequality holds since $\frac{B}{\sigma} \sqrt{\frac{K\mu_1 \mu_2 \log d}{NJ}} \lesssim \frac{\sigma}{\sqrt{KM^2 B (\log d)^{3.5}}} \ll 1$ by (S.2).

3. For the third term, with probability at least $1 - O(d^{-10})$ we have

$$\|\mathbf{U}^{*\top} \mathbf{U}^\perp\| \|\mathbf{E}\| \|\mathbf{V}^{*\top} \mathbf{U}^\perp\| \lesssim \frac{(\tilde{\sigma} \sqrt{N} + \sigma \sqrt{J})^3}{\sigma_K^{*2}} \lesssim \frac{\tilde{\sigma}^2 N + \sigma^2 J}{\sigma_K^*},$$

since $\max\{\|\mathbf{U}^{*\top} \mathbf{U}^\perp\|, \|\mathbf{V}^{*\top} \mathbf{U}^\perp\|\} \leq \frac{\tilde{\sigma} \sqrt{N} + \sigma \sqrt{J}}{\sigma_K^*}$ with probability $1 - O(d^{-10})$.

Putting these pieces together,

$$\|\mathbf{R}_U^\top \mathbf{A} \mathbf{R}_V - \mathbf{\Lambda}^*\| \lesssim \frac{\kappa^*(\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*} + \sigma \sqrt{KM \log d}$$

holds with probability $1 - O(d^{-10})$. Therefore, (S.56) becomes

$$\begin{aligned} & \left\| \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \right\|_\infty \\ & \lesssim \left(\xi'_1 + \left(\sqrt{\frac{\mu_1 K}{N}} + \xi'_1 \right) \frac{\tilde{\sigma}^2 N + \sigma^2 J}{\sigma_K^{*2}} \right) \left(\sqrt{\frac{\mu_2 K}{J}} + \xi'_2 \right) (\sigma_1^* + \tilde{\sigma} \sqrt{N} + \sigma \sqrt{J}) \\ & \quad + \left(\xi'_2 + \left(\sqrt{\frac{\mu_2 K}{J}} + \xi'_2 \right) \frac{\tilde{\sigma}^2 N + \sigma^2 J}{\sigma_K^{*2}} \right) \sqrt{\frac{\mu_1 K}{N}} (\sigma_1^* + \tilde{\sigma} \sqrt{N} + \sigma \sqrt{J}) \\ & \quad + \left(\frac{\kappa^*(\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*} + \sigma \sqrt{KM \log d} \right) \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \\ & \stackrel{(i)}{\lesssim} \xi'_1 \left(\sqrt{\frac{\mu_2 K}{J}} + \xi'_2 \right) \sigma_1^* + \xi'_2 \sqrt{\frac{\mu_1 K}{N}} \sigma_1^* + \frac{\kappa^*(\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \\ & \quad + \sqrt{KM} \sigma \sqrt{\log d} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \\ & \stackrel{(ii)}{\lesssim} \xi'_1 \xi'_2 \sigma_1^* + \sqrt{\frac{\mu_2 K}{J}} \sigma_1^* \xi'_1 + \sqrt{\frac{\mu_1 K}{N}} \sigma_1^* \xi'_2 + \frac{\kappa^*(\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \\ & \quad + \sqrt{KM} \sigma \sqrt{\log d} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}}, \end{aligned} \tag{S.58}$$

which holds with probability at least $1 - O(d^{-10})$. Here (i) follows from the fact that $\tilde{\sigma} \sqrt{N} + \sigma \sqrt{J} \lesssim \sigma_K^* \leq \sigma_1^*$ by (S.1) and the definition of ξ'_1 , and (ii) holds by (S.1) and the definitions of ξ'_1, ξ'_2 .

Plugging in the definitions of ξ'_1, ξ'_2 , one has

$$\begin{aligned} & \xi'_1 \xi'_2 \sigma_1^* \\ & \stackrel{(iii)}{\lesssim} \sigma_1^* \left(\frac{\sigma \sqrt{MN+J}}{\sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{MB \log d}{\kappa^* \sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} \right) \\ & \quad \left(\frac{\sigma \sqrt{MN+J}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{B \log d}{\kappa^* \sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \right) \\ & \stackrel{(iv)}{\lesssim} \frac{\kappa^* \sigma^2 (MN+J)}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} + \frac{\mu_1 K}{N} \cdot \frac{\sigma B \log d \sqrt{MN+J}}{\sigma_K^*} + \frac{\mu_2 K}{J} \cdot \frac{\sigma MB \log d \sqrt{MN+J}}{\sigma_K^*} \\ & \quad + \frac{\kappa^* \sigma \sqrt{K \log d (MN+J)}}{\sigma_K^*} \left(\sigma \sqrt{\frac{\mu_1 K}{N}} + \tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} \right) + \frac{\kappa^* \tilde{\sigma} \sigma K \log d}{\sigma_K^*} \end{aligned}$$

$$\begin{aligned}
& \stackrel{(v)}{\lesssim} \frac{\kappa^* K \sigma^2 \sqrt{M \mu_1 \mu_2} \log d}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}} \\
& + \frac{K \sigma M B \log d}{\sigma_K^*} \left(\frac{\mu_1}{\sqrt{MN}} \sqrt{1 + \frac{r}{M}} + \frac{\mu_2}{\sqrt{J}} \sqrt{1 + \frac{M}{r}} \right). \tag{S.59}
\end{aligned}$$

Here to derive (iii), we employ the following simplification to ξ'_1 and ξ'_2 :

$$\begin{aligned}
\xi'_1 &= \underbrace{\frac{\tilde{\sigma} \sqrt{N} \sigma_K^* + \kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}}}_{\lesssim \frac{\sigma \sqrt{MN+J}}{\sigma_K^*}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} + \underbrace{\frac{\sigma M B \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}}}_{\lesssim \frac{MB \log d}{\kappa^* \sigma_K^*}}, \\
\xi'_2 &= \underbrace{\frac{\sigma \sqrt{J} \sigma_K^* + \kappa^* \sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}}}_{\lesssim \frac{\sigma \sqrt{MN+J}}{\sigma_K^*}} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} + \underbrace{\frac{\sigma B \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}}}_{\lesssim \frac{B \log d}{\kappa^* \sigma_K^*}},
\end{aligned}$$

and (iv) uses (S.2). To obtain (v), we invoke the following facts that

$$\begin{aligned}
\frac{\kappa^* \sigma^2 (MN+J)}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 K^2}{N J}} &= \frac{\kappa^* \sigma^2 M K}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 N}{J}} + \frac{\kappa^* \sigma^2 K}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 J}{N}} \\
&\lesssim \frac{\kappa^* \sigma^2 \sqrt{M} K}{\sigma_K^*} \sqrt{\left(\frac{r}{M} + \frac{M}{r}\right) \mu_1 \mu_2}, \\
\frac{\mu_1 K}{N} \cdot \frac{\sigma B \log d \sqrt{MN+J}}{\sigma_K^*} &= \frac{\mu_1 K \sigma M B \log d}{\sigma_K^* \sqrt{MN}} \sqrt{1 + \frac{r}{M}}, \\
\frac{\mu_2 K}{J} \cdot \frac{\sigma M B \log d \sqrt{MN+J}}{\sigma_K^*} &= \frac{\mu_2 K \sigma M B \log d}{\sigma_K^* \sqrt{J}} \sqrt{\frac{M}{r} + 1},
\end{aligned}$$

and

$$\begin{aligned}
& \frac{\kappa^* \sigma \sqrt{K \log d (MN+J)}}{\sigma_K^*} \left(\sigma \sqrt{\frac{\mu_1 K}{N}} + \tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} \right) \\
& \leq \frac{\kappa^* K \sigma^2 \sqrt{M \log d} \sqrt{\mu_1 \mu_2}}{\sigma_K^*} \left(\sqrt{1 + \frac{r}{M}} + \sqrt{1 + \frac{M}{r}} \right) \\
& \lesssim \frac{\kappa^* K \sigma^2 \sqrt{M \log d} \sqrt{\mu_1 \mu_2}}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}}.
\end{aligned}$$

In addition,

$$\begin{aligned}
& \sqrt{\frac{\mu_2 K}{J}} \sigma_1^* \xi'_1 + \sqrt{\frac{\mu_1 K}{N}} \sigma_1^* \xi'_2 \\
& \lesssim \sqrt{\frac{\mu_2 K}{J}} \sigma_1^* \left(\frac{\sigma \sqrt{MN+J}}{\sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{MB \log d}{\kappa^* \sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*} \right)
\end{aligned}$$

$$\begin{aligned}
& + \sqrt{\frac{\mu_1 K}{N}} \sigma_1^* \left(\frac{\sigma \sqrt{MN+J}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{B \log d}{\kappa^* \sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*} \right) \\
& \lesssim \kappa^* K \sigma \sqrt{MN+J} \sqrt{\frac{\mu_1 \mu_2}{NJ}} + \frac{\mu_2 K}{J} MB \log d + \frac{\mu_1 K}{N} B \log d \\
& \quad + \kappa^* \sqrt{K \log d} \left(\tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} + \sigma \sqrt{\frac{\mu_1 K}{N}} \right) \\
& = \kappa^* K \sigma \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + \frac{\mu_2 K}{J} MB \log d + \frac{\mu_1 K}{N} B \log d \\
& \quad + \kappa^* \sqrt{K \log d} \left(\tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} + \sigma \sqrt{\frac{\mu_1 K}{N}} \right), \tag{S.60}
\end{aligned}$$

Moreover,

$$\frac{\kappa^* (\tilde{\sigma}^2 N + \sigma^2 J)}{\sigma_K^*} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \leq \frac{\sigma^2 \kappa^* K \sqrt{M \mu_1 \mu_2}}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}}. \tag{S.61}$$

To the end, plugging (S.59), (S.60), and (S.61) into (S.58) yields that

$$\begin{aligned}
& \left\| \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \right\|_\infty \\
& \lesssim \frac{\kappa^* K \sigma^2 \sqrt{M \mu_1 \mu_2} \log d}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}} + \frac{K \sigma MB \log d}{\sigma_K^*} \left(\frac{\mu_1}{\sqrt{MN}} \sqrt{1 + \frac{r}{M}} + \frac{\mu_2}{\sqrt{J}} \sqrt{1 + \frac{M}{r}} \right) \\
& \quad + \kappa^* K \sigma \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + \frac{\mu_2 K}{J} MB \log d + \frac{\mu_1 K}{N} B \log d \\
& \quad + \kappa^* \sqrt{K \log d} \left(\tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} + \sigma \sqrt{\frac{\mu_1 K}{N}} \right) \\
& \quad + \frac{\sigma^2 \kappa^* K \sqrt{M \mu_1 \mu_2}}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}} + \sqrt{KM} \sigma \sqrt{\log d} \sqrt{\frac{\mu_1 \mu_2 K^2}{NJ}} \\
& \stackrel{(vi)}{\lesssim} \frac{\kappa^* K \sigma^2 \sqrt{M \mu_1 \mu_2} \log d}{\sigma_K^*} \sqrt{\frac{r}{M} + \frac{M}{r}} \\
& \quad + \kappa^* K \sigma \sqrt{\mu_1 \mu_2 \log d} \sqrt{\frac{M}{J} + \frac{1}{N}} + \frac{\mu_2 K}{J} MB \log d + \frac{\mu_1 K}{N} B \log d \\
& \stackrel{(vii)}{\lesssim} \kappa^* K \sigma \log d \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + \frac{\mu_2 K}{J} MB \log d + \frac{\mu_1 K}{N} B \log d =: \xi_3,
\end{aligned}$$

where (vi) follows from

$$\frac{\sigma}{\sigma_K^*} \sqrt{1 + \frac{r}{M}} \lesssim \frac{1}{\sqrt{MN}}, \quad \frac{\sigma}{\sigma_K^*} \sqrt{1 + \frac{M}{r}} \lesssim \frac{1}{\sqrt{J}}$$

with (S.1) in Assumption S.7, and

$$\begin{aligned} \kappa^* \sqrt{K \log d} \left(\tilde{\sigma} \sqrt{\frac{\mu_2 K}{J}} + \sigma \sqrt{\frac{\mu_1 K}{N}} \right) &\leq \kappa^* K \sigma \sqrt{\log d} \left(\sqrt{\frac{M \mu_2}{J}} + \sqrt{\frac{\mu_1}{N}} \right) \\ &\leq \kappa^* K \sigma \sqrt{\mu_1 \mu_2 \log d} \sqrt{\frac{M}{J} + \frac{1}{N}}. \end{aligned}$$

And (vii) also used (S.1). $\square$

Remark 5. To ensure the $\ell_{2,\infty}$ -consistency of the singular subspaces and the ℓ_∞ -consistency of the full matrix, we can impose further assumptions in addition to those in Theorem 1. For $\frac{\|\mathbf{U}\mathbf{U}^\top \mathbf{U}^* - \mathbf{U}^*\|_{2,\infty}}{\|\mathbf{U}^*\|_{2,\infty}} = o(1)$ to hold, an easy-to-verify condition is $\sigma \sqrt{MN \log d} + \kappa^* J = o(\sigma_K^*)$, $\frac{MB}{\sigma} \lesssim \log d$, and $\frac{\mu_2}{\mu_1} \lesssim \frac{J}{(\log d)^3}$. Similarly, a sufficient condition for $\frac{\|\mathbf{V}\mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^*\|_{2,\infty}}{\|\mathbf{V}^*\|_{2,\infty}} = o(1)$ to hold is $\sigma \sqrt{J \log d} + \kappa^* MN = o(\sigma_K^*)$, $\frac{B}{\sigma} \lesssim \log(N \vee J)$, and $\frac{\mu_1}{\mu_2} \lesssim \frac{N}{(\log d)^3}$.

S.1.4 Verifying Assumption S.5 for Sub-Exponential Noise

It is evident from Assumption S.5 that our theory accommodates a broad class of noise distributions. In what follows we shall spell out the corresponding B satisfying Assumption S.5 for sub-exponential distributions, which is common in statistical scenarios. Assume that every entry $E_{i,j}$ is sub-exponential with parameters (v, α) , i.e.,

$$\mathbb{E}[\exp(\lambda E_{i,j})] \leq \exp\left(\frac{v^2 \lambda^2}{2}\right), \text{ for } i \in [N], j \in [J], \text{ and all } |\lambda| \leq \frac{1}{\alpha}.$$

The following lemma allows us to treat the original sub-exponential $\mathbf{E}$ as an entrywise-bounded and centered random matrix $\mathbf{E}'$ with high probability.

Lemma S.7. *There exists a random matrix $\mathbf{E}' = (E'_{i,j}) \in \mathbb{R}^{N \times J}$ such that*

$$\mathbb{P}(\mathbf{E} = \mathbf{E}') \geq 1 - O(d^{-20}),$$

$|E'_{i,j}| \leq 80\alpha \log d$ and $\mathbb{E}[E'_{i,j}] = 0$. Furthermore, we have

$$\max_{i \in [N], l \in [L]} \|\text{Cov}(\mathbf{E}'_{i, S_l})\| \leq \tilde{\sigma}^2 + O(\alpha d^{-8} \sigma + \alpha^2 d^{-19}).$$

Proof of Lemma S.7. Let $C := 80\alpha \log d$, then for each entry $E_{i,j}$ one has

$$\max\{\mathbb{P}(E_{i,j} \geq C), \mathbb{P}(E_{i,j} \leq -C)\} \leq \exp(-40 \log d) = d^{-40}$$

and

$$\begin{aligned} & \mathbb{E}[|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\}] \\ &= \int_0^\infty \mathbb{P}(|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\} \geq x) dx \\ &= \int_0^C \mathbb{P}(|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\} \geq x) dx + \int_C^\infty \mathbb{P}(|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\} \geq x) dx \\ &\leq C \cdot \mathbb{P}(|E_{i,j}| \geq C) + \int_C^\infty \mathbb{P}(|E_{i,j}| \geq x) dx \\ &\leq 2C \exp(-\frac{C}{2\alpha}) + 2 \int_C^\infty \exp(-\frac{x}{2\alpha}) dx \\ &\leq 2(C + 2\alpha)d^{-40} = (160 \log d + 4)\alpha d^{-40} \end{aligned} \tag{S.62}$$

by (Wainwright, 2019, Proposition 2.9). Let $\widehat{E}_{i,j} := E_{i,j} \mathbb{1}\{|E_{i,j}| < C\}$. Then $\widehat{E}_{i,j} = E_{i,j}$ with probability at least $1 - 2d^{-40}$. Note that $\widehat{E}_{i,j}$ is not centered, we thus make a subtle adjustment by letting $E'_{i,j} := \widehat{E}_{i,j} - \alpha \cdot \text{sign}(\mathbb{E}[\widehat{E}_{i,j}])\beta_{i,j}$, where $\beta_{i,j}$ are independent of $\widehat{E}_{i,j}$ and are independent Bernoulli random variables with

$$\mathbb{P}(\beta_{i,j} = 1) = \frac{1}{\alpha} |\mathbb{E}[\widehat{E}_{i,j}]| \leq \frac{1}{\alpha} \mathbb{E}[|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\}] \leq (160 \log d + 4)d^{-40},$$

where the first inequality holds since $\mathbb{E}[\widehat{E}_{i,j}] + \mathbb{E}[E_{i,j} \mathbb{1}\{|E_{i,j}| \geq C\}] = \mathbb{E}[E_{i,j}] = 0$ and the second inequality follows from (S.62). From the definition of $E'_{i,j}$, it is not hard to see that $\mathbb{E}[E'_{i,j}] = 0$, $|E'_{i,j}| \leq 2C$, and $E'_{i,j} = E_{i,j}$ with probability at least $1 - (160 \log d + 5)d^{-40}$ for $i \in [N], j \in [J]$. Taking a union bound over all possible indices yields that $\mathbf{E} = \mathbf{E}'$ with probability at least $1 - O(d^{-20})$.

In the end, we need to control the spectral norm of $\text{Cov}(\mathbf{E}'_{i,S_l})$ for $l \in [L]$. For $j_1, j_2 \in S_l$, it follows from the Cauchy inequality that

$$|\text{Cov}(E'_{i,j_1}, E'_{i,j_2}) - \mathbb{E}[E_{i,j_1} E_{i,j_2}]|$$

$$\begin{aligned}
&= \left| \mathbb{E}[\widehat{E}_{i,j_1} \widehat{E}_{i,j_2}] - \mathbb{E}[\widehat{E}_{i,j_1}] \mathbb{E}[\widehat{E}_{i,j_2}] - \mathbb{E}[E_{i,j_1} E_{i,j_2}] \right| \\
&= |\mathbb{E}[E_{i,j_1} E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| < C\}] - \mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| \geq C\} E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| < C\}] \\
&\quad - \mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| \geq C\}] \mathbb{E}[E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| \geq C\}] - \mathbb{E}[E_{i,j_1} E_{i,j_2}]| \\
&= |\mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| \geq C\} E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| \geq C\}] + \mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| < C\} E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| \geq C\}] \\
&\quad + \mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| \geq C\} E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| < C\}] \\
&\quad + \mathbb{E}[E_{i,j_1} \mathbb{1}\{|E_{i,j_1}| \geq C\}] \mathbb{E}[E_{i,j_2} \mathbb{1}\{|E_{i,j_2}| \geq C\}]| \\
&\leq \mathbb{E}[|E_{i,j_1}|^2 \mathbb{1}\{|E_{i,j_1}| \geq C\}]^{\frac{1}{2}} \mathbb{E}[|E_{i,j_2}|^2 \mathbb{1}\{|E_{i,j_2}| \geq C\}]^{\frac{1}{2}} \\
&\quad + \mathbb{E}[|E_{i,j_1}|^2 \mathbb{1}\{|E_{i,j_1}| < C\}]^{\frac{1}{2}} \mathbb{E}[|E_{i,j_2}|^2 \mathbb{1}\{|E_{i,j_2}| \geq C\}]^{\frac{1}{2}} \\
&\quad + \mathbb{E}[|E_{i,j_2}|^2 \mathbb{1}\{|E_{i,j_1}| \geq C\}]^{\frac{1}{2}} \mathbb{E}[|E_{i,j_2}|^2 \mathbb{1}\{|E_{i,j_1}| < C\}]^{\frac{1}{2}} \\
&\quad + \max_{j \in S_l} (\mathbb{E}[|E_{i,j}| \mathbb{1}\{|E_{i,j}| \geq C\}])^2, \tag{S.63}
\end{aligned}$$

where we make use of the independence of β_{i,j_1} and β_{i,j_2} , $\mathbb{E}[\widehat{E}_{i,j}] + \mathbb{E}[E_{i,j} \mathbb{1}\{|E_{i,j}| \geq C\}] = 0$, and the definition of $E_{i,j}$.

Regarding the second moments of the truncated versions in (S.63), the tail inequality on a sub-exponential random variable gives that

$$\begin{aligned}
&\mathbb{E}[|\widetilde{E}_{i,j_1}|^2 \mathbb{1}\{|\widetilde{E}_{i,j_1}| \geq C\}] \leq 2C^2 \mathbb{P}[|\widetilde{E}_{i,j_1}| \geq C] + 2 \int_C^\infty x \mathbb{P}[|\widetilde{E}_{i,j_1}| \geq x] dx \\
&\leq 4C^2 d^{-40} + 4 \int_C^\infty x \exp(-\frac{x}{2\alpha}) dx \leq 4C^2 d^{-40} + 4\alpha \int_C^\infty \exp(-\frac{x}{4\alpha}) dx \\
&\leq 4C^2 d^{-40} + 16\alpha^2 d^{-20},
\end{aligned}$$

since $\exp(\frac{x}{4\alpha}) \geq \frac{x}{\alpha}$ for $x \geq C$. Besides, we have $\mathbb{E}[|E_{i,j}|^2 \mathbb{1}\{|E_{i,j}| < L\}] \leq \sigma_0^2$ for $i \in [N], j \in [J]$. Combining these pieces together with (S.62) gives that

$$|\text{Cov}(E'_{i,j_1}, E'_{i,j_2}) - \mathbb{E}[E_{i,j_1} E_{i,j_2}]| \leq O(\alpha d^{-9} \sigma_0 + \alpha^2 d^{-20}).$$

Now we are ready to upper-bound $\left\| \text{Cov}(\mathbf{E}_{i,S_j}) - \text{Cov}(\mathbf{E}'_{i,S_j}) \right\|$ provided the entrywise control above. Invoking the elementary fact that $\|\mathbf{A}\| \leq \|\mathbf{A}\|_F \leq \|\mathbf{A}\|_\infty \sqrt{n_1 n_2}$ for a matrix

$\mathbf{A} \in \mathbb{R}^{n_1 \times n_2}$, we have

$$\left\| \text{Cov}(\mathbf{E}_{i,S_j}) - \text{Cov}(\mathbf{E}'_{i,S_j}) \right\| \leq O(\alpha d^{-8} \sigma_0 + \alpha^2 d^{-19}),$$

since $M \leq d$. □

S.1.5 Additional Lemmas

Lemma S.8. *Suppose Assumptions S.5, S.7 hold. Then with probability at least $1 - O(d^{-10})$,*

$$\begin{aligned} \|\mathbf{E}\mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} &\lesssim \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^*}, \\ \|\mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} &\lesssim \frac{\sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma \sqrt{K \log d}}{\sigma_K^*}. \end{aligned}$$

Proof of Lemma S.8.

$$\begin{aligned} \|\mathbf{E}\mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} &= \max_{i \in [N]} \|\mathbf{E}_{i,:} \mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-2}\| \\ &\leq \max_{i \in [N]} \left(\|\mathbf{E}_{i,:}\|^2 \|\mathbf{U}^*\|_{2,\infty} / \sigma_K^{*2} + \|\mathbf{E}_{i,:} \mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^* \mathbf{\Lambda}^{*-2}\|_2 \right) \\ &\stackrel{(i)}{\lesssim} \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \max_{i \in [N]} \left(\tilde{\sigma} \sqrt{\log d} \|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^* \mathbf{\Lambda}^{*-2}\|_F + MB \log d \|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} \right) \\ &\lesssim \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^{*2}} \max_{i \in [N]} \|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^*\| + \frac{MB \log d}{\sigma_K^{*2}} \max_{i \in [N]} \|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^*\|_{2,\infty} \\ &\stackrel{(ii)}{\lesssim} \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\sigma_K^{*2}} \sigma \sqrt{J} + \frac{MB \log d}{\sigma_K^{*2}} (\sigma \sqrt{K \log d} + B \log d \sqrt{\frac{\mu_1 K}{N}}) \\ &\stackrel{(iii)}{\lesssim} \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\tilde{\sigma} \sigma \sqrt{K J \log d}}{\sigma_K^{*2}} \\ &\stackrel{(iv)}{\lesssim} \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\tilde{\sigma} \sqrt{K \log d}}{\kappa^* \sigma_K^*}, \end{aligned}$$

where (i) used (S.9) in Lemma S.5 and (S.6) in Lemma S.4 conditional on $\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^* \mathbf{\Lambda}^{*-2}$, (ii) used a similar treatment to (S.12) in Lemma S.5 on $\|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^*\|$ and (S.7) in Lemma S.4 on $\|\mathcal{P}_{:, -i}(\mathbf{E}^\top) \mathbf{U}^*\|_{2,\infty}$, (iii) used (S.2) in Assumption S.7, and (iv) is by (S.1) in Assumption S.7. Similarly,

$$\|\mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} = \max_{l \in [L]} \|\mathbf{E}_{:, S_l}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2}\|$$

$$\begin{aligned}
&\leq \max_{l \in [S_l]} \left(\|\mathbf{E}_{:,S_l}\|^2 \|\mathbf{V}^*\|_{2,\infty} / \sigma_K^{*2} + \|\mathbf{E}_{:,S_l}^\top \mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-2}\| \right) \\
&\stackrel{(v)}{\lesssim} \frac{\sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \sigma \sqrt{M \log d} \max_{l \in [L]} \|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-2}\|_F \\
&\quad + \sqrt{MB} \log d \max_{l \in [L]} \|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-2}\|_{2,\infty} \\
&\lesssim \frac{\sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma \sqrt{MK \log d}}{\sigma_K^{*2}} \max_{l \in [L]} \|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^*\| + \frac{\sqrt{MB} \log d}{\sigma_K^{*2}} \max_{l \in [L]} \|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^*\|_{2,\infty} \\
&\stackrel{(vi)}{\lesssim} \frac{\sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma \sqrt{MK \log d}}{\sigma_K^{*2}} \tilde{\sigma} \sqrt{N} + \frac{\sqrt{MB} \log d}{\sigma_K^{*2}} (\tilde{\sigma} \sqrt{K \log d} + MB \log d \sqrt{\frac{\mu_2 K}{J}}) \\
&\stackrel{(vii)}{\lesssim} \frac{\sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\tilde{\sigma} \sigma \sqrt{MKN \log d}}{\sigma_K^{*2}} \\
&\stackrel{(viii)}{\lesssim} \frac{\sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma \sqrt{K \log d}}{\kappa^* \sigma_K^*},
\end{aligned}$$

where (v) used (S.10) in Lemma S.5 and (S.7) in Lemma S.4 conditional on $\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^* \mathbf{\Lambda}^{*-2}$, (vi) used a similar treatment to (S.11) in Lemma S.5 on $\|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^*\|$ and (S.6) in Lemma S.4 on $\|\mathcal{P}_{:,-S_l}(\mathbf{E}) \mathbf{V}^*\|_{2,\infty}$, (vii) used (S.2) in Assumption S.7, and (viii) is by (S.1) in Assumption S.7. $\square$

S.2 Lemmas and Proofs for Theorem 2

Assumption S.8.

$$\sqrt{K} \kappa^3(\mathbf{\Pi}^*) \max \left\{ \frac{\tilde{\sigma} \sqrt{K \mu_1 \log d}}{\sigma_K(\mathbf{\Theta}^*)}, \frac{\kappa^* \sigma^2 J \sqrt{\mu_1 K}}{\sqrt{N} \sigma_K(\mathbf{\Pi}^*) \sigma_K(\mathbf{\Theta}^*)^2}, \frac{\sigma MB \log d \sqrt{(MN+J) \mu_2 K}}{\sqrt{J} \sigma_K(\mathbf{\Pi}^*) \sigma_K(\mathbf{\Theta}^*)^2} \right\} \ll 1.$$

Assumption S.8 is a sufficient condition for $\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi_1 \ll 1$ to hold. To see this,

$$\begin{aligned}
&\sqrt{K} \sigma_1(\mathbf{\Pi}^*) \cdot \kappa^2(\mathbf{\Pi}^*) \cdot \xi_1 \\
&= \sqrt{K} \sigma_1(\mathbf{\Pi}^*) \kappa^2(\mathbf{\Pi}^*) \frac{\tilde{\sigma} \sqrt{N \log d}}{\sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \sqrt{K} \sigma_1(\mathbf{\Pi}^*) \kappa^2(\mathbf{\Pi}^*) \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} \\
&\quad + \sqrt{K} \sigma_1(\mathbf{\Pi}^*) \kappa^2(\mathbf{\Pi}^*) \frac{\sigma MB \log d \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} \\
&\leq \sqrt{K} \kappa^3(\mathbf{\Pi}^*) \frac{\tilde{\sigma} \sqrt{K \mu_1 \log d}}{\sigma_K(\mathbf{\Theta}^*)} + \sqrt{K} \kappa^3(\mathbf{\Pi}^*) \frac{\kappa^* \sigma^2 J \sqrt{\mu_1 K}}{\sqrt{N} \sigma_K(\mathbf{\Pi}^*) \sigma_K(\mathbf{\Theta}^*)^2}
\end{aligned}$$

$$+ \sqrt{K} \kappa^3(\mathbf{\Pi}^*) \frac{\sigma M B \log d \sqrt{(MN + J) \mu_2 K}}{\sqrt{J} \sigma_K(\mathbf{\Pi}^*) \sigma_K(\mathbf{\Theta}^*)^2}.$$

Lemma S.9. $1/\sqrt{K} \leq \|\mathbf{\Pi}^*\|_{2,\infty} \leq 1$, $\|\mathbf{\Theta}^*\|_{2,\infty} \leq \sqrt{K}$, $\sigma_1(\mathbf{\Pi}^*) \leq \sqrt{N}$, $\sigma_1(\mathbf{\Theta}^*) \leq \sqrt{JK}$, and $\kappa(\mathbf{R}^*) \leq \kappa(\mathbf{\Theta}^*) \kappa(\mathbf{\Pi}^*)$.

Proof of Lemma S.9. The first inequality is because $\frac{1}{\sqrt{K}} \leq \|\boldsymbol{\pi}_i\| \leq \|\boldsymbol{\pi}_i\|_1 = 1$ by Cauchy-Schwarz inequality. The second inequality directly follows by the fact that each row of $\mathbf{\Theta}^*$ is a length K Bernoulli probability vector. The third inequality holds because $\sigma_1(\mathbf{\Pi}^*) \leq \|\mathbf{\Pi}^*\|_F \leq \sqrt{N} \|\mathbf{\Pi}^*\|_{2,\infty} \leq \sqrt{N}$ with Lemma S.2. The fourth inequality is because $\sigma_1(\mathbf{\Theta}^*) \leq \|\mathbf{\Theta}^*\|_F \leq \sqrt{JK}$. The last inequality is because $\sigma_1(\mathbf{R}^*) \leq \sigma_1(\mathbf{\Theta}^*) \sigma_1(\mathbf{\Pi}^*)$ and $\sigma_K(\mathbf{R}^*) \geq \sigma_K(\mathbf{\Theta}^*) \sigma_K(\mathbf{\Pi}^*)$. $\square$

Proof of Lemma 1. According to the proof of Theorem 2 in [Chen and Gu \(2024\)](#), Assumption 2 is sufficient for the generalized-GoM to be identifiable. Take the rows indices $\mathbf{S}$ for both sides of $\mathbf{R} = \mathbf{U}^* \mathbf{\Lambda} \mathbf{V}^{*\top}$, then

$$\mathbf{U}_{\mathbf{S},:}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} = \mathbf{R}_{\mathbf{S},:}^* = \mathbf{\Pi}_{\mathbf{S},:}^* \mathbf{\Theta}^{*\top} = \mathbf{\Theta}^{*\top}, \quad (\text{S.64})$$

which gives Eq (7). Further note that

$$\mathbf{U}^* = \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-1} = \mathbf{\Pi}^* \mathbf{\Theta}^{*\top} \mathbf{V}^* \mathbf{\Lambda}^{*-1}. \quad (\text{S.65})$$

Plug (S.64) into (S.65), and we have

$$\mathbf{U}^* = \mathbf{\Pi}^* \mathbf{U}_{\mathbf{S},:}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \mathbf{V}^* \mathbf{\Lambda}^{*-1} = \mathbf{\Pi}^* \mathbf{U}_{\mathbf{S},:}^*. \quad (\text{S.66})$$

Eq (S.66) indicates that $\mathbf{U}_{\mathbf{S},:}^*$ is full rank since both $\mathbf{U}^*$ and $\mathbf{\Pi}^*$ have rank K . Therefore, $\mathbf{\Pi}^* = \mathbf{U}^* \mathbf{U}_{\mathbf{S},:}^{*-1}$ and we have proved the first equation of (7). Furthermore, since $\mathbf{U}^{*\top} \mathbf{U}^* = \mathbf{I}_K$, Eq (S.66) also leads to $\mathbf{U}_{\mathbf{S},:}^{*\top} \mathbf{\Pi}^{*\top} \mathbf{\Pi}^* \mathbf{U}_{\mathbf{S},:}^* = \mathbf{I}_K$ and

$$\sigma_1(\mathbf{U}_{\mathbf{S},:}^*) = \frac{1}{\sigma_K(\mathbf{\Pi}^*)}, \quad \sigma_K(\mathbf{U}_{\mathbf{S},:}^*) = \frac{1}{\sigma_1(\mathbf{\Pi}^*)}, \quad \kappa(\mathbf{U}_{\mathbf{S},:}^*) = \kappa(\mathbf{\Pi}^*). \quad (\text{S.67})$$

$\square$

Lemma S.10 (Theorem 3 in [Gillis and Vavasis \(2013\)](#)). Let $\mathbf{M}' = \mathbf{M} + \mathbf{N} \in \mathbb{R}^{m \times n}$ where $\mathbf{M} = \mathbf{W}\mathbf{Z} = \mathbf{W}[\mathbf{I}_r | \mathbf{Z}']$, $\mathbf{W} \in \mathbb{R}^{m \times r}$ has rank r , $\mathbf{Z} \in \mathbb{R}_+^{r \times n}$, and $\sum_{k=1}^r \mathbf{Z}'_{kj} \leq 1, \forall j$ and $r \geq 2$. Let J be the index set extracted by SPA. If $\|\mathbf{N}\|_{\infty,2} \leq \xi$ with

$$\left(1 + 80 \frac{\|\mathbf{W}\|_{\infty,2}^2}{\sigma_r^2(\mathbf{W})}\right) \xi < \min \left\{ \frac{1}{2\sqrt{r-1}}, \frac{1}{4} \right\} \sigma_r(\mathbf{W}), \quad (\text{S.68})$$

then there exists a permutation P of $\{1, \dots, r\}$ such that

$$\max_{1 \leq j \leq r} \|\mathbf{M}'_{:,J(j)} - \mathbf{W}_{:,P(j)}\| \leq \bar{\xi} := \xi \left(1 + 80 \frac{\|\mathbf{W}\|_{\infty,2}^2}{\sigma_r^2(\mathbf{W})}\right).$$

Lemma S.11 (Modified from Theorem G.2 of [Mao et al. \(2021\)](#)). Let $\hat{\mathbf{S}}$ be the estimated simplex vertex index set returned by SPA. Suppose Assumptions [S.5](#), [S.6](#), [S.7](#) hold, and further assume that $\sqrt{K}\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1 \ll 1$. Then there exists a permutation matrix $\mathbf{P} \in \mathbb{R}^{K \times K}$ such that

$$\|\mathbf{U}_{\hat{\mathbf{S}},:} - \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U}\|_{2,\infty} = O(\kappa^2(\mathbf{\Pi}^*)\xi_1)$$

with probability at least $1 - O((N \vee J)^{-10})$.

Proof of Lemma S.11. In order to use Lemma S.10, let $r = K$, $\mathbf{M}' = \mathbf{U}^{*\top} \mathbf{U} \mathbf{U}^\top$, $\mathbf{M} = \mathbf{U}^{*\top}$, $\mathbf{W} = \mathbf{U}_{\mathbf{S},:}^{*\top}$, and $\mathbf{Z} = \mathbf{\Pi}^{*\top}$. Note that $\mathbf{U}^{*\top} = \mathbf{U}_{\mathbf{S},:}^{*\top} \mathbf{\Pi}^{*\top} = \mathbf{U}_{\mathbf{S},:}^{*\top} (\mathbf{I}_K | \mathbf{\Pi}_{(K+1):N}^{*\top})$. According to Theorem 1, $\mathbf{N} = \mathbf{M}' - \mathbf{M}$ satisfies $\|\mathbf{N}\|_{\infty,2} \leq \xi_1$ with probability at least $1 - O((N \vee J)^{-10})$. Notice that $\|\mathbf{U}_{\mathbf{S},:}^*\|_{2,\infty} \leq \|\mathbf{U}_{\mathbf{S},:}^*\|$ with Lemma S.2, thus

$$\frac{\|\mathbf{W}\|_{\infty,2}^2}{\sigma_K^2(\mathbf{W})} \leq \frac{\sigma_1^2(\mathbf{U}_{\mathbf{S},:}^*)}{\sigma_K^2(\mathbf{U}_{\mathbf{S},:}^*)} = \kappa^2(\mathbf{U}_{\mathbf{S},:}^*) = \kappa^2(\mathbf{\Pi}^*),$$

where the last equality is from (S.67). Therefore, $(1 + 80 \frac{K(\mathbf{W})^2}{\sigma_K^2(\mathbf{W})})\xi_1 \lesssim \kappa^2(\mathbf{\Pi}^*)\xi_1$. Furthermore, since $\sigma_K(\mathbf{W}) = \sigma_K(\mathbf{U}_{\mathbf{S},:}^*) = \frac{1}{\sigma_1(\mathbf{\Pi}^*)}$ with (S.67), inequality (S.68) is satisfied since $\sqrt{K}\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1 \ll 1$. With Lemma S.10, we know that there exists a permutation matrix $\mathbf{P} \in \mathbb{R}^{K \times K}$ such that

$$\|\mathbf{U}_{\hat{\mathbf{S}},:} \mathbf{U}^\top \mathbf{U}^* - \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^*\|_{2,\infty} = O(\kappa^2(\mathbf{\Pi}^*)\xi_1),$$

with probability at least $1 - O((N \vee J)^{-10})$. □

Lemma S.12. (Modified based on Lemma G.3 of [Mao et al. \(2021\)](#)). Suppose Assumptions S.5, S.6, S.7 hold. Let $\hat{\mathbf{S}}$ be the estimated simplex vertex index set returned by SPA. Then we have

$$\left\| \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U} \left(\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - (\mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U})^{-1} \right) \right\|_{2,\infty} = O(\sigma_1(\mathbf{\Pi}^*) \kappa^2(\mathbf{\Pi}^*) \xi_1)$$

with probability at least $1 - O((N \vee J)^{-10})$.

Proof of Lemma S.12. Define $\mathbf{F} = \mathbf{U}^{*\top} \mathbf{U}$, $\mathbf{W} = \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{F}$. For every $i \in [N]$,

$$\begin{aligned} & \left\| \mathbf{e}_i^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U} \left(\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - (\mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U})^{-1} \right) \right\| \\ &= \left\| \mathbf{e}_i^\top \mathbf{U}^* \mathbf{F} (\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - \mathbf{W}^{-1}) \right\| = \left\| \mathbf{e}_i^\top \mathbf{U}^* \mathbf{F} \mathbf{W}^{-1} (\mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:}) \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| \\ &= \left\| \mathbf{e}_i^\top \mathbf{U}^* \mathbf{F} \mathbf{F}^{-1} \mathbf{U}_{\mathbf{S},:}^{*-1} \mathbf{P} (\mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:}) \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| = \left\| \mathbf{e}_i^\top \mathbf{\Pi}^* \mathbf{P} (\mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:}) \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| \quad (\text{S.69}) \\ &\stackrel{(i)}{\leq} \left\| \mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:} \right\|_{2,\infty} \left\| \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| = \left\| \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U} - \mathbf{U}_{\hat{\mathbf{S}},:} \right\|_{2,\infty} \left\| \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| \\ &\stackrel{(ii)}{\leq} O(\kappa^2(\mathbf{\Pi}^*) \xi_1') \left\| \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\| \quad \text{with probability at least } 1 - O((N \vee J)^{-10}). \end{aligned}$$

Here (i) holds because $\mathbf{\Pi}^* \mathbf{P}$ is non-negative and has rows with l_1 norm 1; $\mathbf{\Pi}^* \mathbf{P} (\mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:})$ is a convex combination of rows of $\mathbf{W} - \mathbf{U}_{\hat{\mathbf{S}},:}$, and the l_2 norm of a convex combination can not exceed the maximum norm of the vertices. Inequality (ii) uses Lemma S.11.

Now we bound $\left\| \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} \right\|$. With Weyl's inequality and $\sigma_k(\mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U}) = \sigma_k(\mathbf{U}_{\mathbf{S},:}^*), \forall k$, we have with probability at least $1 - O((N \vee J)^{-10})$

$$\begin{aligned} |\sigma_k(\mathbf{U}_{\hat{\mathbf{S}},:}) - \sigma_k(\mathbf{U}_{\mathbf{S},:}^*)| &\leq \left\| \mathbf{U}_{\hat{\mathbf{S}},:} - \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U} \right\| \\ &\stackrel{(iii)}{\leq} \sqrt{K} \left\| \mathbf{U}_{\hat{\mathbf{S}},:} - \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U} \right\|_{2,\infty} \\ &\stackrel{(iv)}{=} O(\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \xi_1'), \end{aligned}$$

for $k = 1, \dots, K$, where (iii) uses Lemma S.2 and (iv) uses Lemma S.11. It immediately follows that

$$\sigma_K(\mathbf{U}_{\hat{\mathbf{S}},:}) \geq \sigma_K(\mathbf{U}_{\mathbf{S},:}^*) \left(1 - O \left(\frac{\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \xi_1'}{\sigma_K(\mathbf{U}_{\mathbf{S},:}^*)} \right) \right)$$

$$= \frac{1}{\sigma_1(\mathbf{\Pi}^*)} \left(1 - O \left(\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \right) \right)$$

with (S.67). Therefore,

$$\|\mathbf{U}_{\hat{\mathbf{S}},:}^{-1}\| = \frac{1}{\sigma_K(\mathbf{U}_{\hat{\mathbf{S}},:})} \leq \frac{\sigma_1(\mathbf{\Pi}^*)}{\left(1 - O \left(\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \right) \right)} = O(\sigma_1(\mathbf{\Pi}^*)), \quad (\text{S.70})$$

since $\sqrt{K} \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \ll 1$. By plugging Eq (S.70) into Eq (S.69), we get

$$\left\| \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U} \left(\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - (\mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U})^{-1} \right) \right\|_{2,\infty} = O \left(\kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \right)$$

with probability at least $1 - O((N \vee J)^{-10})$. $\square$

Proof of Theorem 2. Recall that $\hat{\mathbf{\Pi}} = \mathbf{U} \mathbf{U}_{\hat{\mathbf{S}},:}^{-1}$. We control the estimation error of each row up to a permutation:

$$\begin{aligned} & \max_{i \in [N]} \|\mathbf{e}_i^\top (\hat{\mathbf{\Pi}} - \mathbf{\Pi}^* \mathbf{P})\|_2 \\ &= \max_{i \in [N]} \|\mathbf{e}_i^\top (\mathbf{U} \mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - \mathbf{U}^* \mathbf{U}_{\mathbf{S},:}^{*-1} \mathbf{P})\|_2 \\ &\leq \max_{i \in [N]} \left(\|\mathbf{e}_i^\top (\mathbf{U} - \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U}) \mathbf{U}_{\hat{\mathbf{S}},:}^{-1}\| + \|\mathbf{e}_i^\top \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U} (\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - (\mathbf{U}^{*\top} \mathbf{U})^{-1} \mathbf{U}_{\mathbf{S},:}^{*-1} \mathbf{P})\|_2 \right) \\ &\leq \|\mathbf{U} - \mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U}\|_{2,\infty} \|\mathbf{U}_{\hat{\mathbf{S}},:}^{-1}\| + \|\mathbf{U}^* \mathbf{U}^{*\top} \mathbf{U} (\mathbf{U}_{\hat{\mathbf{S}},:}^{-1} - (\mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U})^{-1})\|_{2,\infty} \\ &\stackrel{(i)}{\lesssim} \sigma_1(\mathbf{\Pi}^*) \xi'_1 + \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \\ &\lesssim \kappa^2(\mathbf{\Pi}^*) \sigma_1(\mathbf{\Pi}^*) \xi'_1 \quad \text{with probability at least } 1 - O((N \vee J)^{-10}), \end{aligned}$$

where (i) uses Lemma S.12 and Eq (S.70).

On the other hand, recall $\hat{\mathbf{\Theta}} = \mathbf{V} \mathbf{\Lambda} \mathbf{U}_{\hat{\mathbf{S}},:}^\top$ and ξ_3 is the upper bound in (8) in Theorem 1.

$$\begin{aligned} & \|\hat{\mathbf{\Theta}} \mathbf{P}^\top - \mathbf{\Theta}^*\|_\infty \\ &= \|\mathbf{P} \mathbf{U}_{\hat{\mathbf{S}},:} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}_{\mathbf{S},:}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top}\|_\infty \\ &\leq \|(\mathbf{P} \mathbf{U}_{\hat{\mathbf{S}},:} - \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U}) \mathbf{\Lambda} \mathbf{V}^\top\|_\infty + \|\mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}_{\mathbf{S},:} \mathbf{\Lambda} \mathbf{V}^\top\|_\infty \\ &\quad + \|\mathbf{U}_{\mathbf{S},:} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}_{\mathbf{S},:}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top}\|_\infty \\ &\stackrel{(i)}{\leq} \|\mathbf{U}_{\hat{\mathbf{S}},:} - \mathbf{P}^\top \mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U}\|_{2,\infty} \cdot \sigma_1(\mathbf{R}) \cdot \|\mathbf{V}\|_{2,\infty} + \|\mathbf{U}_{\mathbf{S},:}^* \mathbf{U}^{*\top} \mathbf{U} - \mathbf{U}_{\mathbf{S},:}\|_{2,\infty} \cdot \sigma_1(\mathbf{R}) \cdot \|\mathbf{V}\|_{2,\infty} \end{aligned}$$

$$\begin{aligned}
& + \|\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top - \mathbf{R}^*\|_\infty \\
& \stackrel{(ii)}{\lesssim} (\kappa^2(\mathbf{\Pi}^*) + 1) \cdot \xi'_1 \cdot \sigma_1(\mathbf{R}) \cdot \|\mathbf{V}\|_{2,\infty} + \xi_3 \\
& \stackrel{(iii)}{\lesssim} \kappa^2(\mathbf{\Pi}^*) \cdot \xi'_1 \cdot \sigma_1(\mathbf{R}) \cdot \left(\sqrt{\frac{K\mu_2}{J}} + \xi'_2\right) + \xi_3 \\
& \stackrel{(iv)}{\lesssim} \kappa^2(\mathbf{\Pi}^*) \sigma_1^* \left(\sqrt{\frac{K\mu_2}{J}} + \xi'_2\right) \xi'_1 + \xi_3 \\
& \stackrel{(v)}{\lesssim} \kappa^2(\mathbf{\Pi}^*) \xi_3,
\end{aligned}$$

with probability at least $1 - O((N \vee J)^{-10})$. Here (i) uses Lemma S.2, (ii) uses Lemma S.11 and the definitions of ξ'_1, ξ_3 , and (iii) is from (S.54) and the definitions of μ_2 and ξ_2 . In addition, (iv) holds since $|\sigma_1(\mathbf{R}) - \sigma_1(\mathbf{R}^*)| \leq \|\mathbf{E}\| \lesssim \sigma\sqrt{J} + \tilde{\sigma}\sqrt{N} \leq \sigma(\sqrt{J} + \sqrt{MN}) \lesssim \sigma_K^* \leq \sigma_1^*$ holds with probability at least $1 - O((N \vee J)^{-20})$, where we use Assumption S.7 and Lemma S.5. In addition, (v) follows since $\xi'_1 \xi'_2 \sigma_1^* + \sqrt{\frac{\mu_2 K}{J}} \sigma_1^* \xi'_1 \lesssim \xi_3$ from (S.58) in the proof of Theorem 1. $\square$

Proof of Corollary 1. Given the additional assumptions, with probability at least $1 - O((N \vee J)^{-10})$ we have

$$\begin{aligned}
& \left\| \hat{\mathbf{\Pi}} - \mathbf{\Pi}^* \mathbf{P} \right\|_{2,\infty} \\
& \lesssim \kappa^2(\mathbf{\Pi}^*) \cdot \sigma_1(\mathbf{\Pi}^*) \cdot \xi'_1 \\
& \lesssim_{\sigma_1(\mathbf{\Pi}^*)} \left(\frac{\tilde{\sigma}\sqrt{\mu_1 K \log d}}{\sigma_K^*} + \frac{\sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma M B \log d \sqrt{MN + J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} \right) \\
& \leq \frac{\tilde{\sigma}\sqrt{\mu_1 K \log d}}{\sigma_K(\mathbf{\Theta}^*)} + \frac{\sigma^2}{\sigma_K(\mathbf{\Pi}^*) \sigma_K^2(\mathbf{\Theta}^*)} \left(J \sqrt{\frac{\mu_1 K}{N}} + \frac{M B \log d \sqrt{MN + J}}{\sigma} \sqrt{\frac{\mu_2 K}{J}} \right) \\
& \leq \frac{\tilde{\sigma}\sqrt{\mu_1 K \log d}}{\sigma_K(\mathbf{\Theta}^*)} + \frac{\sigma^2 \sqrt{JK}}{\sigma_K(\mathbf{\Pi}^*) \sigma_K^2(\mathbf{\Theta}^*)} \left(\sqrt{r\mu_1} + \sqrt{1 + \frac{M}{r} \frac{M B \log d}{\sigma \sqrt{J}}} \sqrt{\mu_2} \right),
\end{aligned}$$

where we use the fact that $\sigma_K^* \geq \sigma_K(\mathbf{\Pi}^*) \sigma_K(\mathbf{\Theta}^*)$.

For the error bound of $\hat{\mathbf{\Theta}}$,

$$\|\hat{\mathbf{\Theta}} \mathbf{P}^\top - \mathbf{\Theta}^*\|_\infty \lesssim K \sigma \log d \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + M B \log d \left(\frac{\mu_1 K}{MN} + \frac{\mu_2 K}{J} \right)$$

with probability at least $1 - O((N \vee J)^{-10})$. $\square$

Lemma S.13. (Lemma 3.6 in [Mao et al. \(2021\)](#).) If $\pi_i^* \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\alpha})$ for each $i \in [N]$

with $\alpha_{\max} = \max_k \alpha_k$, $\alpha_{\min} = \min_k \alpha_k$, $\alpha_0 = \sum_k \alpha_k$ and $\nu := \alpha_0 / \alpha_{\min}$, then

$$\begin{aligned} \mathbb{P} \left(\sigma_K^2(\mathbf{\Pi}^*) \geq \frac{N}{2\nu(1+\alpha_0)} \right) &\geq 1 - K \exp \left(-\frac{N}{36\nu^2(1+\alpha_0)^2} \right), \\ \mathbb{P} \left(\sigma_1^2(\mathbf{\Pi}^*) \leq \frac{3N(\alpha_{\max} + \|\boldsymbol{\alpha}\|^2)}{2\alpha_0(1+\alpha_0)} \right) &\geq 1 - K \exp \left(-\frac{N}{36\nu^2(1+\alpha_0)^2} \right), \\ \mathbb{P} \left(\kappa^2(\mathbf{\Pi}^*) \leq 3 \frac{\alpha_{\max} + \|\boldsymbol{\alpha}\|^2}{\alpha_{\min}} \right) &\geq 1 - 2K \exp \left(-\frac{N}{36\nu^2(1+\alpha_0)^2} \right). \end{aligned} \quad (\text{S.71})$$

Lemma S.14. If $(\theta_{M(l-1)+c,k}^*)_{c=1}^M \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\beta})$ for each $l \in [L]$, $k \in [K]$ with $\beta_0 =$

$\sum_{c=1}^M \beta_c$, $c_1 = \frac{\beta_0^2 - \|\boldsymbol{\beta}\|^2}{\beta_0^2(1+\beta_0)}$, and $c_2 = K \frac{\|\boldsymbol{\beta}\|^2}{\beta_0^2} + c_1$, then

$$\begin{aligned} \mathbb{P} \left(\sigma_K^2(\boldsymbol{\Theta}^*) \geq \frac{c_1 L}{2} \right) &\geq 1 - K \exp \left(-\frac{c_1^2 L}{80K^2} \right), \\ \mathbb{P} \left(\sigma_1^2(\boldsymbol{\Theta}^*) \leq \frac{3c_2 L}{2} \right) &\geq 1 - K \exp \left(-\frac{c_1^2 L}{80K^2} \right), \\ \mathbb{P} \left(\kappa^2(\boldsymbol{\Theta}^*) \leq \frac{3c_2}{c_1} \right) &\geq 1 - 2K \exp \left(-\frac{c_1^2 L}{80K^2} \right). \end{aligned} \quad (\text{S.72})$$

Proof of Lemma S.14 (Modified based on the proof of Lemma 3.6 in [Mao et al. \(2021\)](#).)

Write

$$\boldsymbol{\Theta}^* = (\boldsymbol{\Theta}_1^{*\top}, \dots, \boldsymbol{\Theta}_L^{*\top})^\top,$$

where $\boldsymbol{\Theta}_l^* = (\boldsymbol{\theta}_{l1}^*, \dots, \boldsymbol{\theta}_{lK}^*) \in [0, 1]^{C \times K}$ with $\boldsymbol{\theta}_{lk}^* = (\theta_{C(l-1)+c,k}^*)_{c=1}^C \stackrel{i.i.d}{\sim} D(\boldsymbol{\beta})$ for each $l \in [L]$ and $k \in [K]$. Denote $\mathbf{M} := \boldsymbol{\Theta}^{*\top} \boldsymbol{\Theta}^*$. Note that $\mathbf{M} - \mathbb{E}[\mathbf{M}] = \sum_l \mathbf{X}_l$, where each $\mathbf{X}_l = \boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^* - \mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]$ is an independent mean zero random $K \times K$ matrix. Since $\|\boldsymbol{\theta}_{lk}^*\| \leq \|\boldsymbol{\theta}_{lk}^*\|_1 = 1$, $\|\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*\| \leq \|\boldsymbol{\Theta}_l^*\|_F^2 \leq K$.

Also note that

$$\mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*] = \frac{\|\boldsymbol{\beta}\|^2}{\beta_0^2} \mathbf{1}\mathbf{1}^\top + \text{diag} \left(\frac{\beta_0^2 - \|\boldsymbol{\beta}\|^2}{\beta_0^2(1+\beta_0)} \right).$$

With Weyl's inequality,

$$\lambda_K(\mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]) = \frac{\beta_0^2 - \|\boldsymbol{\beta}\|^2}{\beta_0^2(1+\beta_0)} =: c_1 < 1,$$

$$\lambda_1(\mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]) = K \frac{\|\boldsymbol{\beta}\|^2}{\beta_0^2} + \frac{\beta_0^2 - \|\boldsymbol{\beta}\|^2}{\beta_0^2(1 + \beta_0)} =: c_2 < K + 1.$$

Therefore,

$$\|\mathbf{X}_l\| \leq \|\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*\| + \|\mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]\| \leq K + c_2 < 2K + 1 \leq 3K.$$

Furthermore, by Jensen's inequality,

$$\|\mathbb{E}[\mathbf{X}_l^2]\| \leq \mathbb{E}[\|\mathbf{X}_l^2\|] \leq \mathbb{E}[\|\mathbf{X}_l\|^2] \leq 9K^2.$$

By the matrix Bernstein inequality in [Tropp \(2012\)](#), for all $\tau > 0$, we have

$$\Pr(\sigma_1(\mathbf{M} - \mathbb{E}[\mathbf{M}]) \geq \tau) = \Pr\left(\sigma_1\left(\sum_{l=1}^L \mathbf{X}_l\right) \geq \tau\right) \leq K \exp\left(-\frac{\tau^2}{18LK^2 + 2K\tau}\right) := \delta_\tau.$$

By Weyl's inequality, with probability at least $1 - \delta_\tau$,

$$\begin{aligned} |\sigma_1(\mathbf{M}) - \sigma_1(\mathbb{E}[\mathbf{M}])| &\leq \sigma_1(\mathbf{M} - \mathbb{E}[\mathbf{M}]) \leq \tau \\ |\sigma_K(\mathbf{M}) - \sigma_K(\mathbb{E}[\mathbf{M}])| &\leq \sigma_1(\mathbf{M} - \mathbb{E}[\mathbf{M}]) \leq \tau. \end{aligned}$$

Also we have

$$\begin{aligned} \sigma_K(\mathbb{E}[\mathbf{M}]) &= \sigma_K\left(\sum_l \mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]\right) \geq Lc_1. \\ \sigma_1(\mathbb{E}[\mathbf{M}]) &= \sigma_1\left(\sum_l \mathbb{E}[\boldsymbol{\Theta}_l^{*\top} \boldsymbol{\Theta}_l^*]\right) \leq Lc_2 \end{aligned}$$

For $\sigma_K(\mathbf{M})$, choose $\tau_1 = \frac{c_1 L}{2}$. Then with probability at least

$$1 - \delta_{\tau_1} = 1 - K \exp\left(-\frac{c_1^2 L}{72K^2 + 4Kc_1}\right) \geq 1 - K \exp\left(-\frac{c_1^2 L}{80K^2}\right),$$

one has

$$\sigma_K^2(\boldsymbol{\Theta}^*) = \sigma_K(\mathbf{M}) \geq \sigma_K(\mathbb{E}[\mathbf{M}]) - |\sigma_K(\mathbf{M}) - \sigma_K(\mathbb{E}[\mathbf{M}])| \geq \tau_2 = \frac{c_2 L}{2}.$$

For $\sigma_1(\mathbf{M})$, choose $\tau_2 = \frac{c_2 L}{2}$. Then with probability at least

$$1 - \delta_{\tau_2} = 1 - K \exp\left(-\frac{c_2^2 L}{72K^2 + 4Kc_2}\right) \geq 1 - K \exp\left(-\frac{c_2^2 L}{80K^2}\right),$$

one has

$$\sigma_1^2(\Theta^*) = \sigma_1(\mathbf{M}) \leq |\sigma_1(\mathbf{M}) - \sigma_1(\mathbb{E}[\mathbf{M}])| + \sigma_1(\mathbb{E}[\mathbf{M}]) \leq 3\tau_2 = \frac{3c_2L}{2}.$$

With probability at least $1 - 2K \exp\left(-\frac{c_1^2L}{80K^2}\right)$, one has

$$\kappa^2(\Theta^*) = \frac{\sigma_1^2(\Theta^*)}{\sigma_K^2(\Theta^*)} \leq \frac{3c_2}{c_1}.$$

□

Proof of Corollary 2. First note that $J = LM, B \leq 1, \sigma \leq 1/2, \tilde{\sigma} \leq \sqrt{M}/2$ for the flattened data matrix with Bernoulli entries. Define the event

$$\Omega := \left\{ (\Pi^*, \Theta^*) : \sigma_K(\Pi^*) \gtrsim \frac{\sqrt{N}}{K}, \sigma_1(\Pi^*) \lesssim \sqrt{\frac{N}{K}}, \kappa(\Pi^*) \lesssim \sqrt{K}, \right. \\ \left. \sigma_K(\Theta^*) \gtrsim \frac{\sqrt{J}}{M}, \sigma_1(\Theta^*) \leq \frac{\sqrt{KJ}}{M}, \kappa(\Theta^*) \leq \sqrt{K} \right\}.$$

With (13), the probabilities on the RHS in (S.71) and (S.72) are in the order of $1 - O(KN^{-3})$ and $1 - O(KL^{-3})$, respectively. Along with the assumptions on α and β , we have

$$\Pr(\Omega) \geq 1 - O(K(N \wedge L)^{-3}).$$

Below we discuss for $(\Pi^*, \Theta^*) \in \Omega$. With the lower bounds on N, J , we have

$$\sigma \kappa^*(M\sqrt{N} + \sqrt{J}) \lesssim K(M\sqrt{N} + \sqrt{J}) \lesssim \frac{\sqrt{NJ}}{MK} \lesssim \sigma_K(\Pi^*)\sigma_K(\Theta^*),$$

and thus (S.1) in Assumption S.7 holds. Now we bound μ_1, μ_2 with Lemmas S.2 and S.9 and (S.67). Since $\mathbf{U}^* = \Pi^* \mathbf{U}_{\mathbf{S},:}^*$, we have

$$\mu_1 = \frac{N}{K} \|\mathbf{U}^*\|_{2,\infty}^2 \leq \frac{N}{K} \|\Pi^*\|_{2,\infty}^2 \|\mathbf{U}_{\mathbf{S},:}^*\|^2 \leq \frac{N}{K\sigma_K^2(\Pi^*)} \lesssim K.$$

Similarly, $\mathbf{V}^* = \Theta^* \mathbf{U}_{\mathbf{S},:}^* (\mathbf{U}_{\mathbf{S},:}^{*\top} \mathbf{U}_{\mathbf{S},:}^*)^{-1} \Lambda^{*-1}$. Thus,

$$\mu_2 = \frac{J}{K} \|\mathbf{V}^*\|_{2,\infty}^2 \leq \frac{J}{K} \|\Theta^*\|_{2,\infty}^2 \frac{\|\mathbf{U}_{\mathbf{S},:}^*\|^2}{\sigma_K^4(\mathbf{U}_{\mathbf{S},:}^*) \sigma_K^2(\Pi^*) \sigma_K^2(\Theta^*)} \leq J \frac{\kappa^4(\Pi^*)}{\sigma_K^2(\Theta^*)} \lesssim K^2 M^2.$$

One can verify that (S.2) in Assumption S.7 holds with the lower bounds on N, J in Assumption 4. Furthermore, we can verify that Assumption S.8 is satisfied with the lower bounds on N, J and (29), thus Assumption (21) is satisfied. Now we have verified that Assumption S.7 holds. By Remark 1, with probability at least $1 - O((N \vee J)^{-10})$ one has

$$\begin{aligned}\xi_1 &\lesssim \frac{M^{3/2}K^2\sqrt{\log d}}{\sqrt{NJ}} + \frac{M^2K^4}{N^{3/2}}, \\ \xi_3 &\lesssim \frac{M^{5/2}K^{7/2}\log d}{\sqrt{J}} + \frac{M^2K^{7/2}\log d}{\sqrt{N}}.\end{aligned}$$

Furthermore, with Theorem 2,

$$\begin{aligned}\|\hat{\Pi} - \Pi^*\mathbf{P}\|_{2,\infty} &\lesssim \frac{M^{3/2}K^{5/2}\sqrt{\log d}}{\sqrt{J}} + \frac{\sqrt{M^2K^{9/2}}}{N}, \\ \|\hat{\Theta} - \Theta^*\mathbf{P}\|_\infty &\lesssim K\xi_3 \lesssim \frac{M^{5/2}K^{9/2}\log d}{\sqrt{J}} + \frac{M^2K^{9/2}\log d}{\sqrt{N}}.\end{aligned}$$

□

S.3 Additional Computational Details

S.3.1 Gibbs Sampling Updating Steps

For simplicity of presentation, we set the number of possible responses as a constant, i.e. $C_j \equiv C$ for all items j ; extensions to fully general cases is straightforward. Here we use slightly different notations. Denote $\boldsymbol{\theta}_{jk} \in \mathbb{R}^C$ as the probabilities for all C response categories for item j and extreme latent profile k . When written in the form of the likelihood function, the GoM model is equivalent to assuming the existence of latent class variables $Z_{ij} \in \{1, \dots, K\}$ for each individual i and each item j .

$$\begin{aligned}R_{ij}|Z_{ij} = k &\stackrel{i.i.d}{\sim} \text{Categorical}([C]; \boldsymbol{\theta}_{jk}); \\ Z_{ij}|\boldsymbol{\pi}_i &\stackrel{i.i.d}{\sim} \text{Categorical}([K]; \boldsymbol{\pi}_i); \\ \boldsymbol{\pi}_i &\stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\alpha}), \boldsymbol{\alpha} \in \mathbb{R}^K;\end{aligned}$$

$$\boldsymbol{\theta}_{jk} \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\beta}), \quad \boldsymbol{\beta} \in \mathbb{R}^C.$$

We write $Z_{ijk} := \mathbb{1}(Z_{ij} = k)$ and $R_{ijc} := \mathbb{1}(R_{ij} = c)$. The likelihood function of the GoM model is then

$$L(\mathbf{R}; \boldsymbol{\Theta}, \mathbf{Z}, \boldsymbol{\Pi}) = \prod_{i=1}^N \prod_{j=1}^J \prod_{k=1}^K \left(\prod_{c=1}^C \theta_{jkc}^{R_{ijc}} \right)^{Z_{ijk}}.$$

- For the conditional distribution of $\boldsymbol{\theta}_{jk}$ we have

$$p(\boldsymbol{\theta}_{jk} | \mathbf{R}, \mathbf{Z}, \boldsymbol{\Pi}) \propto \prod_{i=1}^N \left(\prod_{c=1}^C \theta_{jkc}^{R_{ijc}} \right)^{Z_{ijk}} \prod_{c=1}^C \theta_{jkc}^{\beta_c - 1} \propto \prod_{c=1}^C \theta_{jkc}^{\sum_{i=1}^N R_{ijc} Z_{ijk} + \beta_c - 1},$$

which leads to

$$\boldsymbol{\theta}_{jk} | \mathbf{R}, \mathbf{Z}, \boldsymbol{\Pi} \sim \text{Dirichlet} \left(\left[\sum_{i=1}^N R_{ijc} Z_{ijk} + \beta_c \right]_c \right).$$

- For $\boldsymbol{\pi}_i$, since

$$p(\mathbf{Z}_i | \boldsymbol{\pi}_i) = \prod_{j=1}^J \prod_{k=1}^K \pi_{ik}^{Z_{ijk}},$$

we have

$$p(\boldsymbol{\pi}_i | \mathbf{Z}) \propto \prod_{k=1}^K \pi_{ik}^{\sum_{j=1}^J Z_{ijk}} \prod_{k=1}^K \pi_{ik}^{\alpha_k - 1} = \prod_{k=1}^K \pi_{ik}^{\sum_{j=1}^J Z_{ijk} + \alpha_k - 1}.$$

Thus

$$\boldsymbol{\pi}_i | \mathbf{Z} \sim \text{Dirichlet} \left(\left[\sum_{j=1}^J Z_{ijk} + \alpha_k \right]_k \right).$$

- For Z_{ij} , with

$$p(R_{ij} | Z_{ij} = k) = \prod_{c=1}^C \theta_{jkc}^{R_{ijc}}$$

and $p(Z_{ij} = k) = \pi_{ik}$, we have

$$p(Z_{ij} = k | R_{ij}) \propto \prod_{c=1}^C \theta_{jkc}^{R_{ijc}} \pi_{ik},$$

and thus

$$p(Z_{ij} = k | \mathbf{R}, \boldsymbol{\Theta}, \boldsymbol{\Pi}) = \frac{\prod_{c=1}^C \theta_{jkc}^{R_{ijc}} \pi_{ik}}{\sum_{k=1}^K \prod_{c=1}^C \theta_{jkc}^{R_{ijc}} \pi_{ik}}.$$

S.3.2 Additional Algorithms and Figures

Algorithm 2 below estimates the GoM model for Poisson data. The difference of Algorithm 2 from Algorithm 1 is the truncation for $\hat{\Theta}$.

Algorithm 2 Poisson-GoM Estimation with Potential Local Dependence

Input: Count matrix $\mathbf{R} \in \mathbb{N}_+^{N \times J}$, number of extreme profiles K .

Output: Estimated $\hat{\mathbf{S}} \in \mathbb{R}^K$, $\hat{\Theta}^{(\text{post})} \in \mathbb{R}^{J \times K}$, $\hat{\Pi}^{(\text{post})} \in \mathbb{R}^{N \times K}$

Get the top K singular value decomposition of $\mathbf{R}$ as $\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top$

$\mathbf{Y} = \mathbf{U}$

for $k \in \{1, \dots, K\}$ **do**

$$\hat{S}_k = \arg \max(\{\|\mathbf{Y}_{i,:}\|_2 : i \in [N] \setminus \hat{\mathbf{P}}\})$$

$$\mathbf{u} = \mathbf{Y}_{\hat{S}_k,:} / \|\mathbf{Y}_{\hat{S}_k,:}\|_2$$

$$\mathbf{Y} = \mathbf{Y}(\mathbf{I}_K - \mathbf{u}\mathbf{u}^\top)$$

end for

$$\hat{\Pi} = \mathbf{U}(\mathbf{U}_{\hat{\mathbf{S}},:})^{-1}$$

$$\hat{\Pi}^{(\text{post})} = \text{diag}(\hat{\Pi}_+ \mathbf{1}_K)^{-1} \hat{\Pi}_+$$

$$\hat{\Theta} = \mathbf{V}\mathbf{\Lambda}\mathbf{U}_{\hat{\mathbf{S}},:}^\top$$

$$\text{Let } \hat{\Theta}^{(\text{post})} = (\hat{\theta}_{j,k}^{(\text{post})}) \text{ with } \hat{\theta}_{j,k}^{(\text{post})} = \begin{cases} \hat{\theta}_{j,k} & \text{if } \hat{\theta}_{j,k} \geq 0 \\ \epsilon & \text{if } \hat{\theta}_{j,k} < 0 \end{cases}$$

Figure S.1 presents the heatmaps of the covariance matrix of the true error matrix $\mathbf{R} - \mathbf{R}^*$ (left) and the estimated error matrix $\mathbf{R} - \hat{\Pi}\hat{\Theta}^\top$ (right) in a simulation study. Whether or not there is local dependence, the covariance matrix of the estimated error matrix empirically approximates that of the ground truth error matrix well. The details of the simulation setting can be found in Section 5.

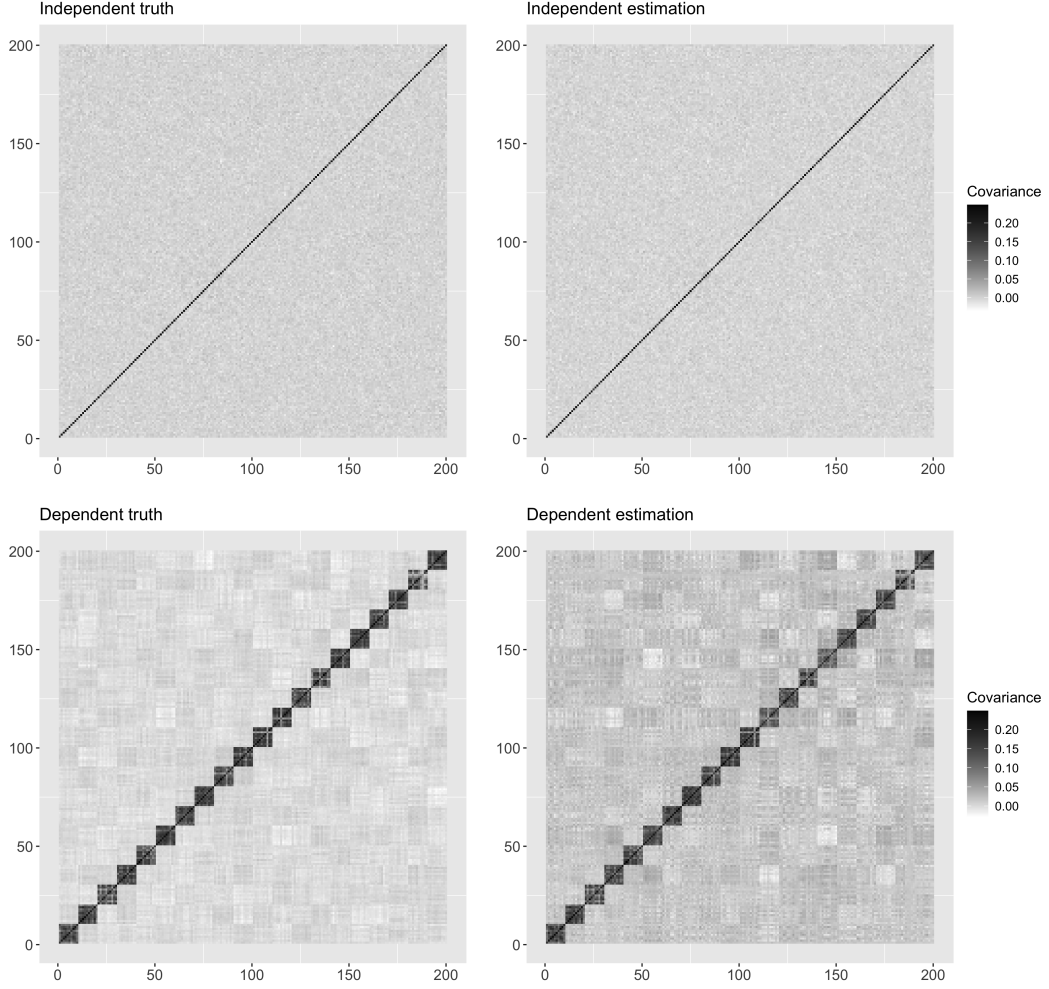


Figure S.1: Heatmaps of the covariance matrices of $\mathbf{R} - \mathbf{R}^*$ (left) and $\mathbf{R} - \hat{\Pi}\hat{\Theta}^\top$ (right), without local dependence (upper) and with local dependence (lower).